

Lecture 3

A 3D visualization of a quantum spin chain, showing a lattice of blue and purple spheres with arrows indicating spin states. Red wavy lines represent excitations or entanglement. Text overlays include "IBM Quantum" and "Slides? will also post on BSS and zlatko-minev.com/blog". Social media icons and handles are at the bottom.

Slides? will also post on BSS
and zlatko-minev.com/blog

Quantum Error Mitigation for Non-Equilibrium Quantum Dynamics

Lecture 2

Probabilistic error cancelation (PEC)

- Summarize one qubit example
- Analogy to random walks
- Error bars & confidence
- Generalize (optional)
- Show unbiased estimator

Learning quantum noise

- Challenge
- Overcoming: sparse model

Lecture 3

Putting it together

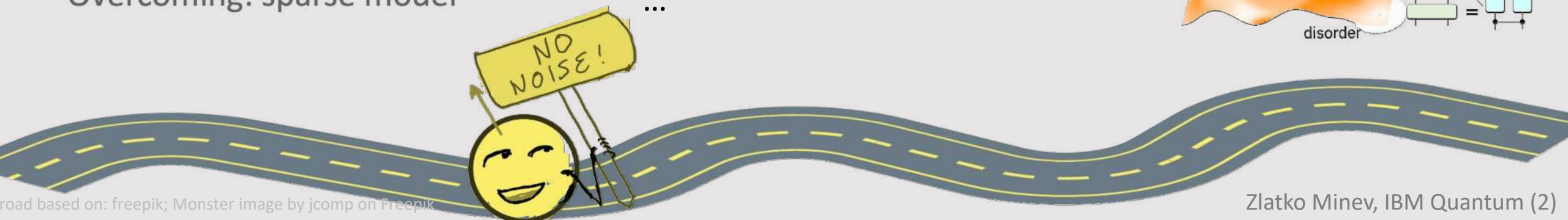
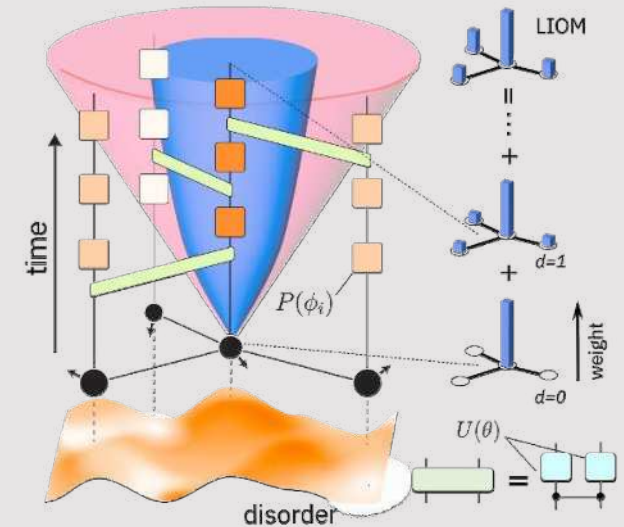
- Simulate Ising time evolution
- Experiments with PEC
- Big picture consequences

Many body experiment example

- State-of-art experiments at the 120Q+, depth 50+: uncovering local integrals of motion

...

Outlook and hardware progress



Simulating transverse-field Ising model time evolution with PEC



Spins

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$

J : exchange coupling between neighbouring spins
 h : transverse magnetic field

Average magnetization
density?

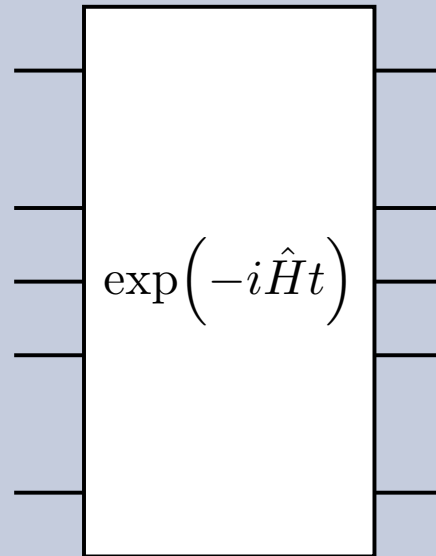
$$\vec{M} := \sum_n (\langle X \rangle_n, \langle Y \rangle_n, \langle Z \rangle_n) / N$$

Step 1: Map to quantum circuit

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$



Time evolution

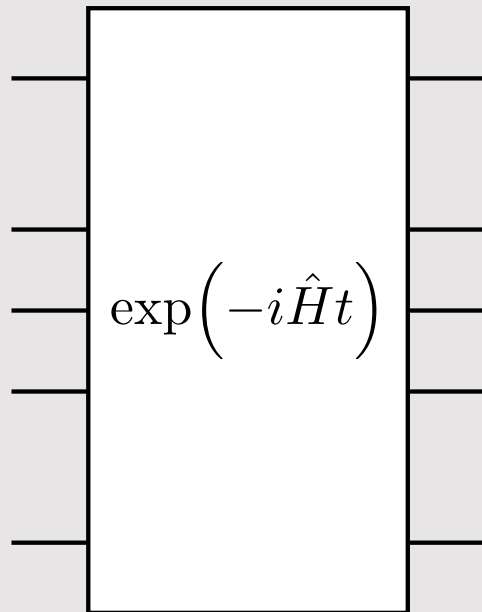


See lecture by
Frank Pollmann

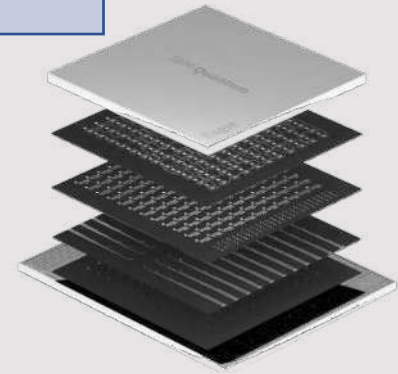
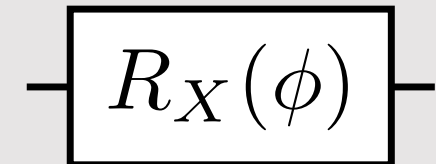
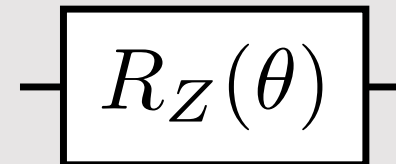
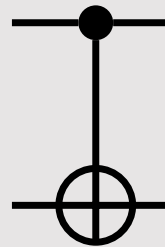


Step 1: Map to quantum circuit

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$

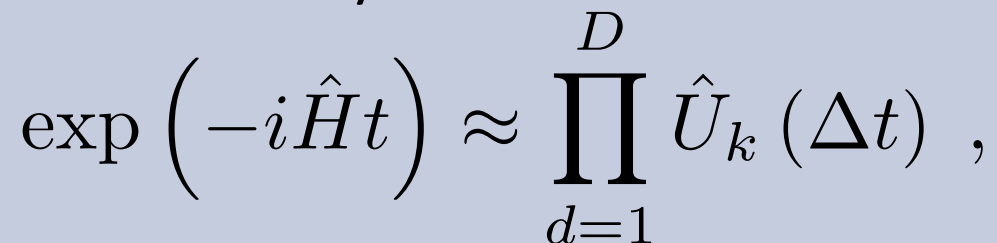


Decompose into native gates that we can actually implement on the QPU





See lecture by Frank


$$\hat{U}_1(\Delta t) = \prod_{j=1}^N e^{-i\hat{H}_j \Delta t},$$

First order expansion

$$\hat{H}_{ZZ} = \sum_j Z_j Z_{j+1}$$

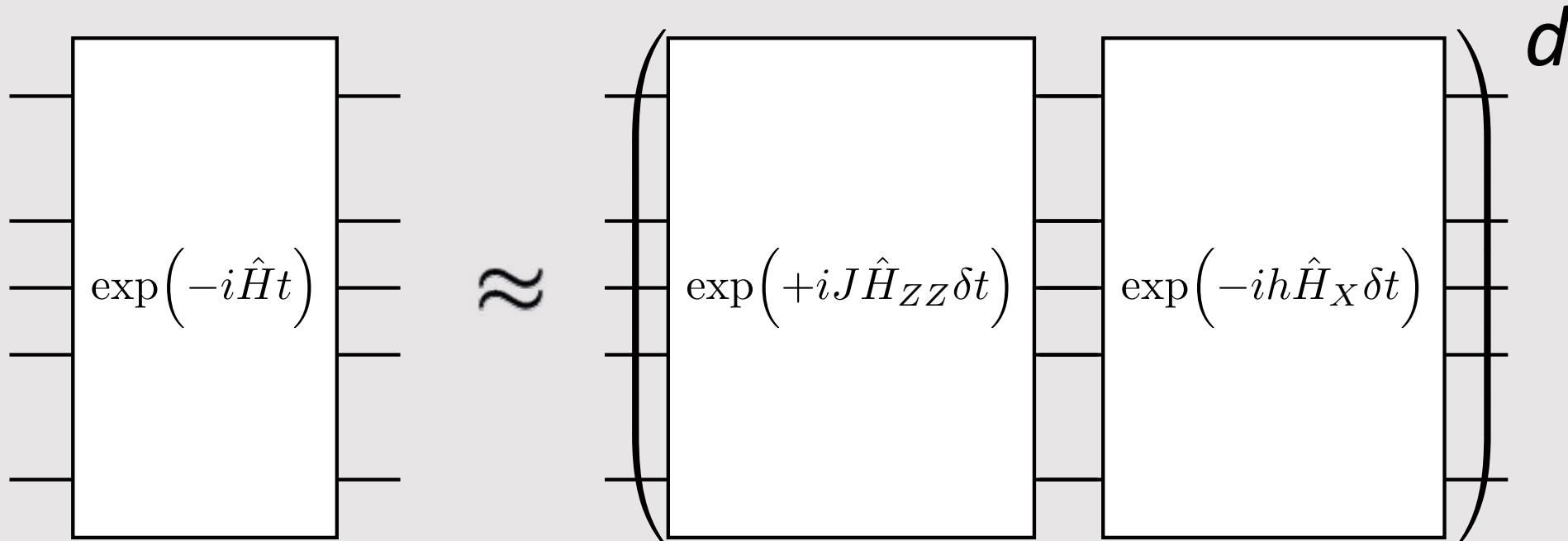
$$\hat{H}_X = \sum_j \hat{X}_j$$

First order Trotter

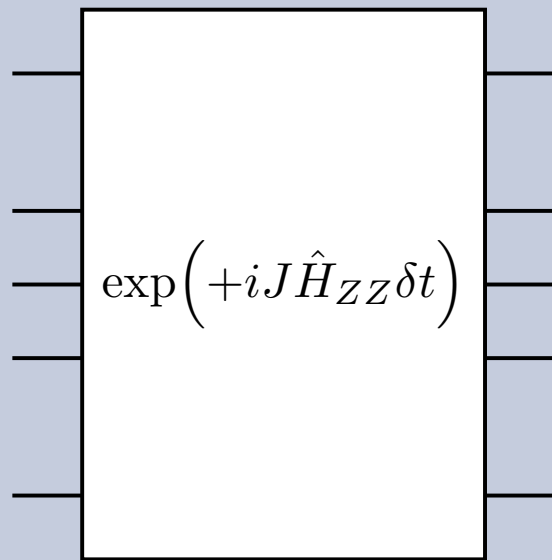
For error bounds, see [arxiv:2302.14592](https://arxiv.org/abs/2302.14592)

Step 1: Map to quantum circuit

$$H = -J \sum_j Z_j Z_{j+1} + h \sum_j X_j$$



Step 1: Trotter circuit (30 s)

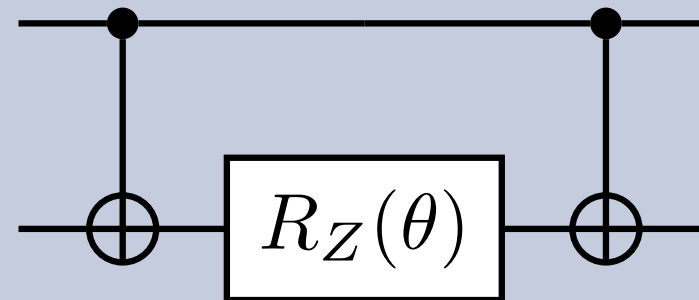
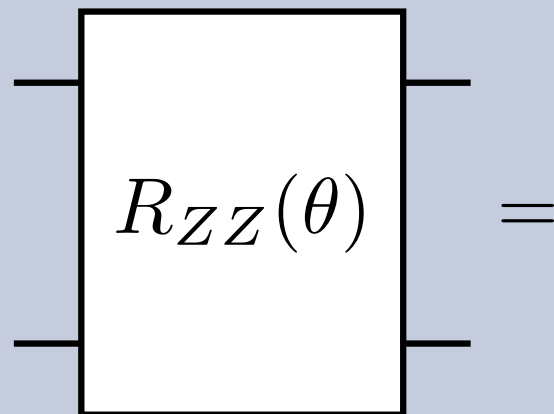


$$\exp\left(-i\frac{\theta}{2}\sum_{j=0}^{n-1}Z_jZ_{j+1}\right) = \quad (\text{use } [Z_jZ_{j+1}, Z_{j'}Z_{j'+1}] = 0)$$

$$\theta := -2J\delta t$$

$$= \prod_{j=0}^{n-1} \exp\left(-i\frac{\theta}{2}Z_jZ_{j+1}\right)$$

$$= \prod_{j=0}^{n-1} R_{ZZ}(\theta)$$

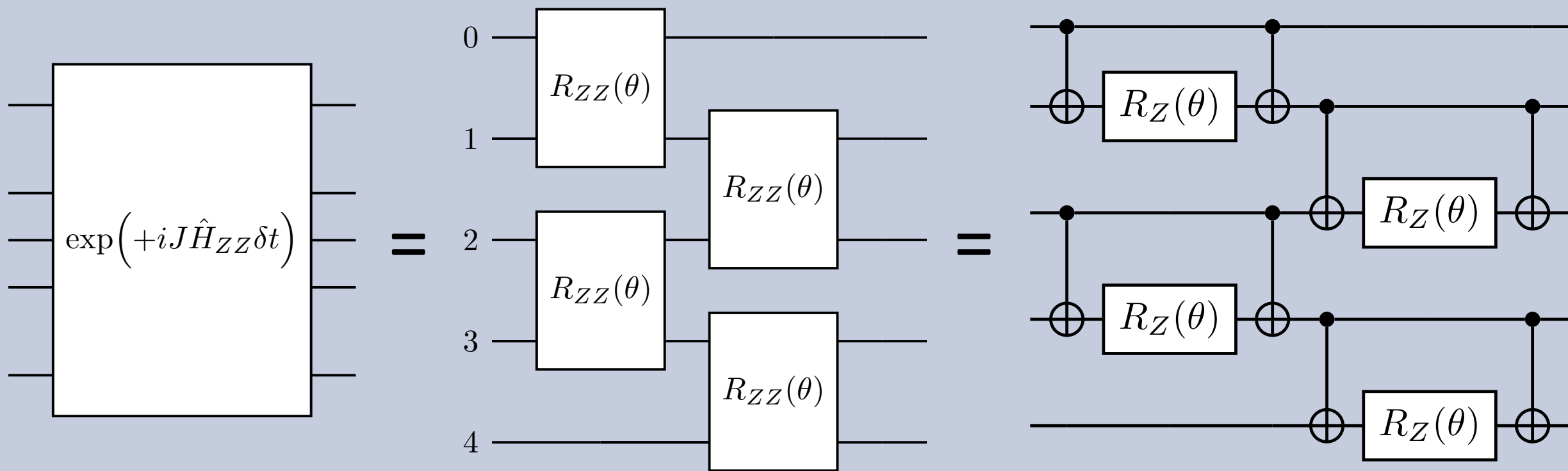


$$R_{ZZ}(\theta) = \exp\left(-i\frac{\theta}{2}ZZ\right)$$

$$= \text{cX}R_Z(\theta)\text{cX}$$

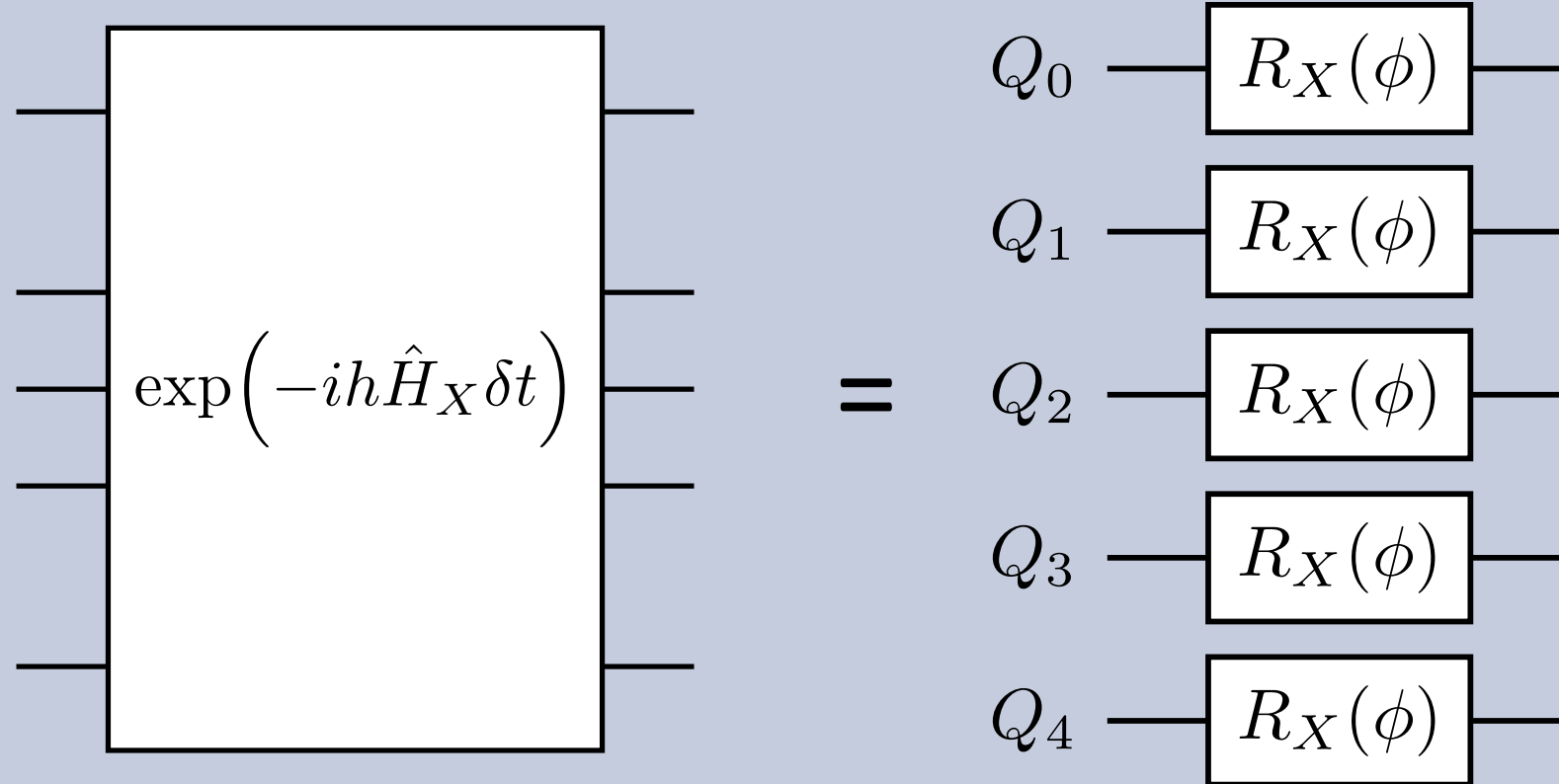


Step 1: Decompose into native gates





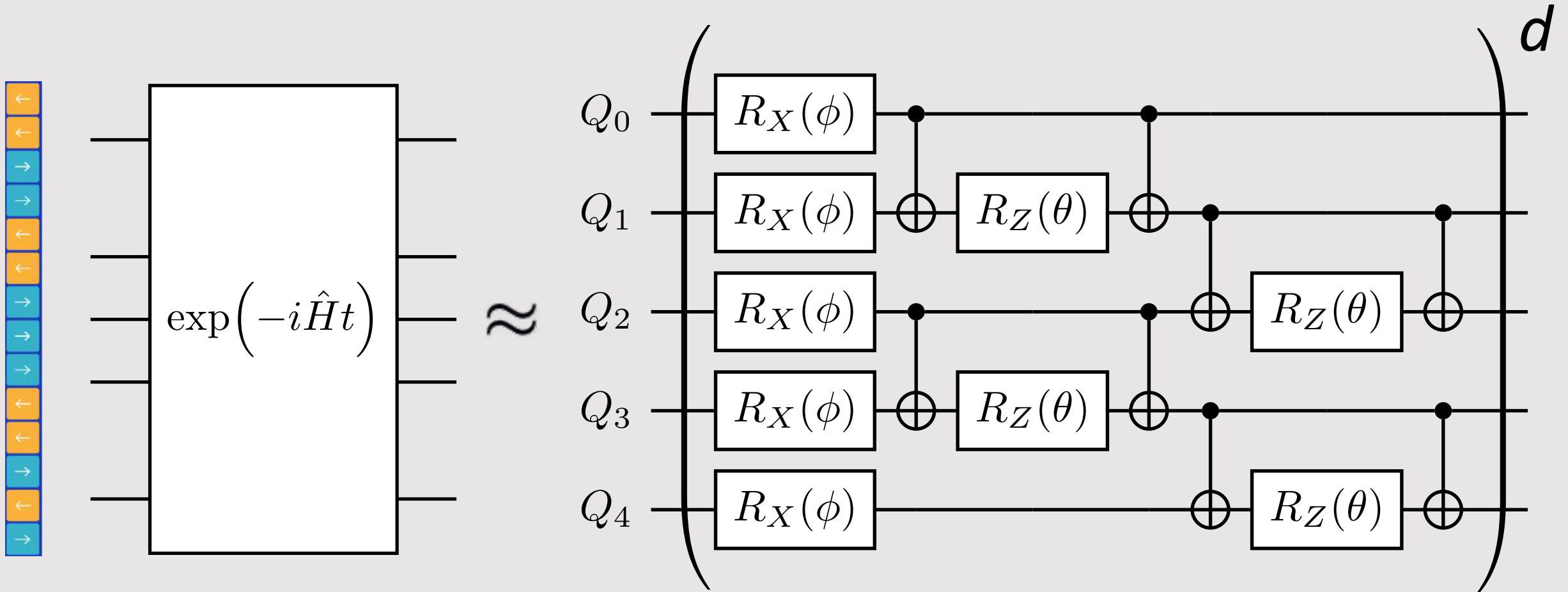
Step 1: Decompose into native gates



$$R_X(\phi) := \exp\left(-\frac{1}{2}\phi X\right)$$

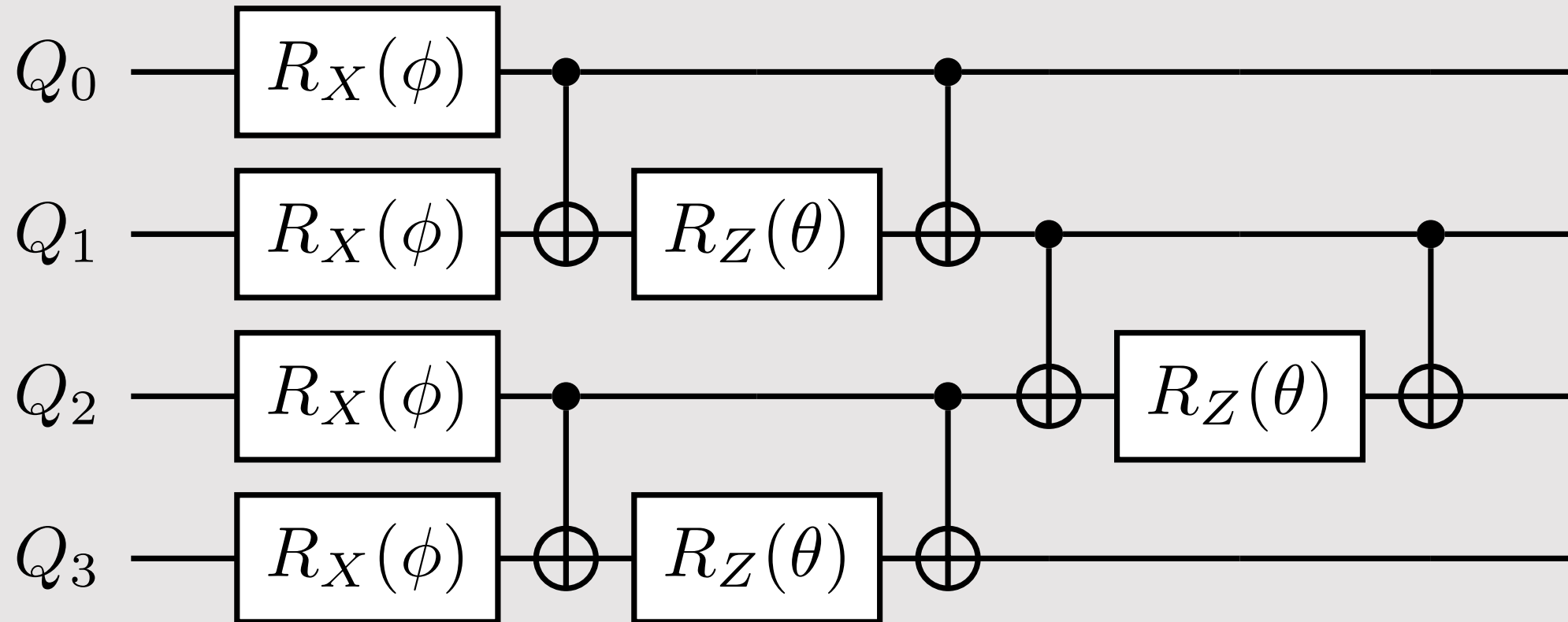
$$\phi := +2h\delta t$$

Step 1: Quantum Hamiltonian time evolution

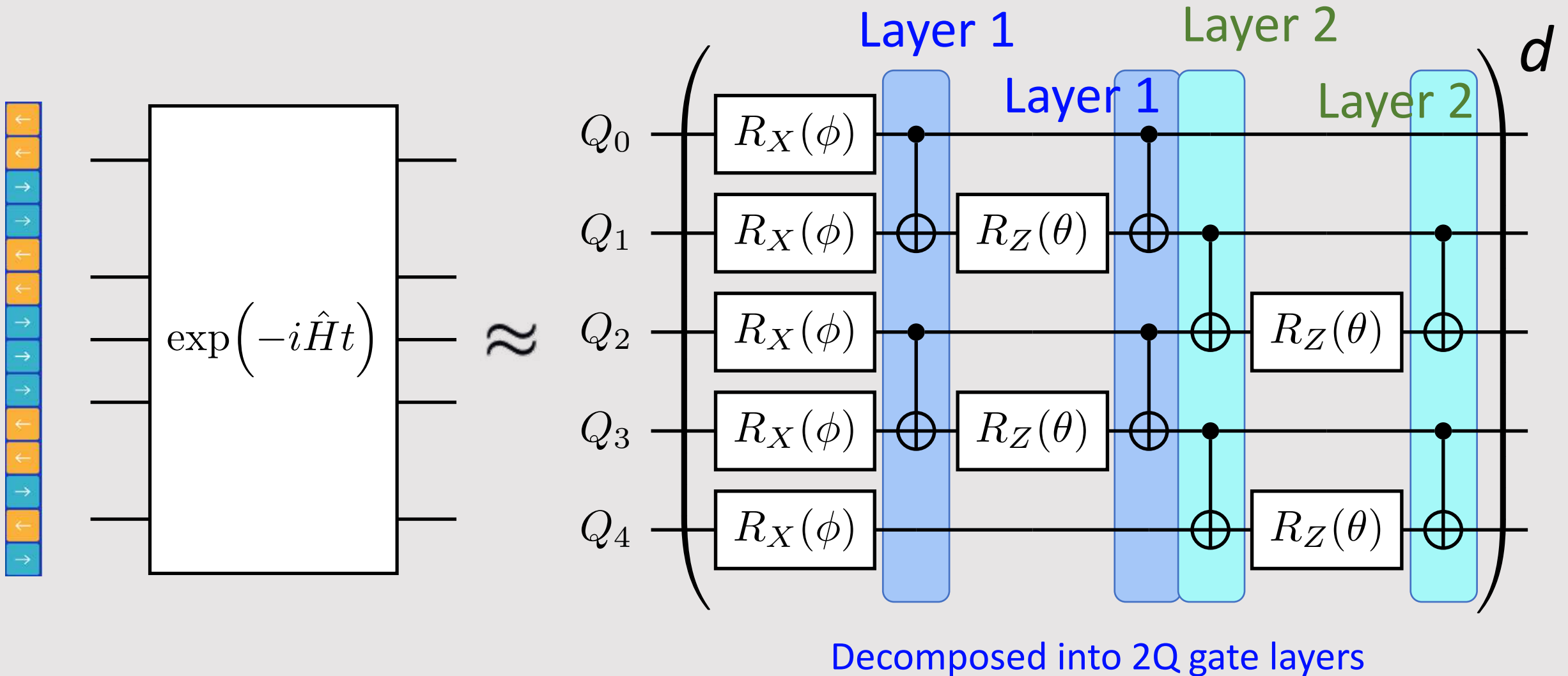


Decomposed into quantum computer
native gates

Make things even simpler



Step 2: Decompose into layers



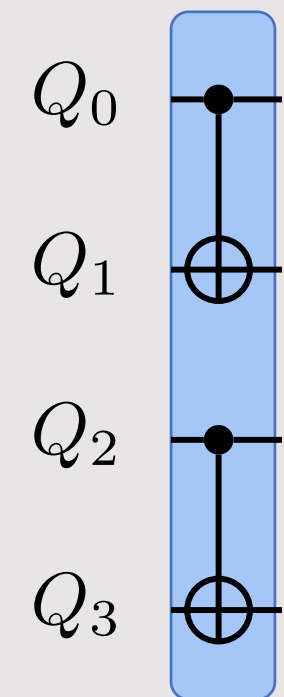
Step 2: Learn the noise on each layer

Let's make even simpler, and make it 4 qubits (instead of 5 for our first example)

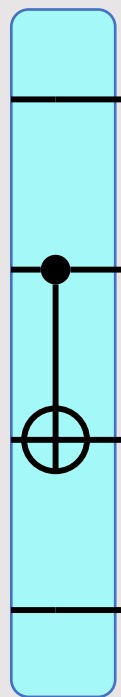
Layer 1

Layer 2

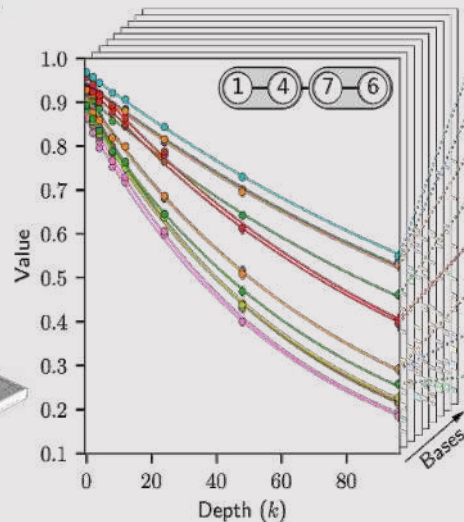
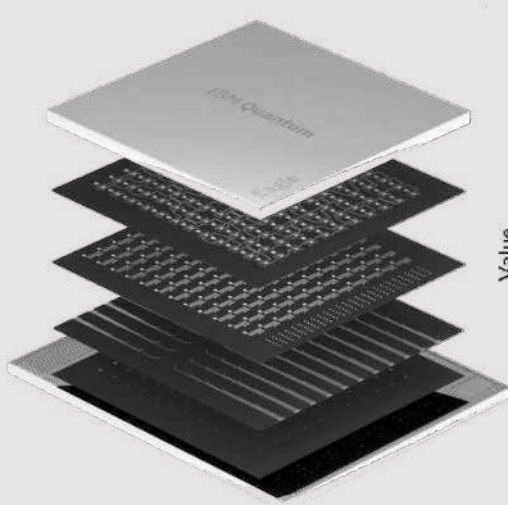
See lecture 2 for learning the noise



Λ_1

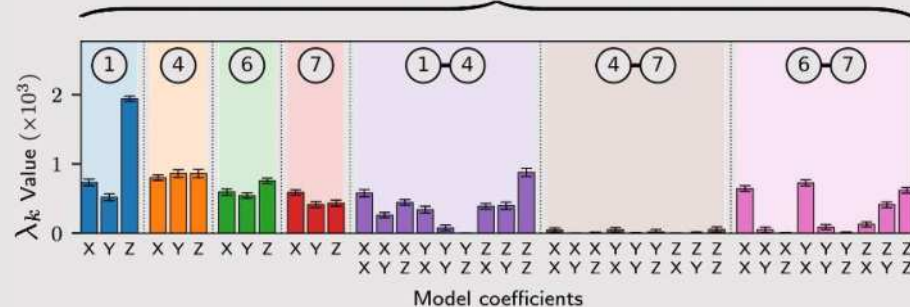


Λ_2



Sparse Lindblad tomogram

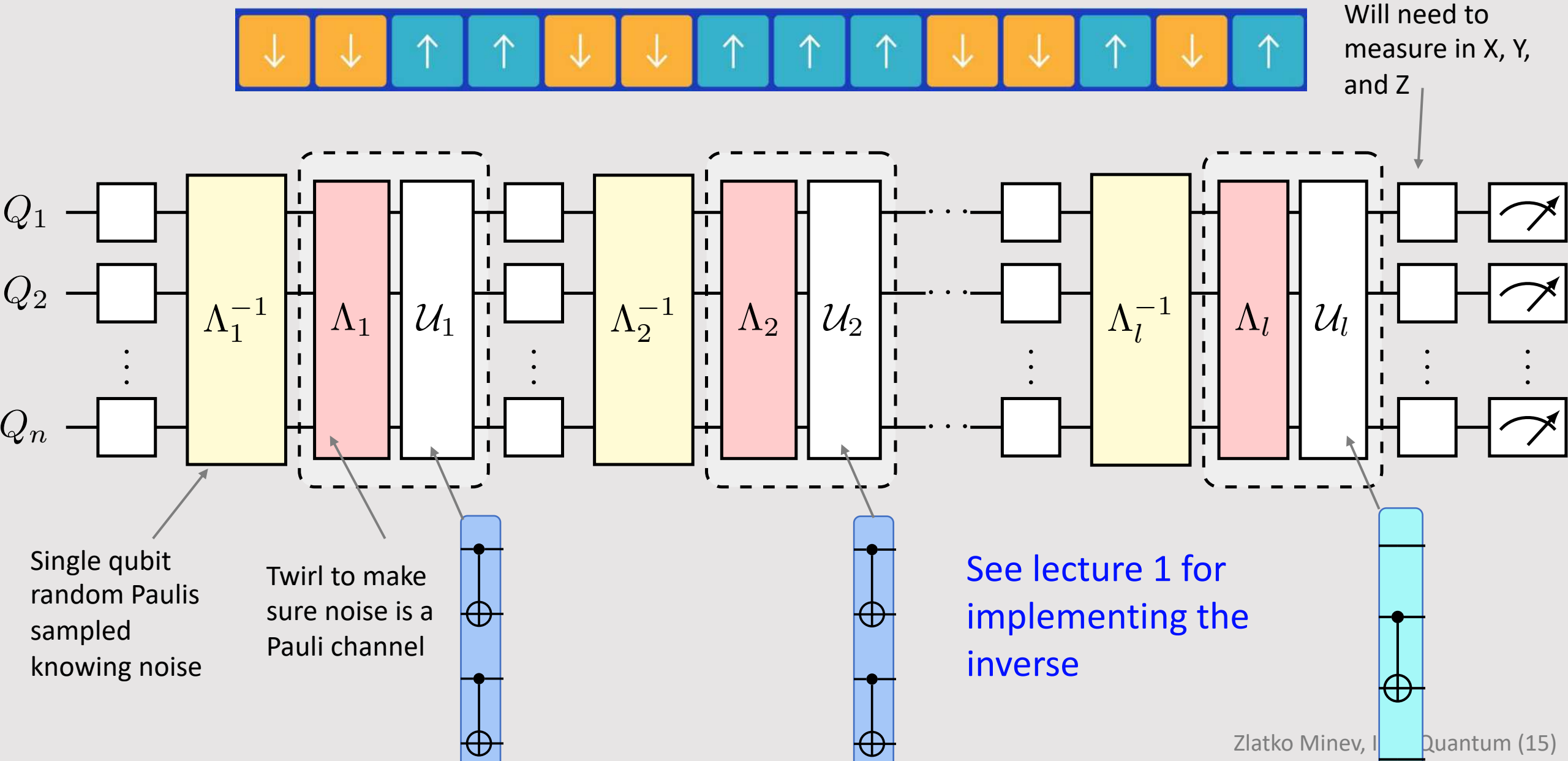
$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k^\dagger - \rho)$$



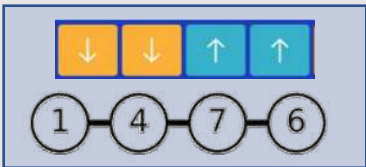
$$\gamma_1 = 1.0309 \pm 8.40 \cdot 10^{-5}$$

$$\gamma_2 = 1.0384 \pm 2.20 \cdot 10^{-4}$$

Step 3: Cancel the noise

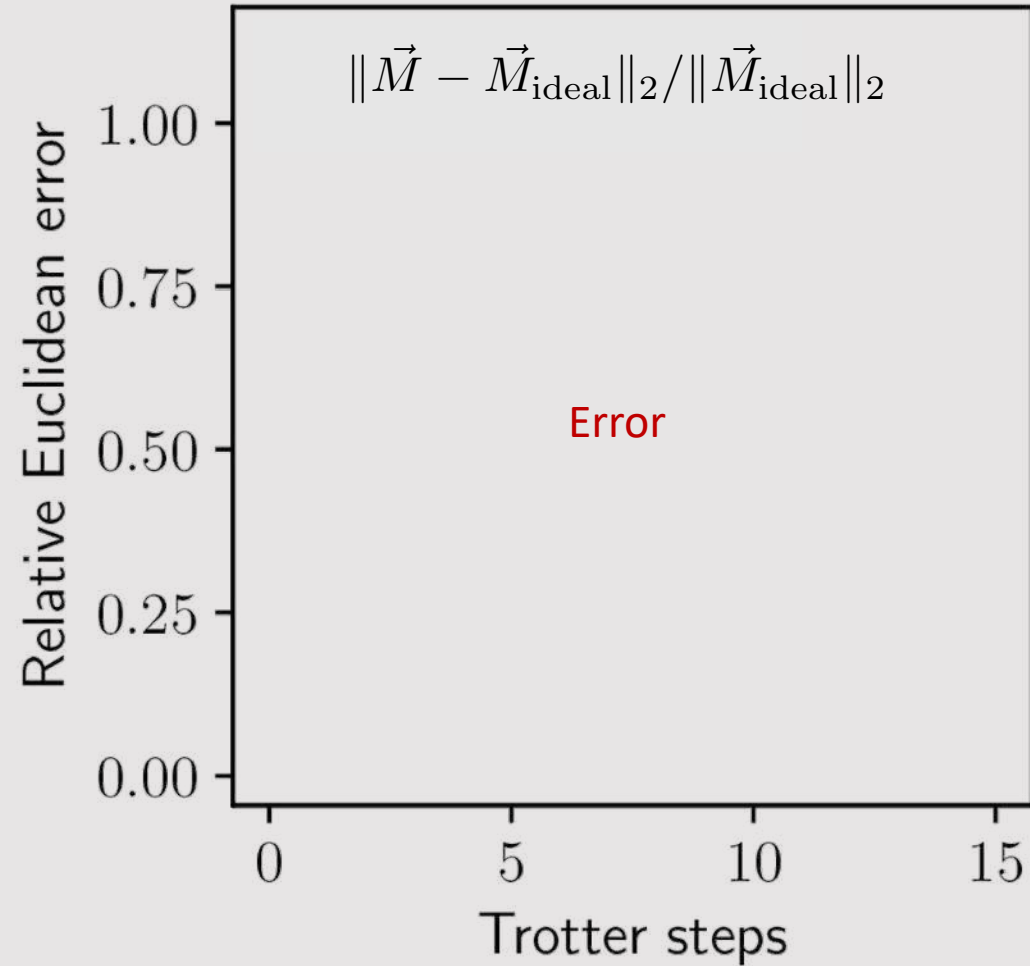
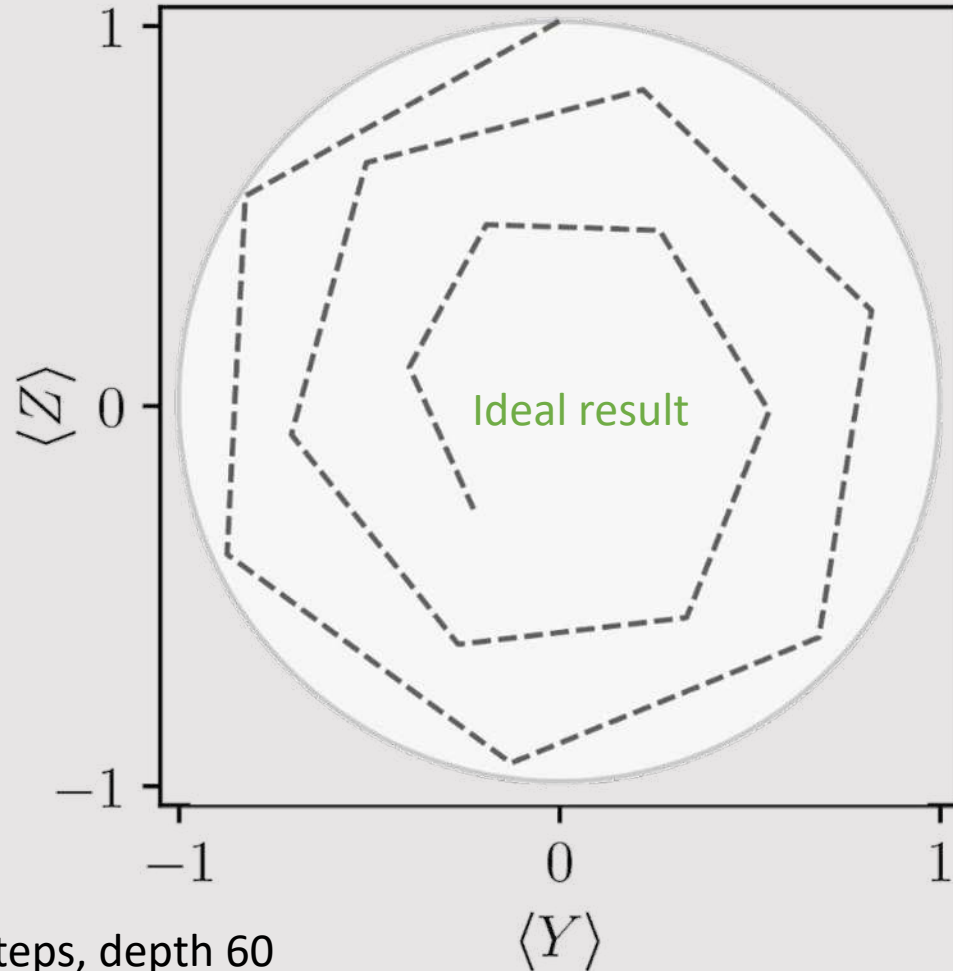


Ideal Ising model evolution



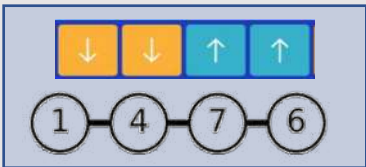
--- Ideal

$$\vec{M} := \sum_n (\langle X \rangle_n, \langle Y \rangle_n, \langle Z \rangle_n) / N$$

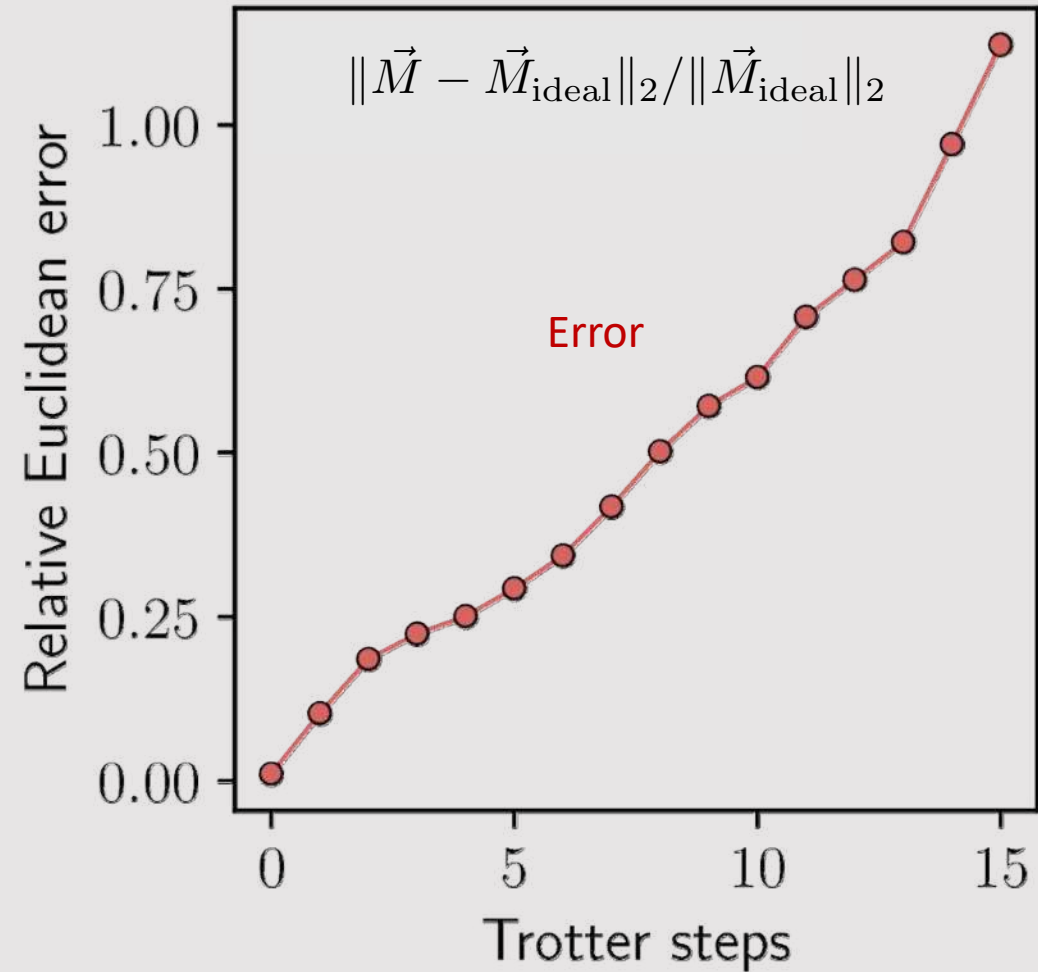
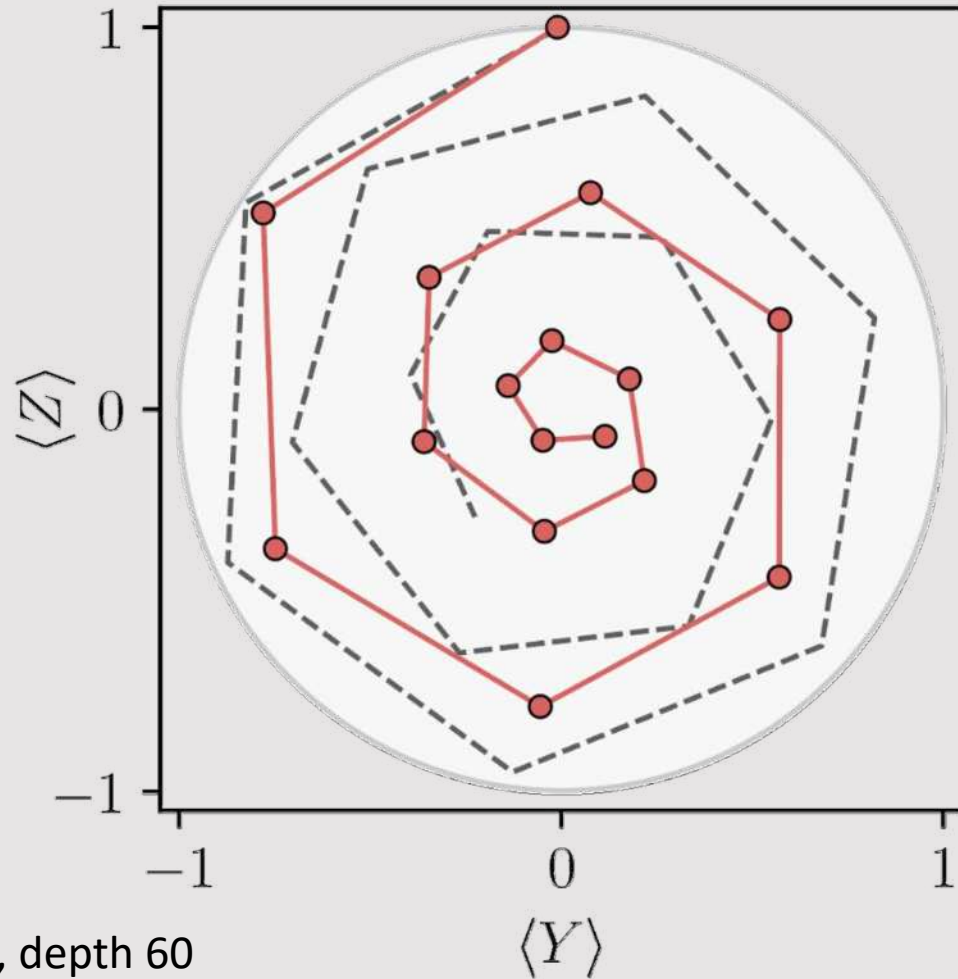


4Q, 15 steps, depth 60
 $h = 1, J = -0.15, \delta t = 1/4$

Without PEC: but with DD & twirl readout mitigation



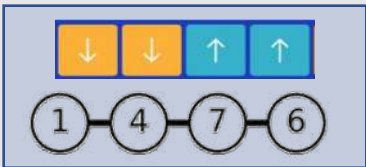
--- Ideal —●— without PEC



15 steps, depth 60

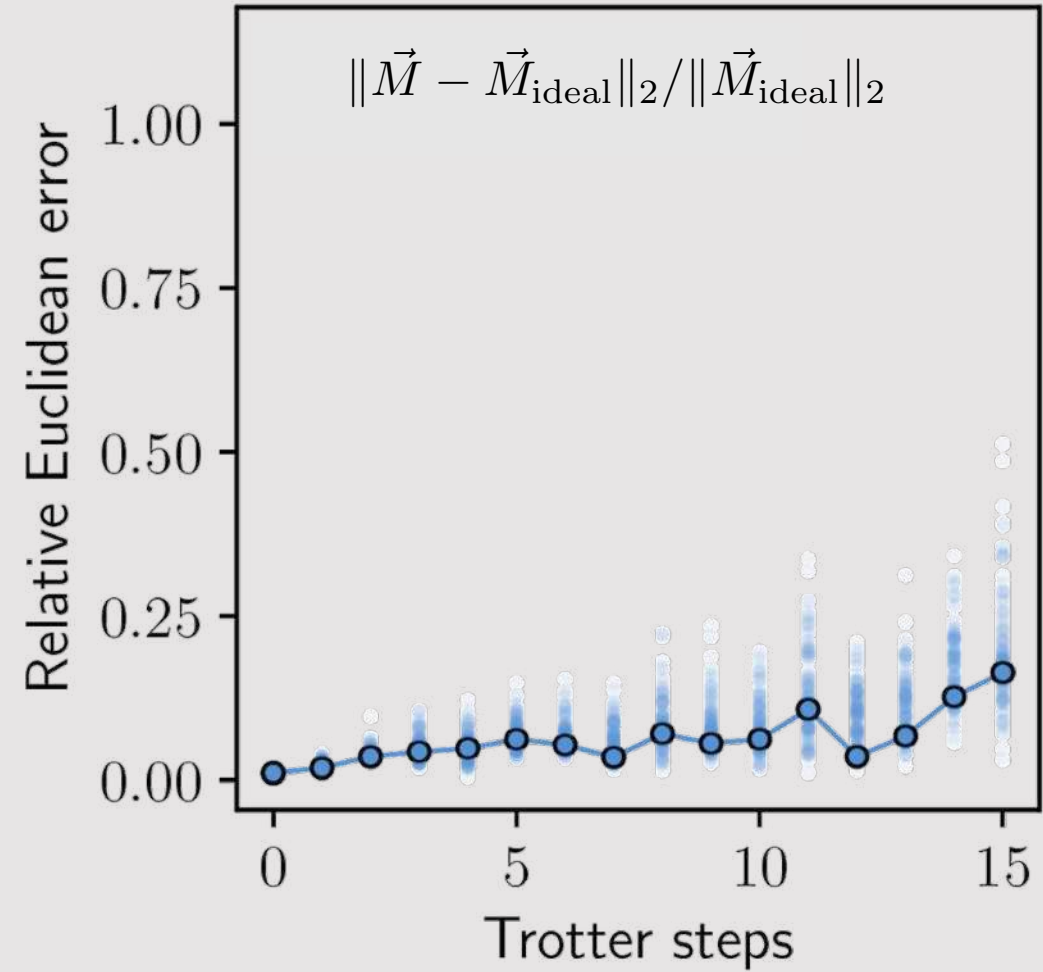
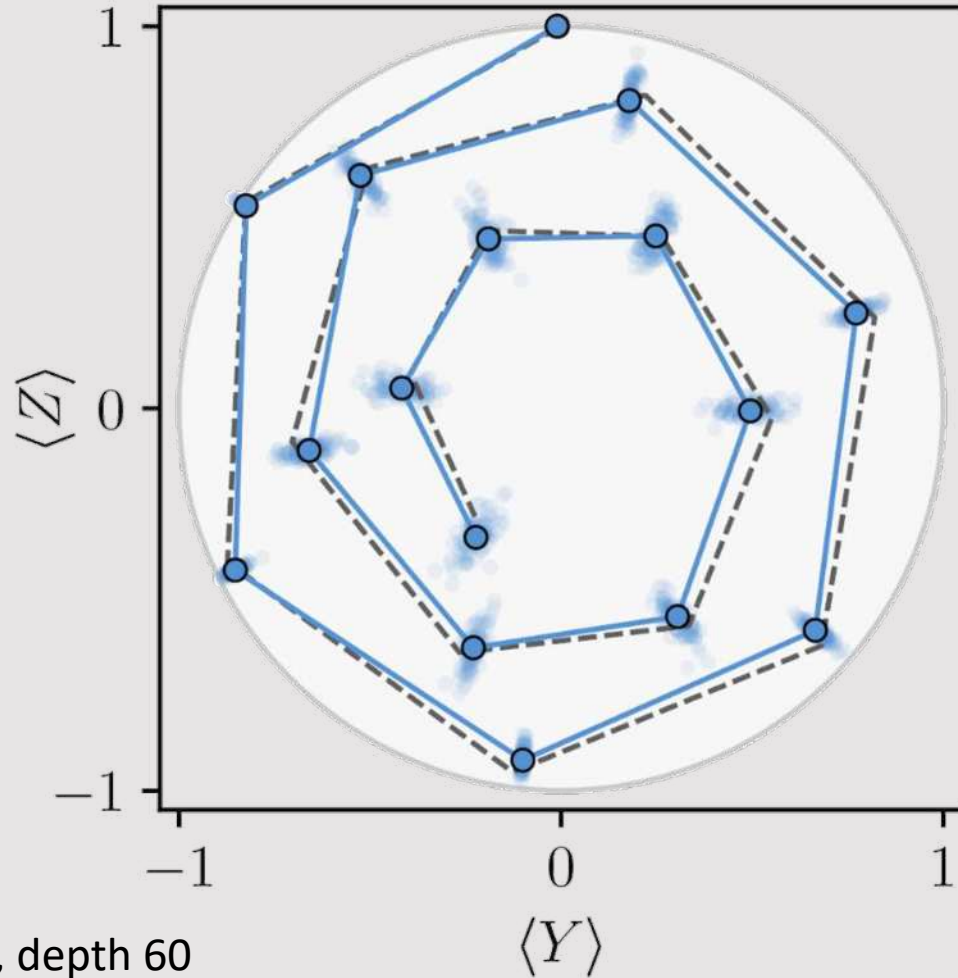
$h = 1, J = -0.15, \delta t = 1/4$

With PEC



--- Ideal

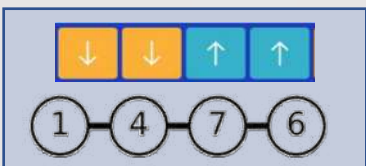
—●— with PEC



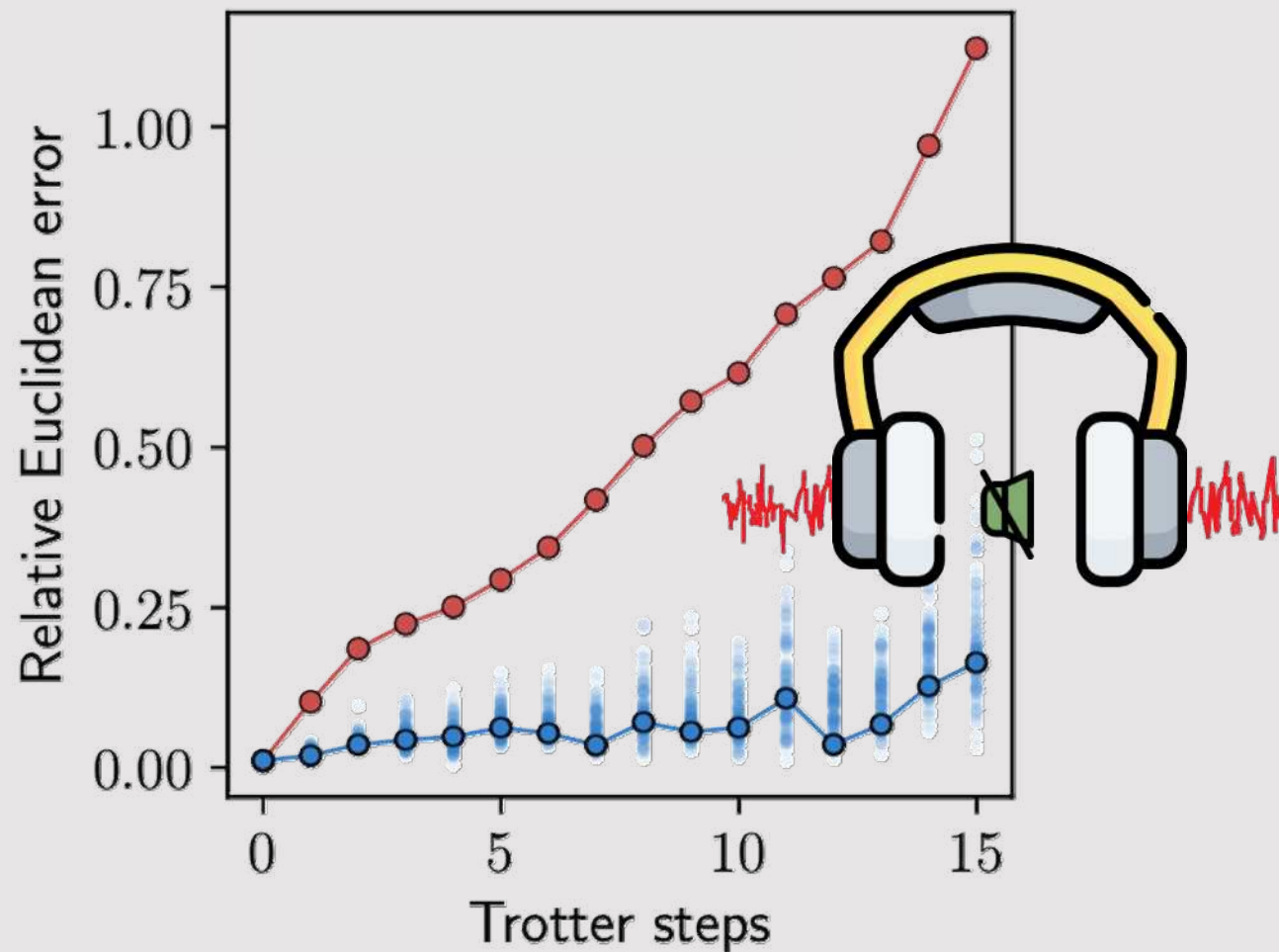
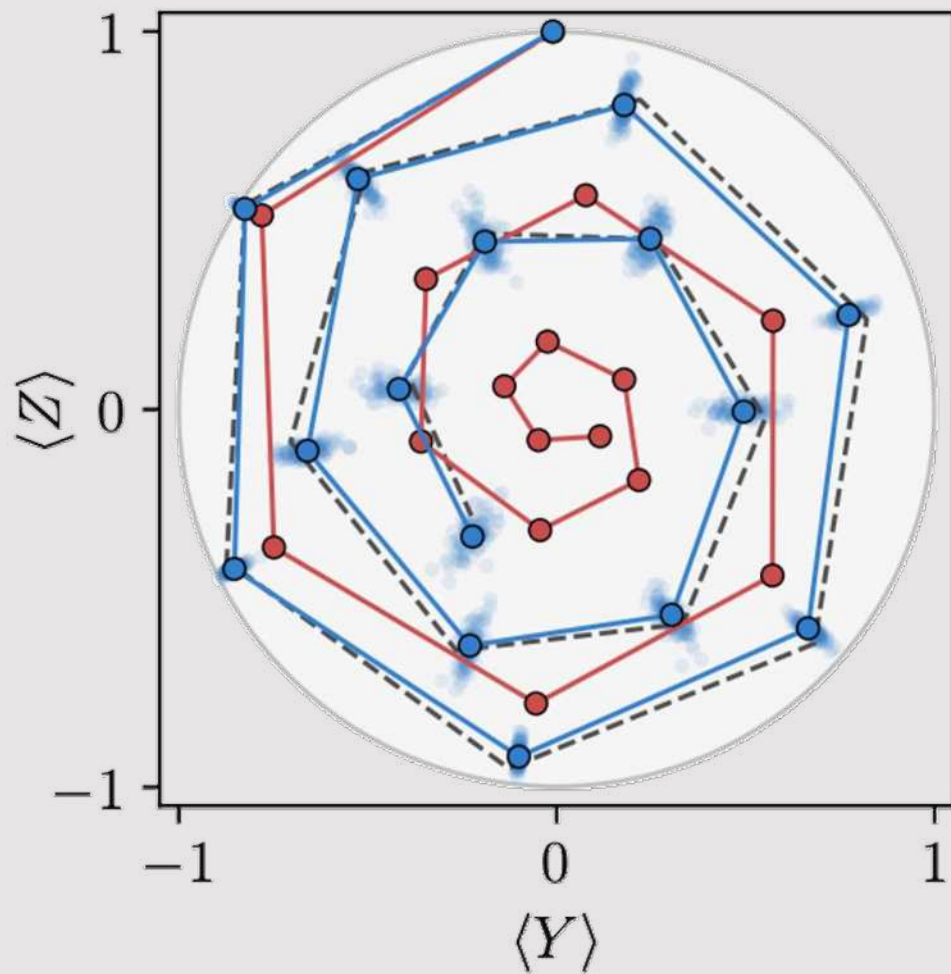
15 steps, depth 60

$h = 1, J = -0.15, \delta t = 1/4$

With vs. without PEC

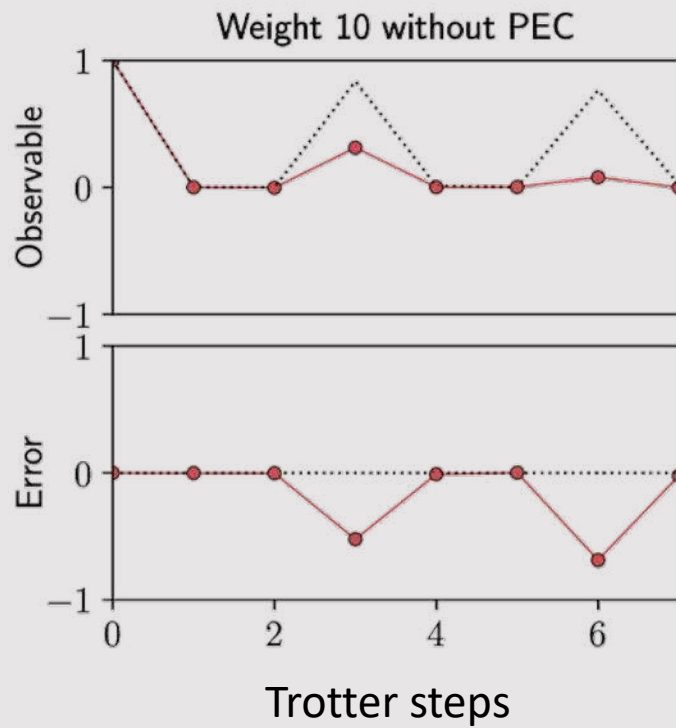
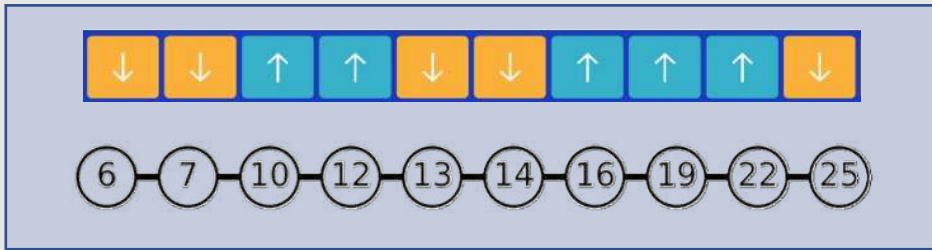


Legend: --- Ideal ● without PEC ● with PEC



$$h = 1, J = -0.15, \delta t = 1/4$$

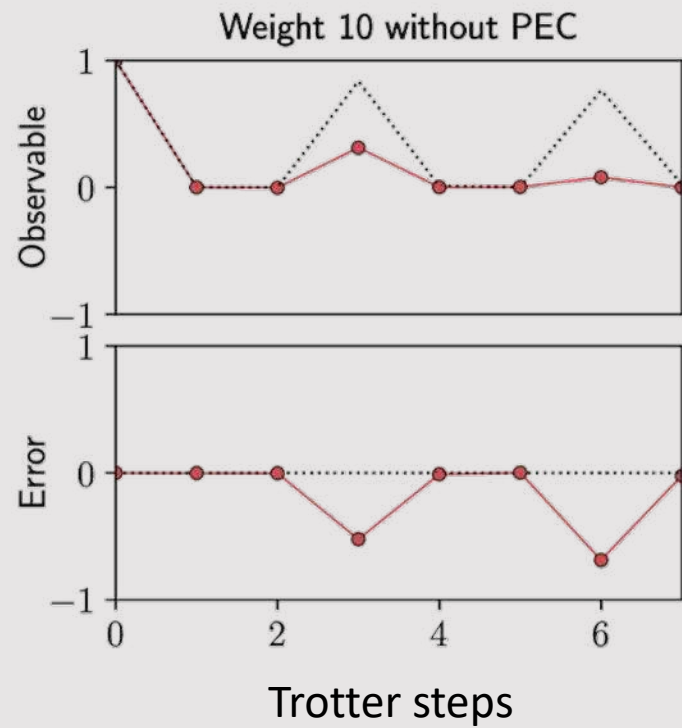
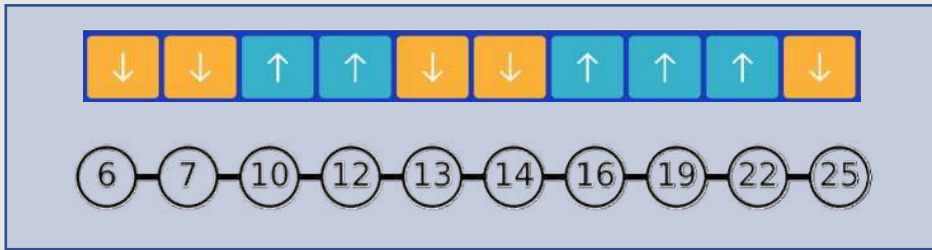
With vs without PEC: 10 qubit high-weight observables



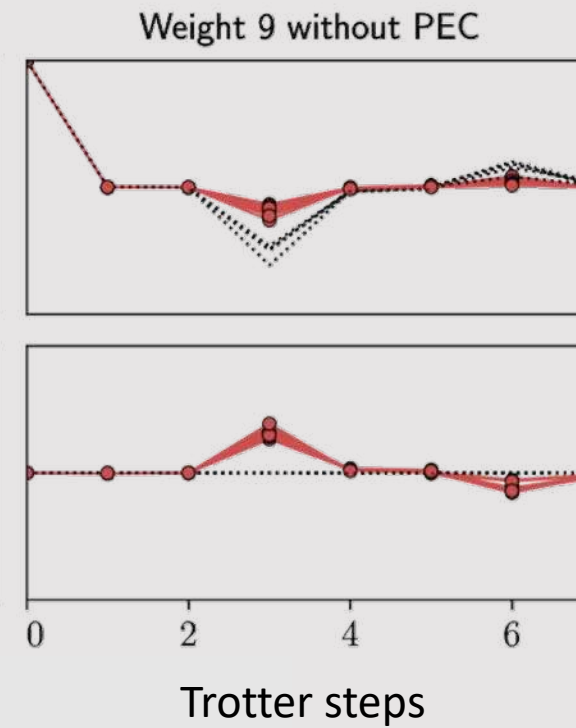
ZZZZZZZZZZZZ

$$h = 1, J = -0.5236, \delta t = 1/4$$

With vs without PEC: 10 qubit high-weight observables



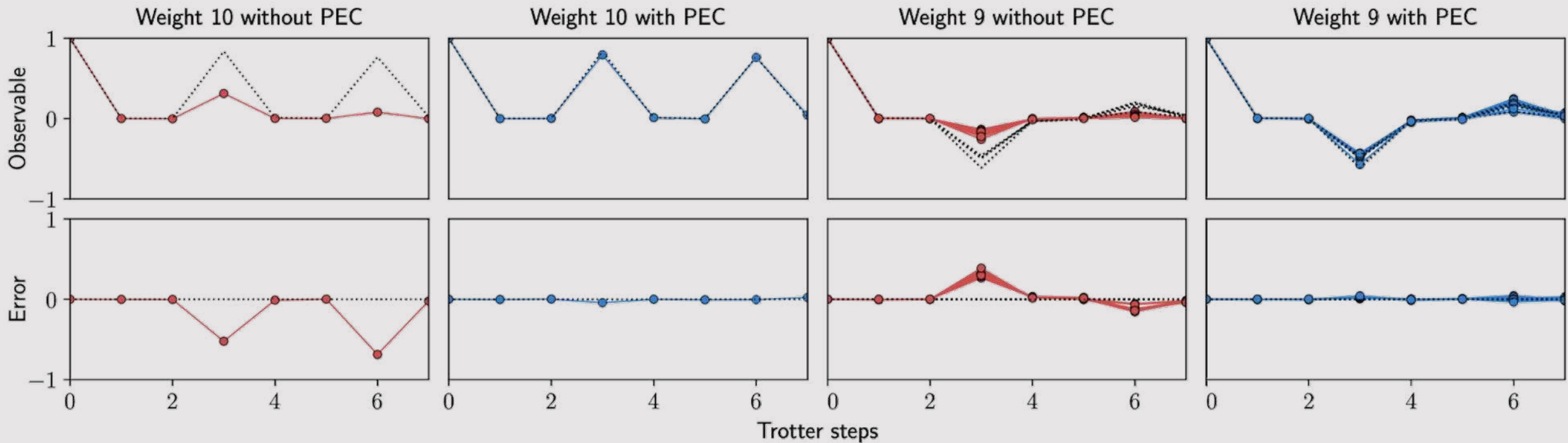
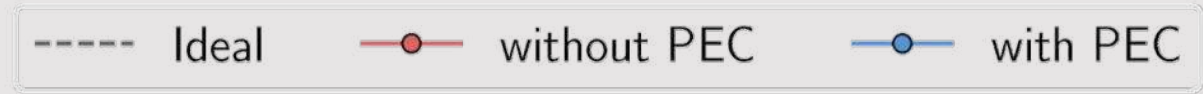
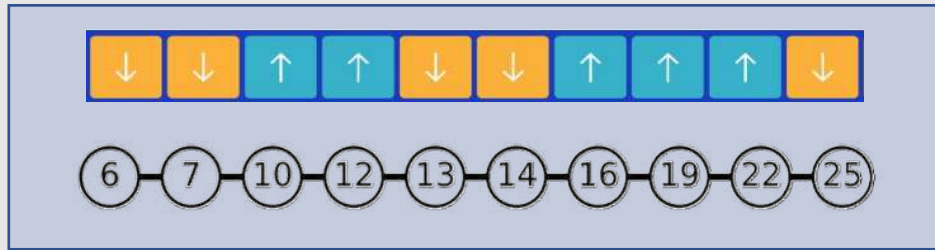
ZZZZZZZZZZZZ



IZZZZZZZZZZZZ
 ZIZZZZZZZZZZZ
 ZZIZZZZZZZZZZZ
 ...
 ZZZZZZZZZZZZI

$$\hbar = 1, J = -0.5236, \delta t = 1/4$$

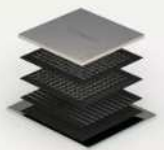
With vs without PEC: 10 qubit high-weight observables



$$\hbar = 1, J = -0.5236, \delta t = 1/4$$

Scaling and error budget

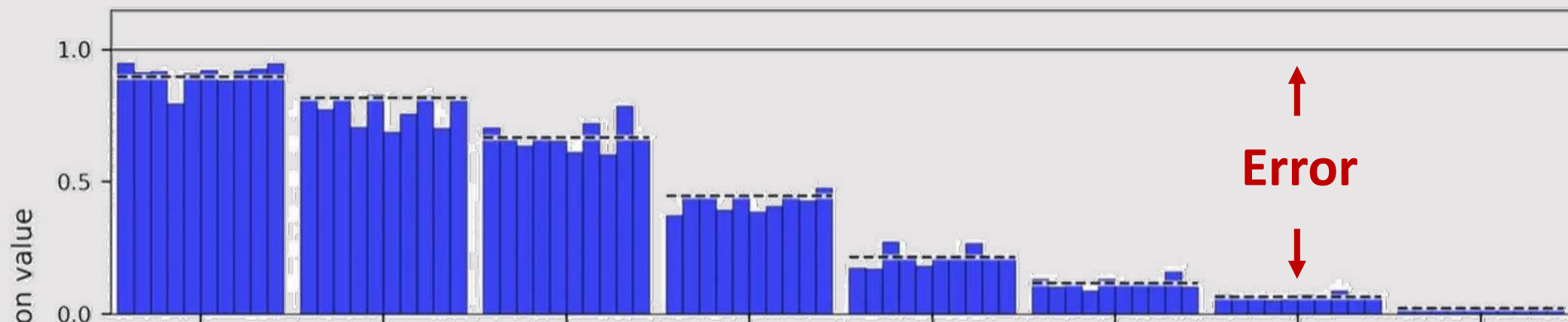




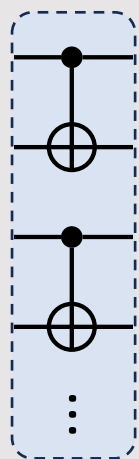
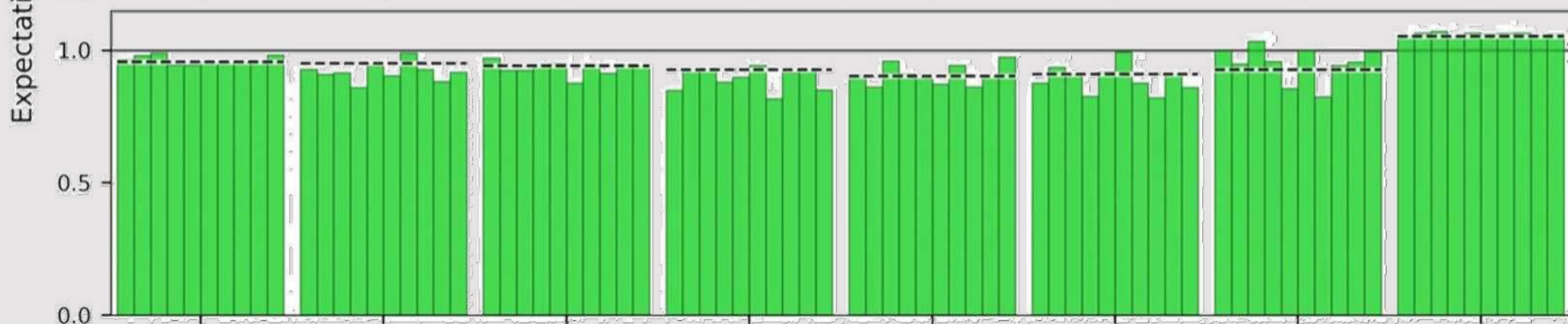
PEC on 50 qubit observables

Z stabilizers of increasing weight

Without PEC



With PEC

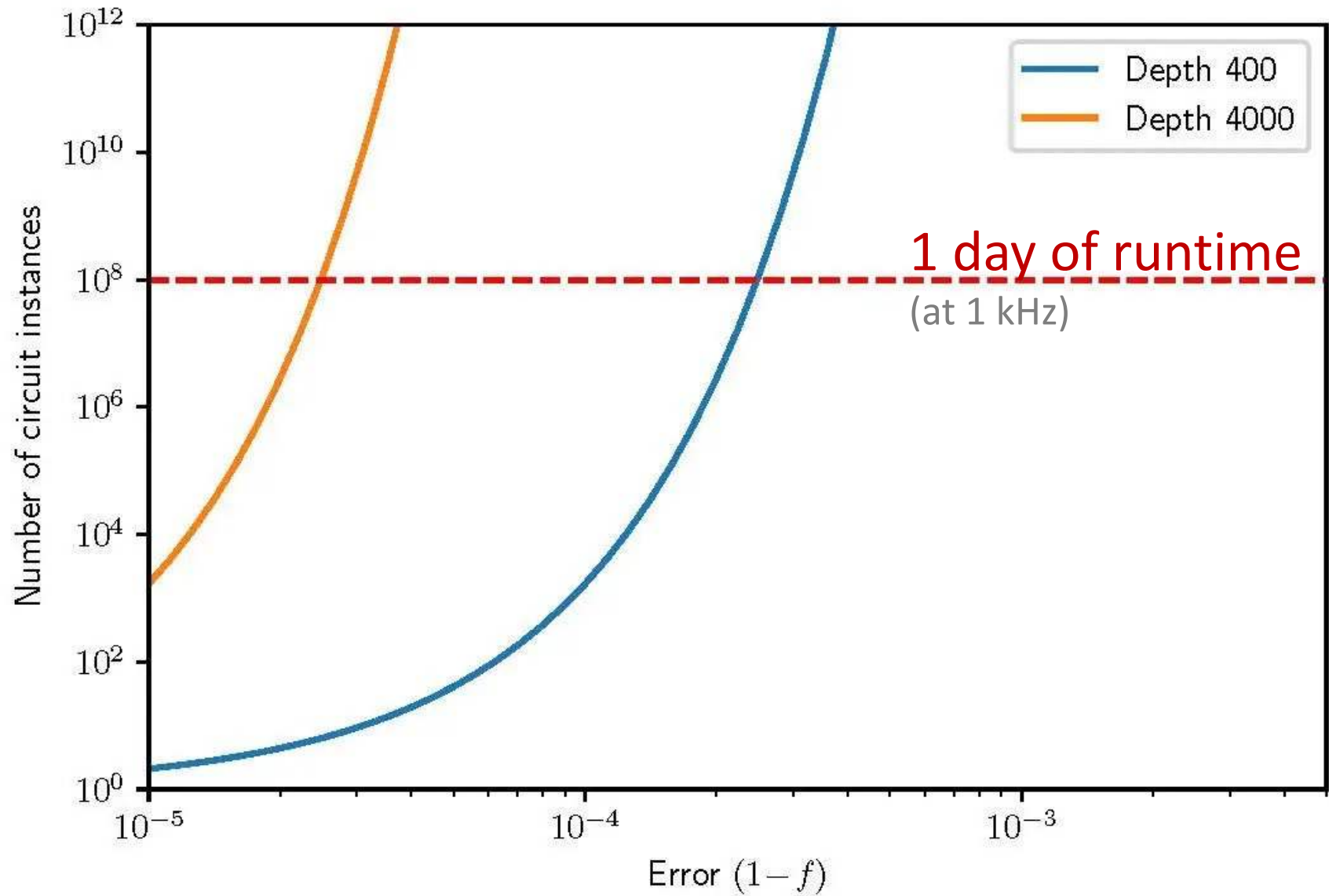


50Q
2 layers of
cNOT gates

Weight of mitigated observable

Path to 100+ qubits?

Estimating
PEC overhead
for Trotter
circuits
comprising
100 qubits



See also on speed: A. Wack, et al., Quality, speed, and scale: three key attributes to measure the performance of near-term quantum computers (2021).

Path to quantum computing

Noise-free estimators can be obtained from noisy quantum computers TODAY, at a runtime cost that is exponential in number of qubits n and circuit depth d

$$\text{Runtime} = \beta d (\bar{\gamma})^{n d} \text{ seconds}$$

d is the depth of the quantum circuit

β is a measure of the time per circuit layer operation (CLOPS) (increase by pushing **speed**)

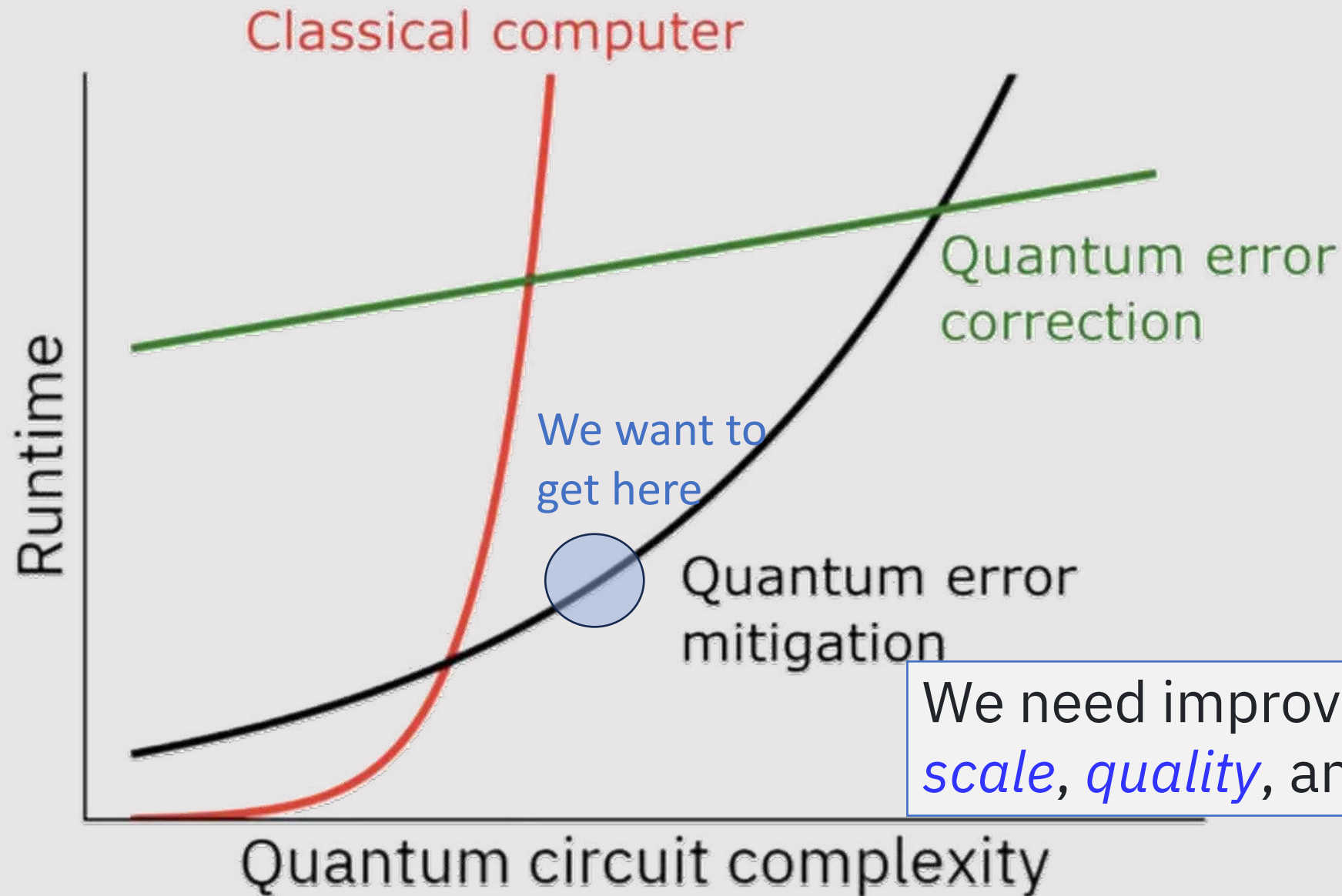
$\bar{\gamma}$ is a measure of the collective quantum noise (increasing **quality** brings it closer to 1)

n is the number of operational qubits (increase by pushing **scale**)

You can further reduce runtime using light cones and other strategies.

	Improvements	$\bar{\gamma}$
Hummingbird r2 (Brooklyn, 65Q)		1.038
Hummingbird r3 (Ithaca, 65Q)	2-3x coherence improvements over r2	1.024
Falcon r10 (Prague, 32Q)	State-of-the-art two-qubit gates, reduced crosstalk	1.012

Path to utility of quantum computers before error correction



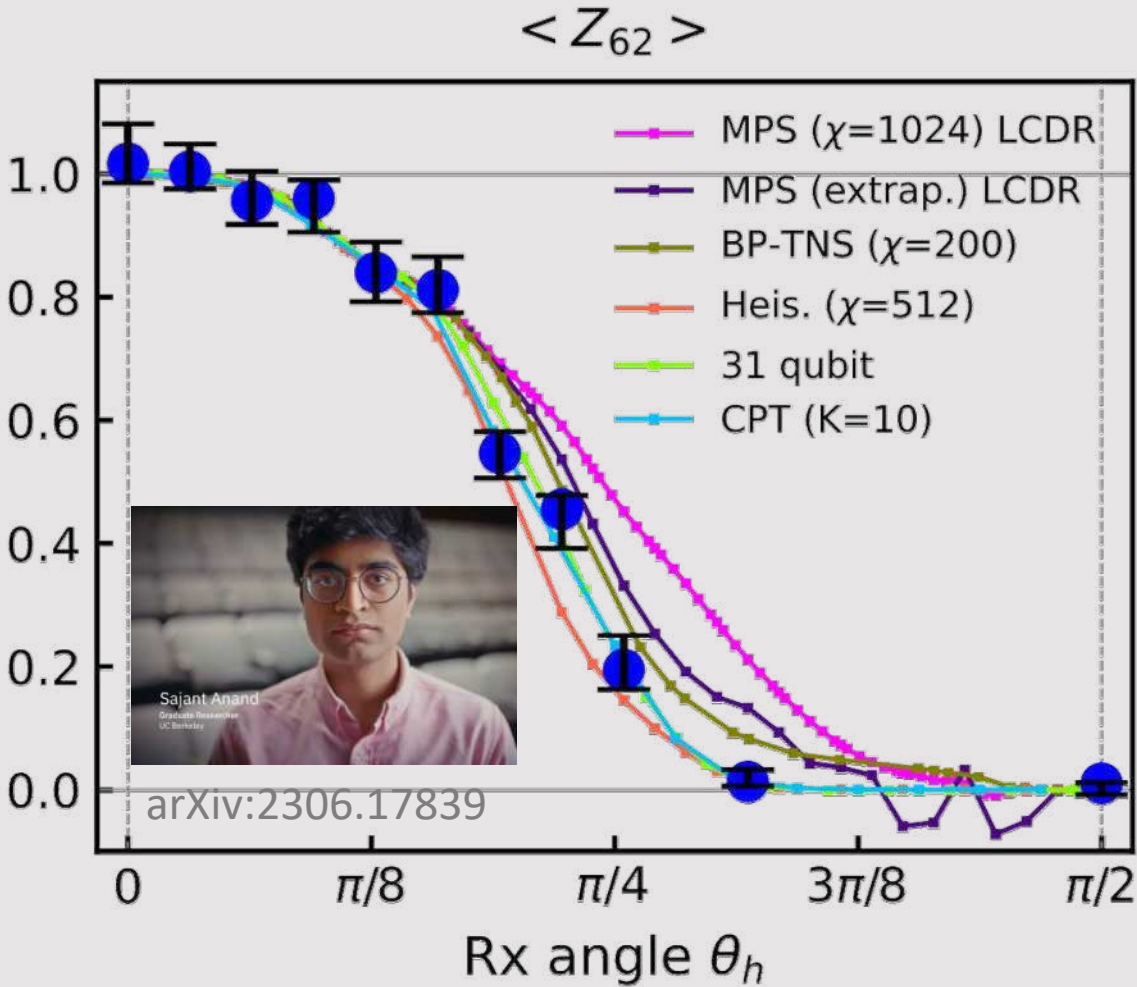
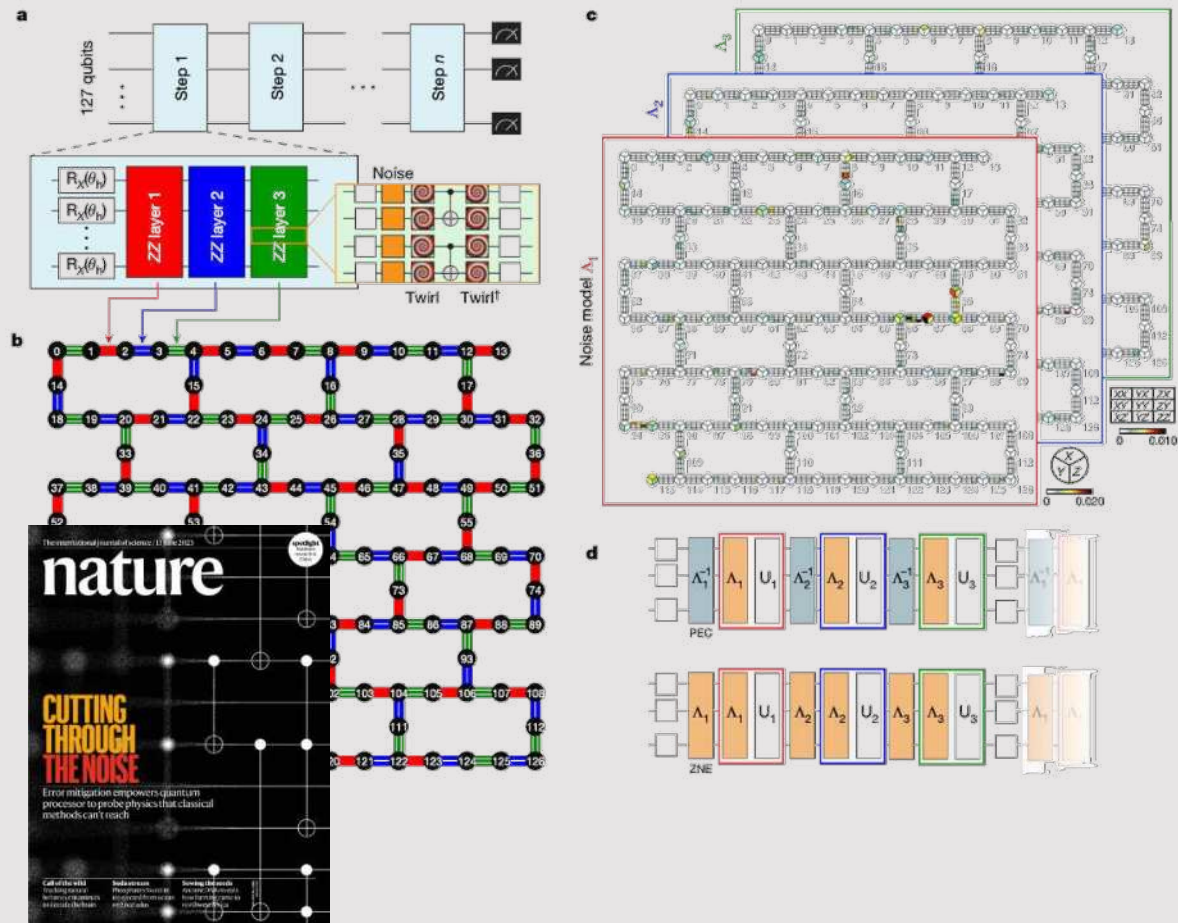
We need improvements in *scale*, *quality*, and *speed*.

PEC + ZNE for 127 qubits

Article

Evidence for the utility of quantum computing before fault tolerance

Youngseok Kim^{1,6}, Andrew Eddins^{2,6}, Sajant Anand³, Ken Xuan Wei¹, Ewout van den Berg¹, Sami Rosenblatt¹, Hasan Nayfeh¹, Yantao Wu^{3,4}, Michael Zaletel^{3,5}, Kristan Temme¹ & Abhinav Kandala¹



How to think about experiments from the point of view of error mitigation?

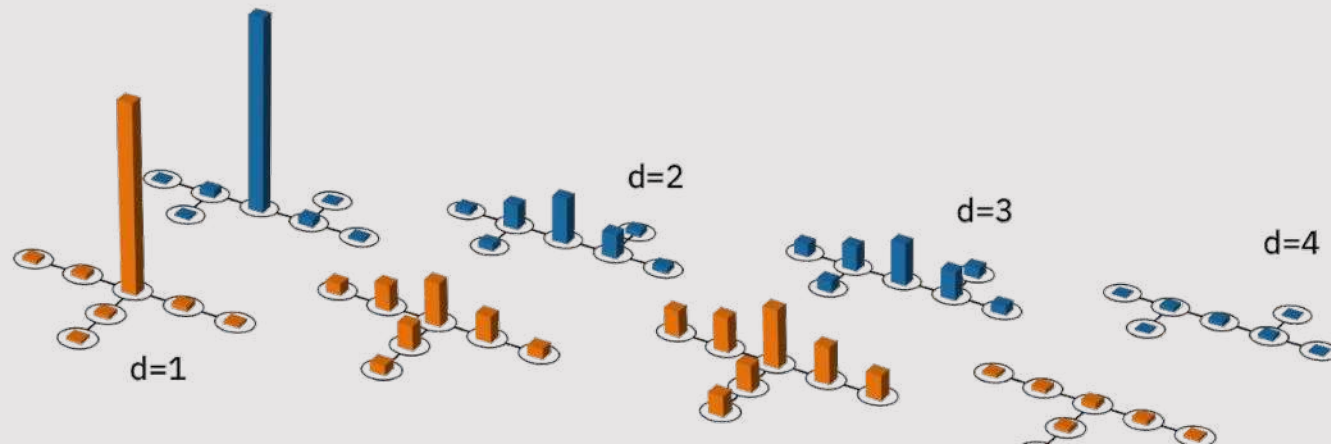
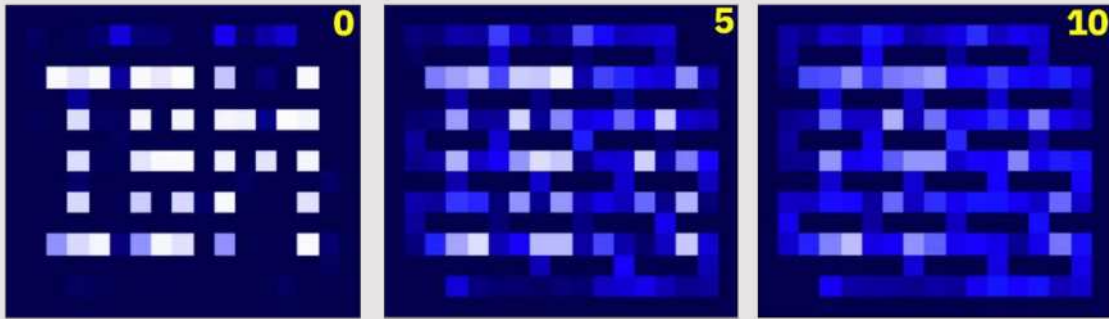
Uncovering Local Integrability in Quantum Many-Body Dynamics

arXiv:2307.07552

A detailed portrait

Oles Shtanko^{*†,1} Derek S. Wang^{*,2} Haimeng Zhang,² Nikhil Harle,^{2,3}
Alireza Seif,² Ramis Movassagh,⁴ and Zlatko Mineev^{&2}

IBM Quantum



Acknowledgements: IBM Quantum team

Uncovering the dynamics of many-body systems

Many-body quantum systems and their dynamics

- fundamental and technological
- but generically difficult to simulate and understand

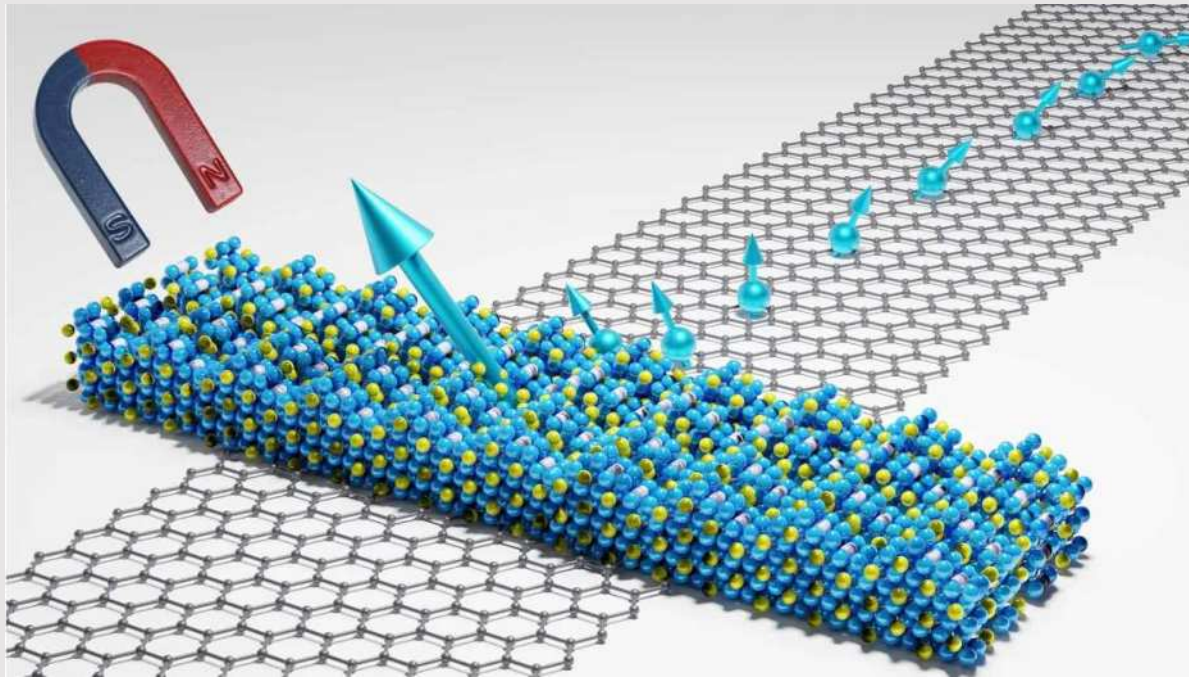


Image: [Chalmers](#)

Symmetries, conservation laws, and integrability

- can unravel intricacies of these complex systems
- but generically difficult to discover

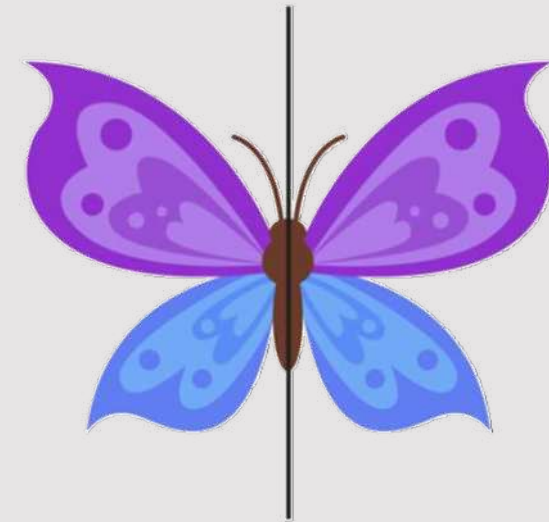


Image: [SuperSimple](#)

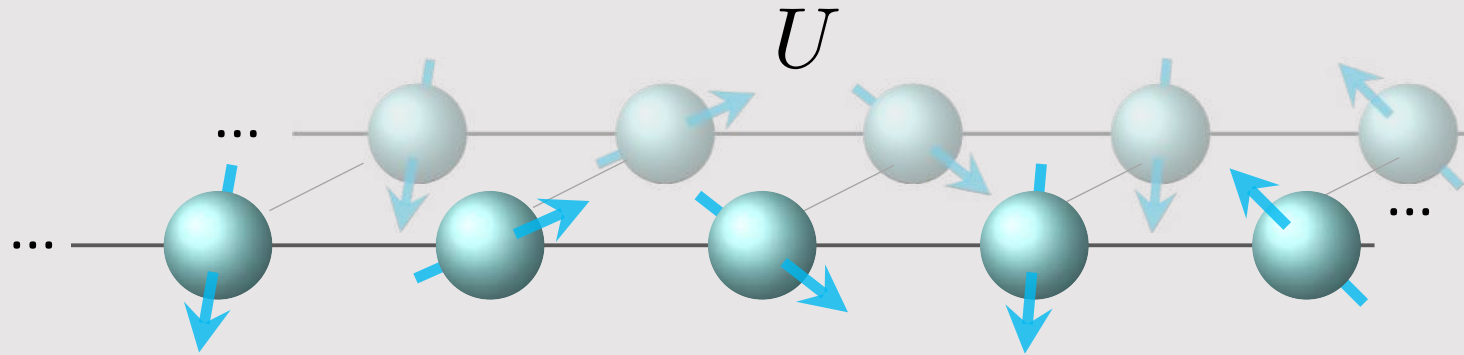
Integrals of motion

Integral of motion L

$$[U, L] = 0 \quad [H, L] = 0$$

$$\langle L \rangle = \text{const}$$

$$L = \sum_{\mu=1}^{4^n-1} a_{\mu} P_{\mu}$$



Integrals of motion (IOM): toy example

Integral of motion L

$$[U, L] = 0 \quad [H, L] = 0$$

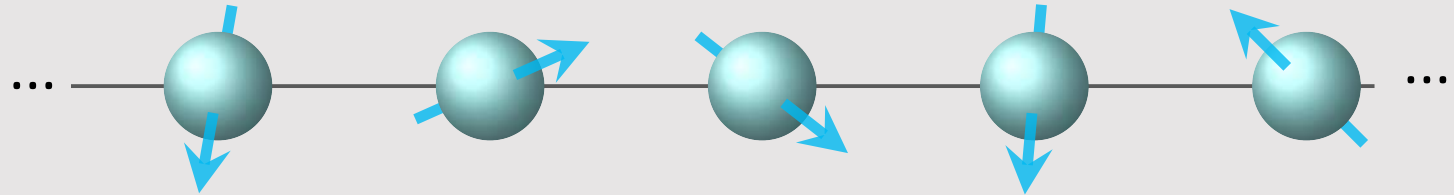
$$\langle L \rangle = \text{const}$$

$$L = \sum_{\mu=1}^{4^n-1} a_{\mu} P_{\mu}$$

n integrals of motion (IOM)

can label eigenstates*

$$H = \sum_{i=0}^{n-1} c_i Z_i + \sum_{i \neq j} c_{ij} Z_i Z_j$$



$$[H, Z_0] = 0$$

$$[H, Z_1] = 0$$

...

$$[H, Z_{n-2}] = 0$$

$$[H, Z_{n-1}] = 0$$

$$L_0 = Z_0, L_1 = Z_1, \dots$$

$$|l_0 l_1 \dots l_{n-1}\rangle = |l_0\rangle_{L_0} \otimes |l_1\rangle_{L_1} \otimes \dots \otimes |l_{n-1}\rangle_{n-1}$$

For this trivial toy model easy, but generically very hard!

* Say if we construct n orthogonal IOMs with eigenvalues ± 1 .

Energy is the only constant of motion in a non-integrable system (time independent).
In general, an **integrable system** has constants of motion other than the energy.

Local integral of motion (LIOM)

Integral of motion L

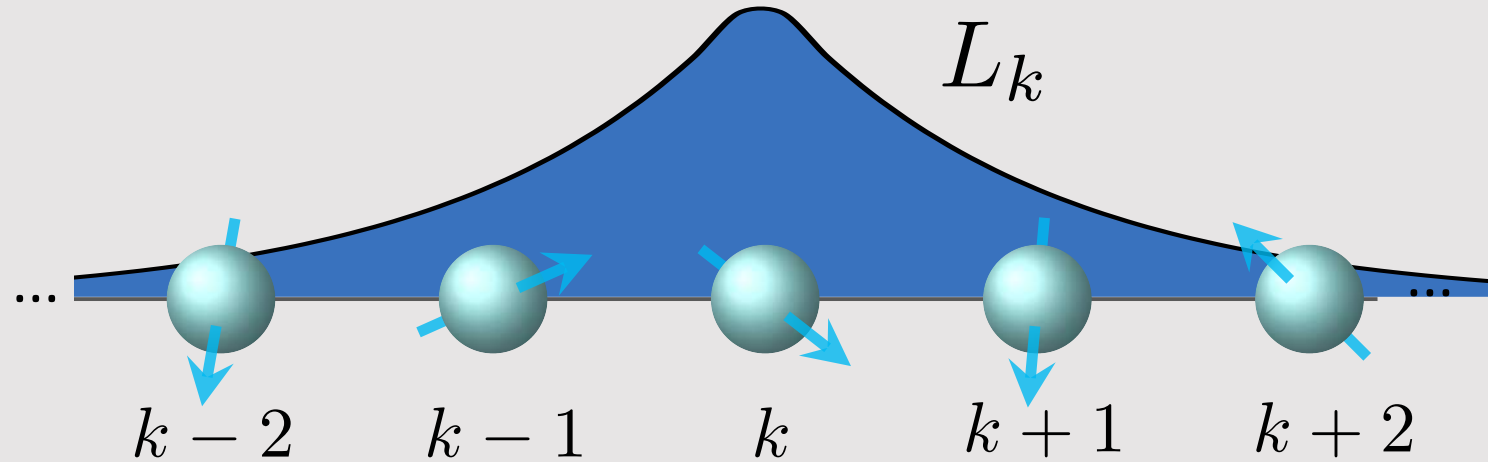
$$[U, L] = 0 \quad [H, L] = 0$$

$$\langle L \rangle = \text{const}$$

Local integral of motion (LIOM) L

$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_\mu P_\mu$$

local neighborhood



Tricky

- Beyond finding ground states
- We will be interested in dynamics and the full Hilbert space and spectrum
- Harder, can't use most methods like subspaces

Connection to pre-thermalization and many-body localization

Many-body localization (MBL)

- the "avalanche"
- many-body resonances.

Anderson 1958, Altshuler, G, KL 1997
Bosko et al 2005, many more

MBL is Anderson localization of many interacting degrees of freedom in states that thermodynamically have nonzero entropy density

Following Anderson 1958, let's do spin models:

Trivial limit: N spin- $1/2$'s

$$H_0 = \sum_n h_n S_{n,z}$$

h_n nondegenerate
random in $[-$

See earlier lectures by David Huse, Vedika, ...

Many-body localization, thermalization, and entanglement



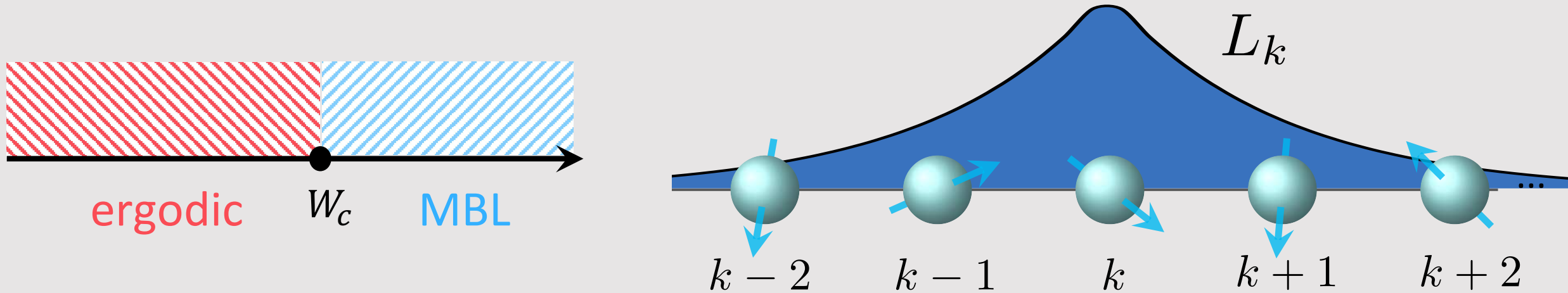
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Colloquium: Many-body localization, thermalization, and entanglement

Dmitry A. Abanin, Ehud Altman, Immanuel Bloch, and Maksym Serbyn
Rev. Mod. Phys. **91**, 021001 – Published 22 May 2019

Connection to pre-thermalization and many-body localization



Hypothesis: In the MBL regime, the original Hamiltonian

$$H = \sum_k \epsilon_k L_k + \sum_{k < j} J_{kj}^{(2)} L_k L_j + \sum_{i < j < k} J_{ijk}^{(3)} L_i L_j L_k + \dots$$

Basko, Aleiner, Altshuler, Ann Phys (2006)
Pal and Huse, RRB (2010)
Serbyn, Papic, Abanin, PRL (2013)
Huse, Nandkishore, Oganesyan PRB (2014)
...

See earlier lectures by David Huse, Vedika, ...

Local integrals of motion (LIOM): Ergodicity breaking and many-body localization

Prototypical phenomenon:
prethermalization and **many-body localization (MBL)**

Goes back to Anderson and **disordered** systems,
but this was single non-interacting particles



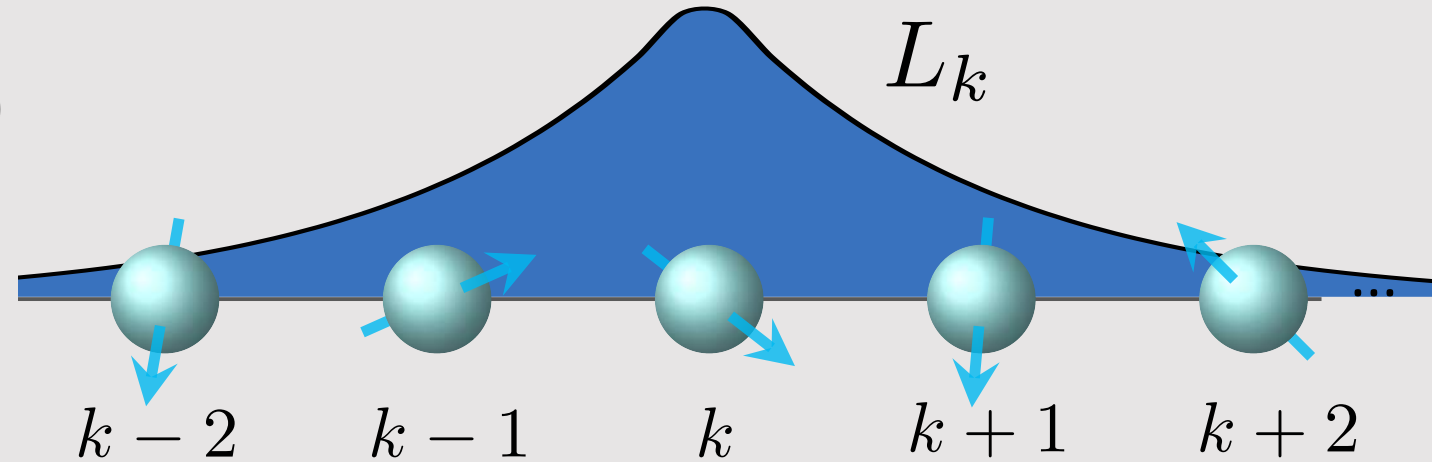
"for their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems"

Existence of MBL phase is under debate.
Most of community agree that MBL exists in 1D.

- **MBL systems** have extensive set of local integrals of motion (LIOMs).
- **Prethermal systems (non-MBL)** can have approximate LIOMs

$$[e^{-iHt}, L_k] \approx 0$$

- We uncover LIOMs in 1D and approximate LIOMs in 2D in 104 and 124 qubit lattices using a digital quantum computer

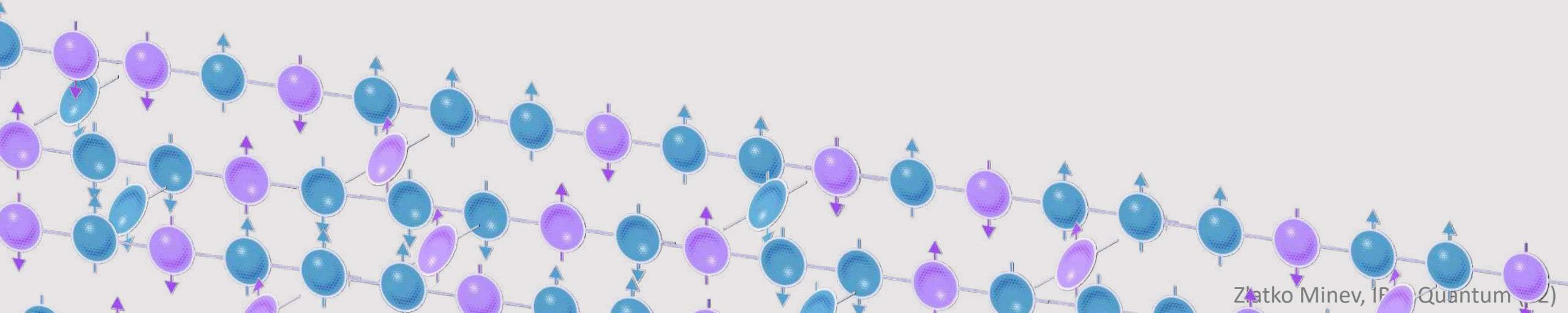


Basko, Aleiner, Altshuler, Ann Phys (2006)
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...

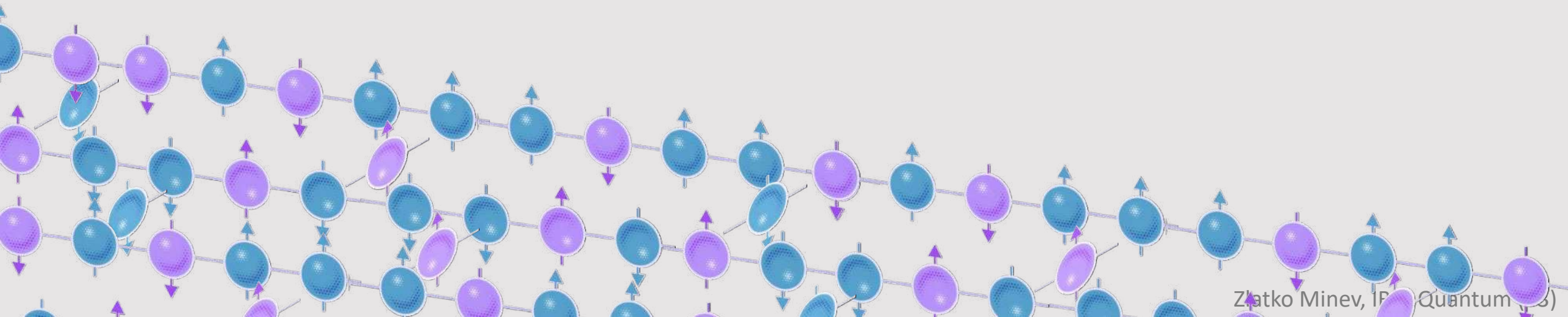
Experiments related to MBL

- M. Schreiber, S. S. Hodgman, P. Bordia, H. P. Lüschén, M. H. Fischer, R. Vosk, E. Altman, U. Schneider, and I. Bloch, Observation of many-body localization of interacting fermions in a quasirandom optical lattice, *Science* 349, 842 (2015).
- J.-y. Choi, S. Hild, J. Zeiher, P. Schauß, A. Rubio-Abadal, T. Yefsah, V. Khemani, D. A. Huse, I. Bloch, and C. Gross, Exploring the many-body localization transition in two dimensions, *Science* 352, 1547 (2016).
- J. Smith, A. Lee, P. Richerme, B. Neyenhuis, P. W. Hess, P. Hauke, M. Heyl, D. A. Huse, and C. Monroe, Many-body localization in a quantum simulator with programmable random disorder, *Nat. Phys.* 12, 907 (2016).
- P. Roushan, C. Neill, J. Tangpanitanon, V. M. Bastidas, A. Megrant, R. Barends, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, et al., Spectroscopic signatures of localization with interacting photons in superconducting qubits, *Science* 358, 1175 (2017).
- P. Bordia, H. Lüschén, U. Schneider, M. Knap, and I. Bloch, Periodically driving a many-body localized quantum system, *Nat. Phys.* 13, 460 (2017).
- P. Bordia, H. Lüschén, S. Scherg, S. Gopalakrishnan, M. Knap, U. Schneider, and I. Bloch, Probing slow relaxation and many-body localization in two-dimensional quasiperiodic systems, *Phys. Rev. X* 7, 041047 (2017).
- M. Rispoli, A. Lukin, R. Schittko, S. Kim, M. E. Tai, J. Léonard, and M. Greiner, Quantum critical behaviour at the many-body localization transition, *Nature* 573, 385 (2019).
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- X. Mi, M. Ippoliti, C. Quintana, A. Greene, Z. Chen, J. Gross, F. Arute, K. Arya, J. Atalaya, R. Babbush, et al., Time-crystalline eigenstate order on a quantum processor, *Nature* 601, 531 (2022).
- J. Léonard, S. Kim, M. Rispoli, A. Lukin, R. Schittko, J. Kwan, E. Demler, D. Sels, and M. Greiner, Probing the onset of quantum avalanches in a many-body localized system, *Nat. Phys.* 19, 481 (2023).
-

Can we uncover the integrals of motion $\{L\}$
of a large, disordered many-body system
using a digital quantum computer?



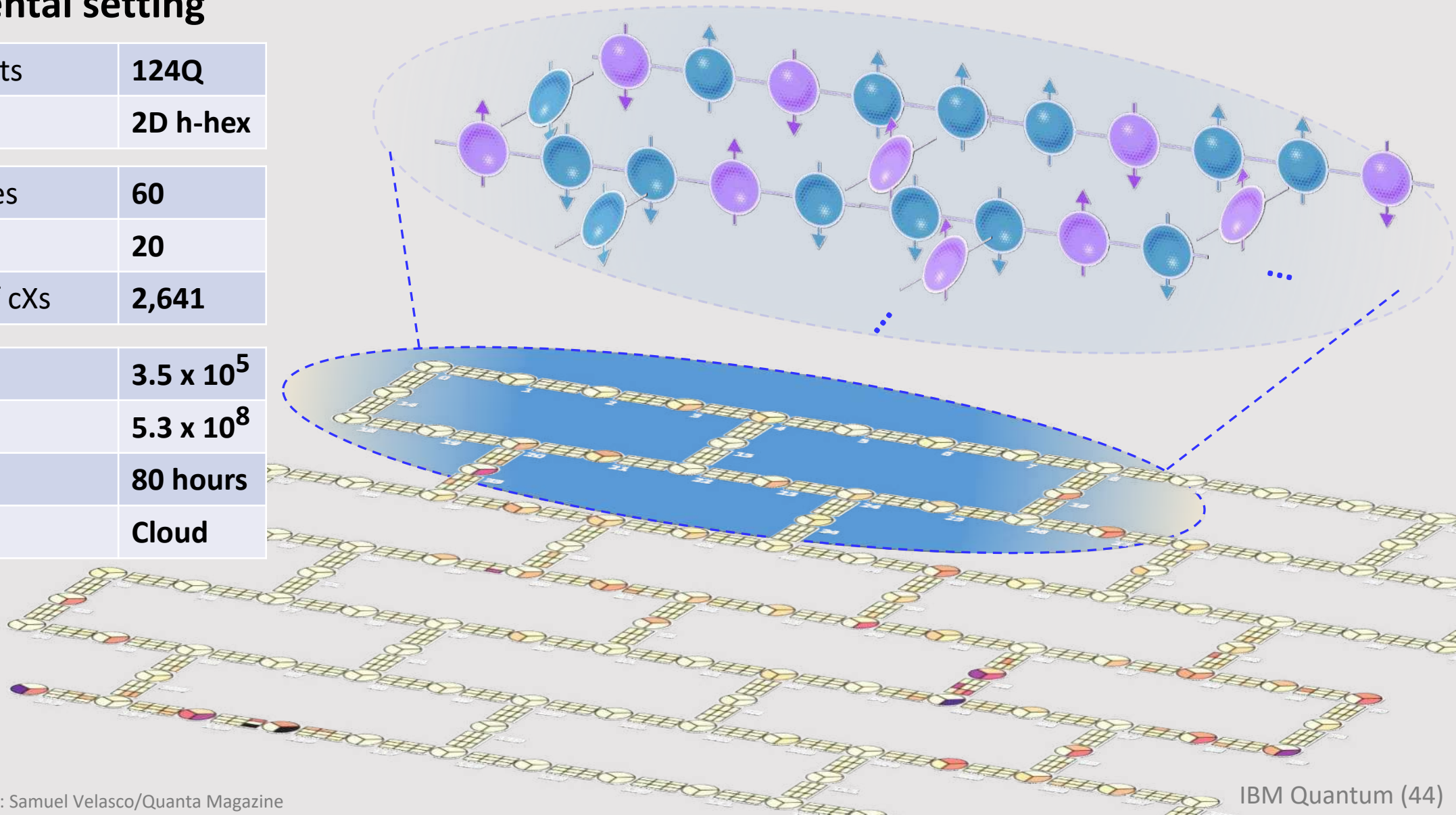
Preview



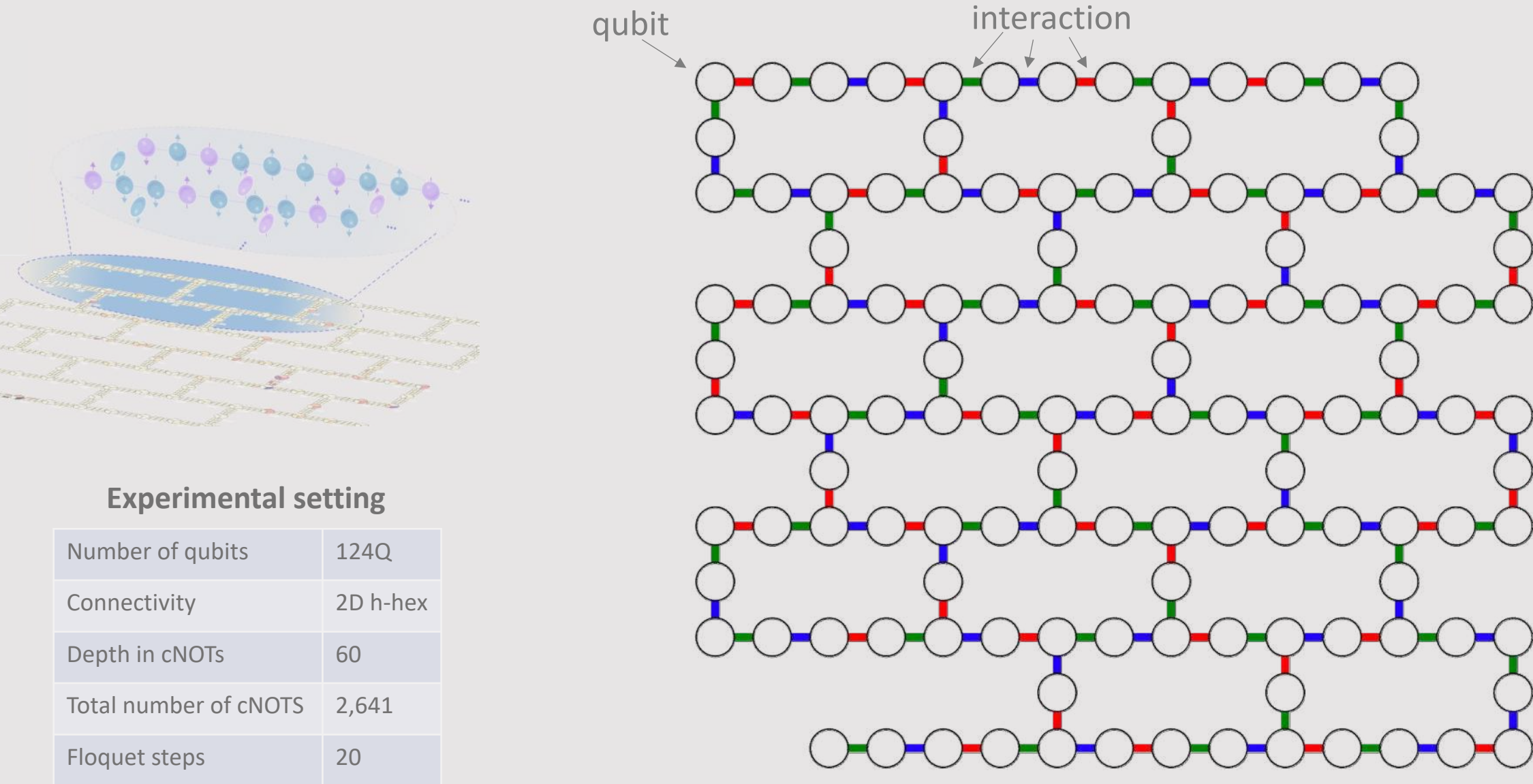
QSim on 100Q+: 2D interacting many-body Floquet system

Experimental setting

Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641
Circuits	3.5×10^5
Shots	5.3×10^8
QPU runtime	80 hours
Environment	Cloud

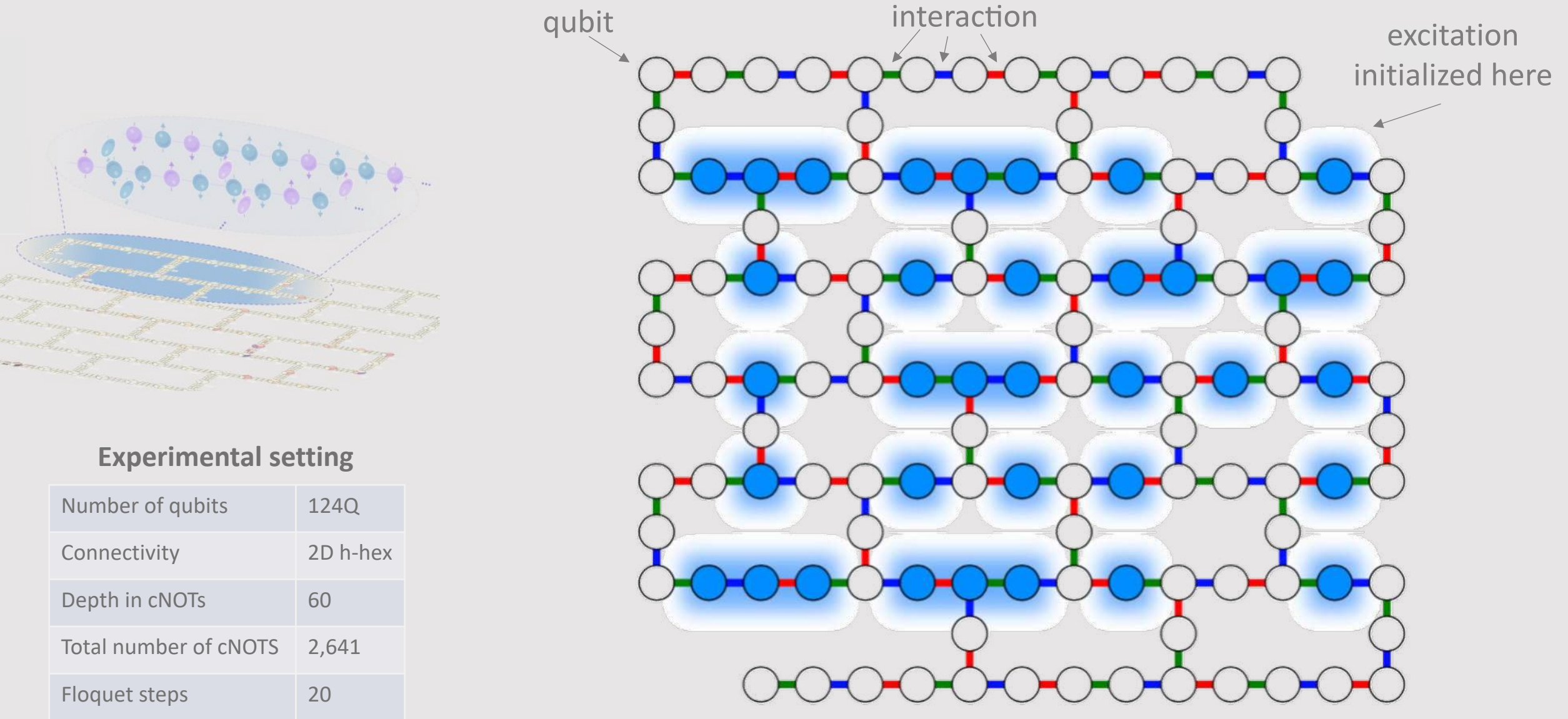


Interaction map and device layers



Color represent charge/spin polarization

Initialize lattice in fun states

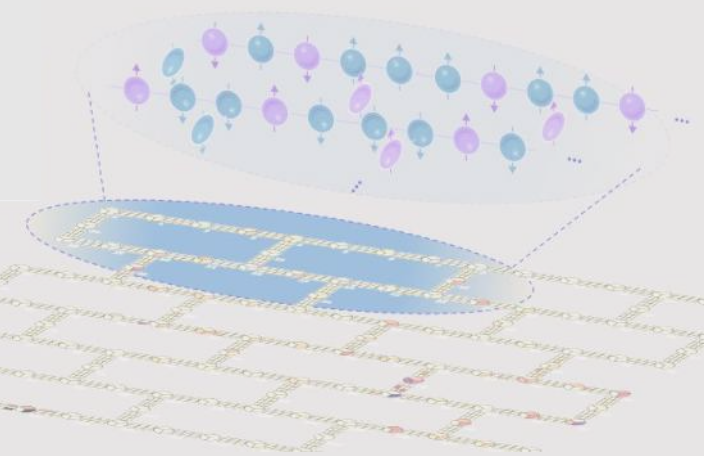


Experimental setting

Number of qubits	124Q
Connectivity	2D h-hex
Depth in cNOTs	60
Total number of cNOTS	2,641
Floquet steps	20

Color represent spin polarization

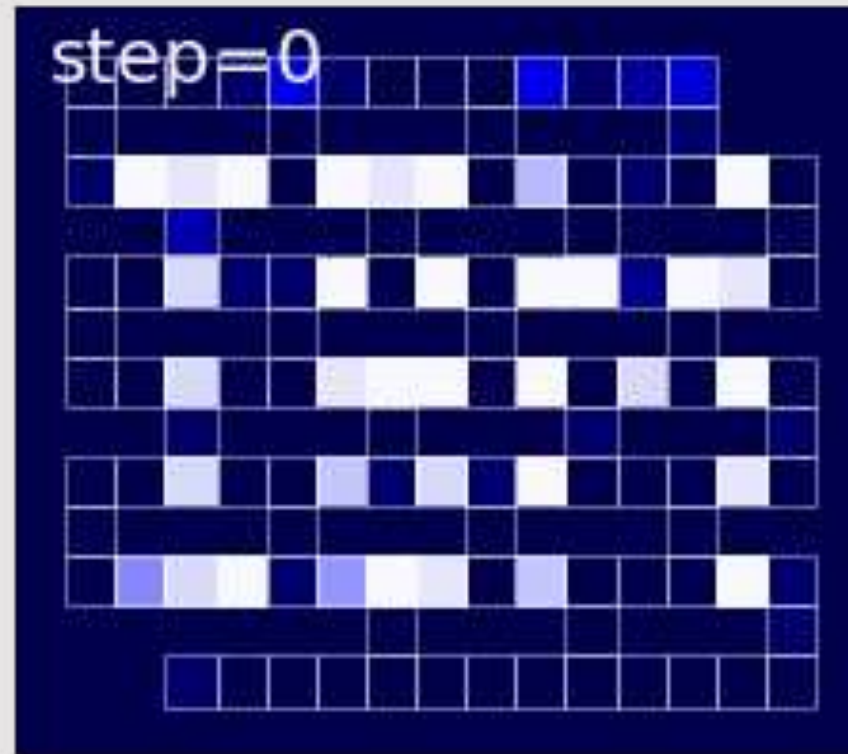
Quantum dynamics in different regimes



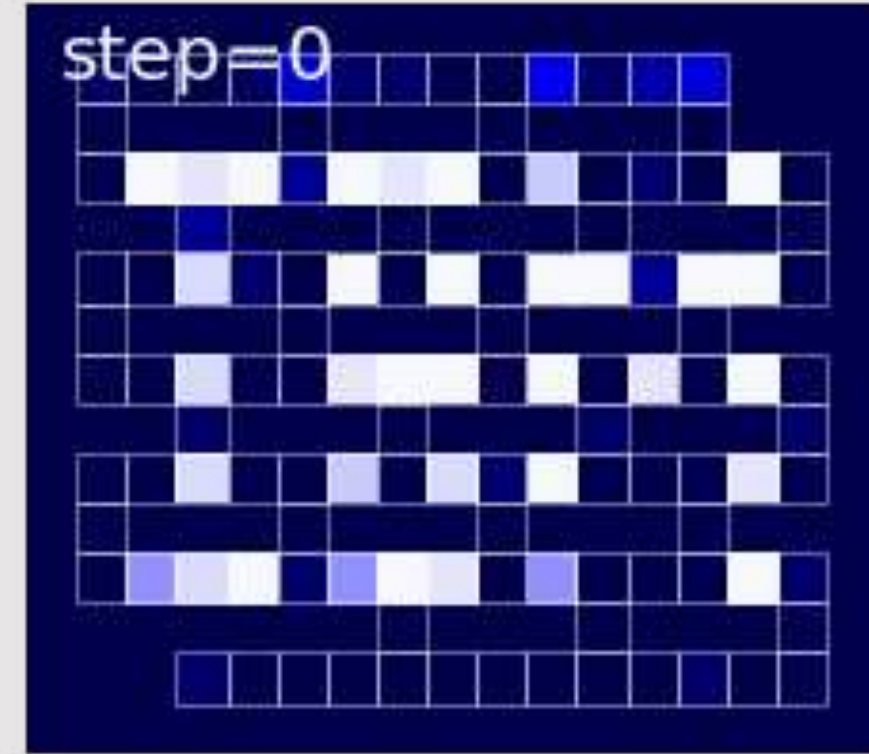
Experimental setting

Number of qubits	124Q
Connectivity	2D h-hex
Depth in cNOTs	60
Total number of cNOTs	2,641
Floquet steps	20

Shtanko, Wang, ..., Mineev (2023)



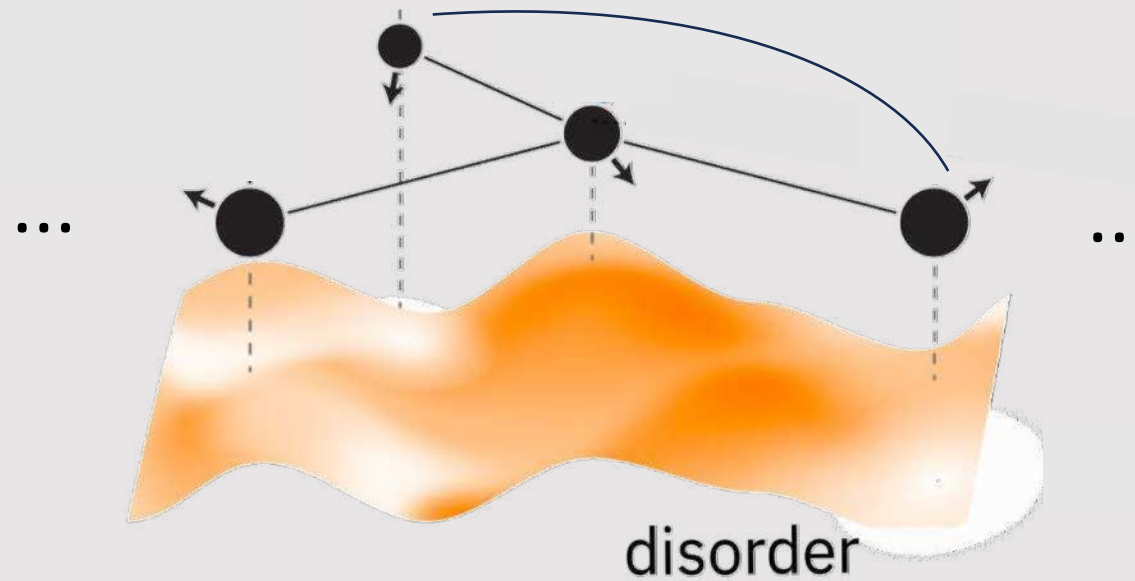
Thermalizing regime



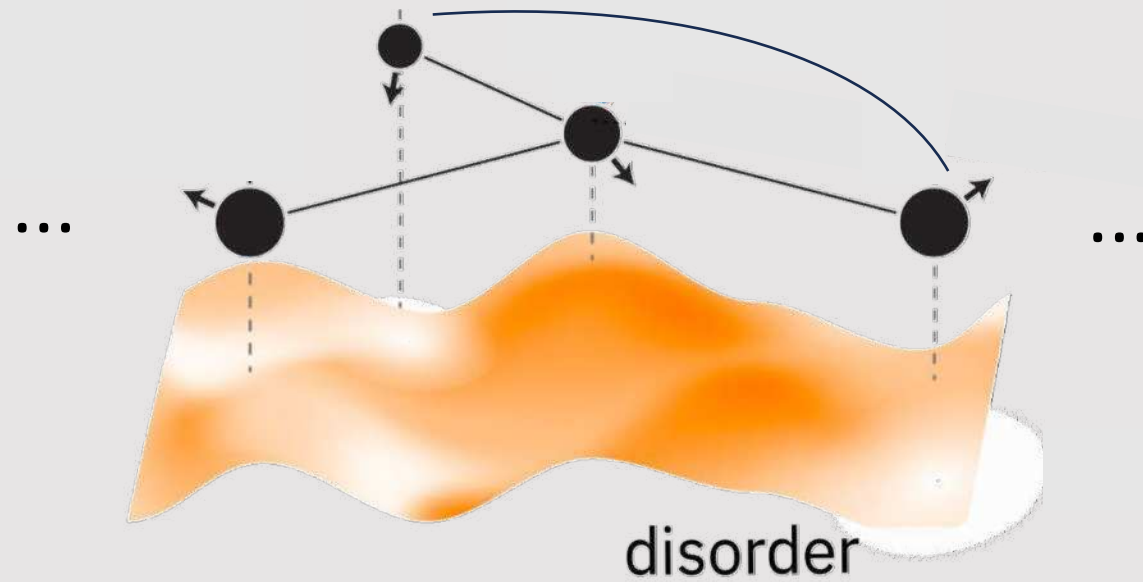
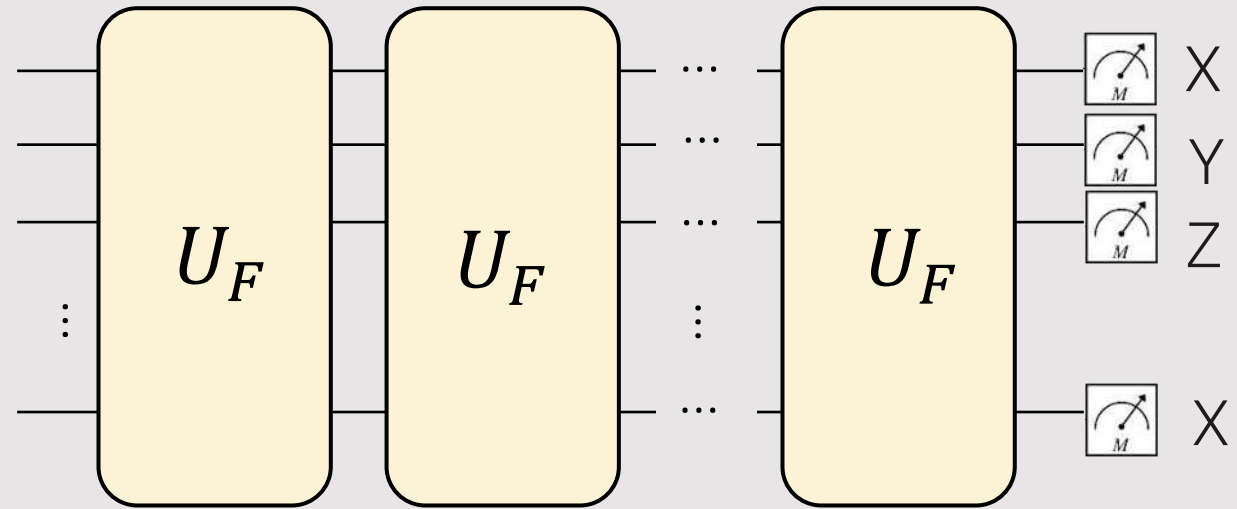
Prethermal regime

Color represent spin polarization

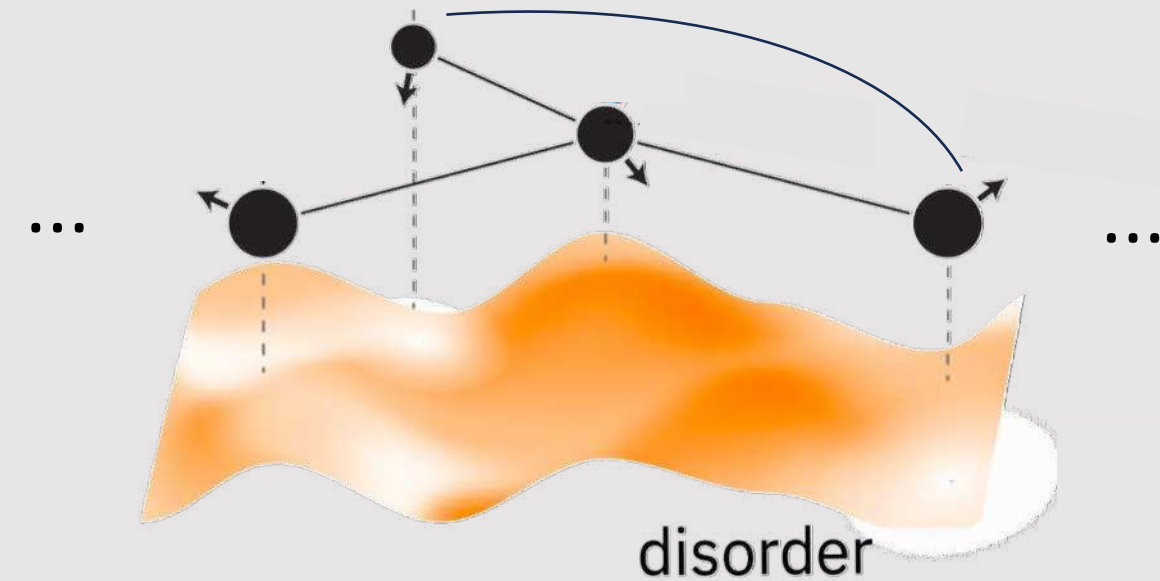
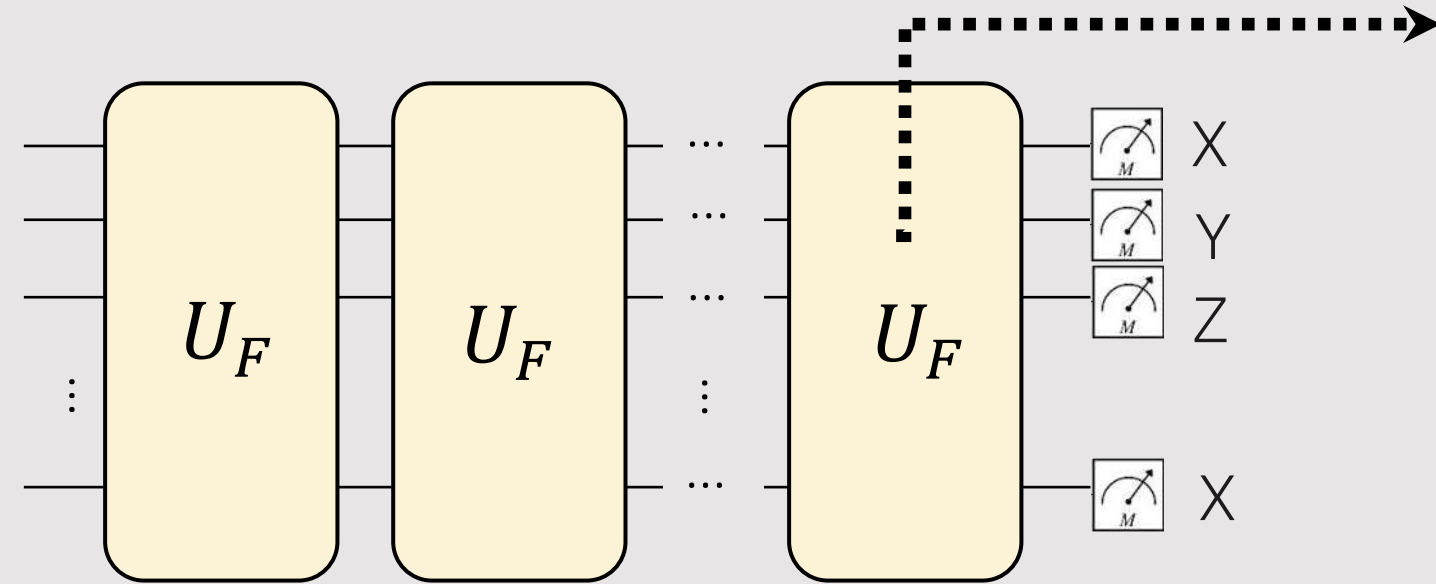
Spin lattice system over some potential



Floquet circuit evolution

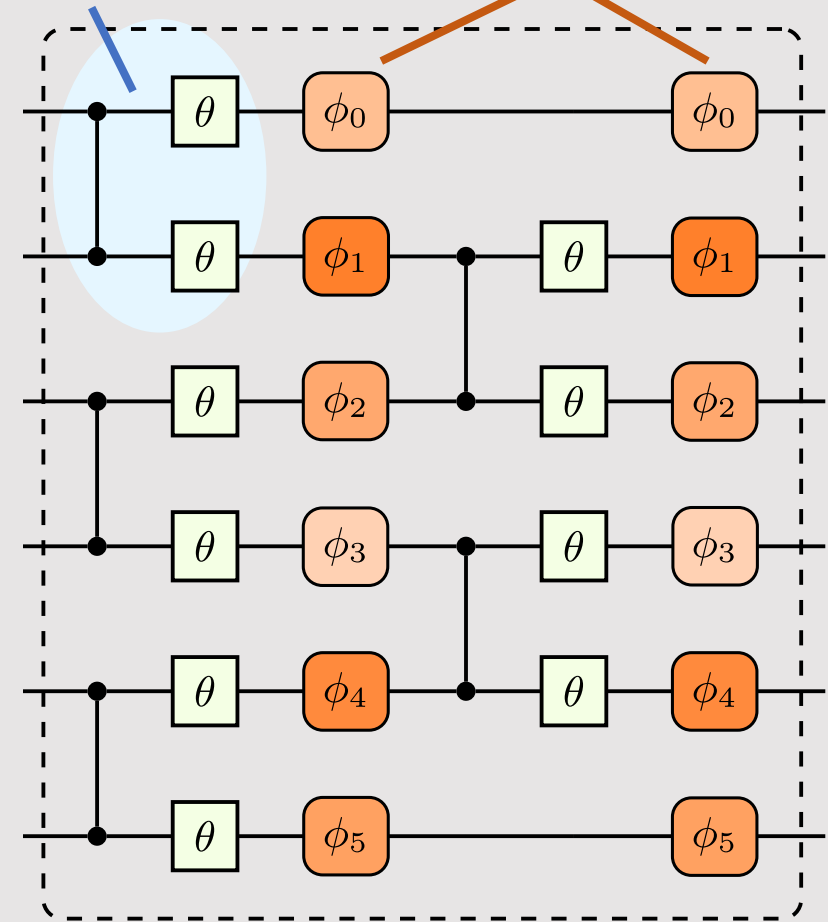


Circuit model



Kinetic term
+ Interactions

Spatial disorder

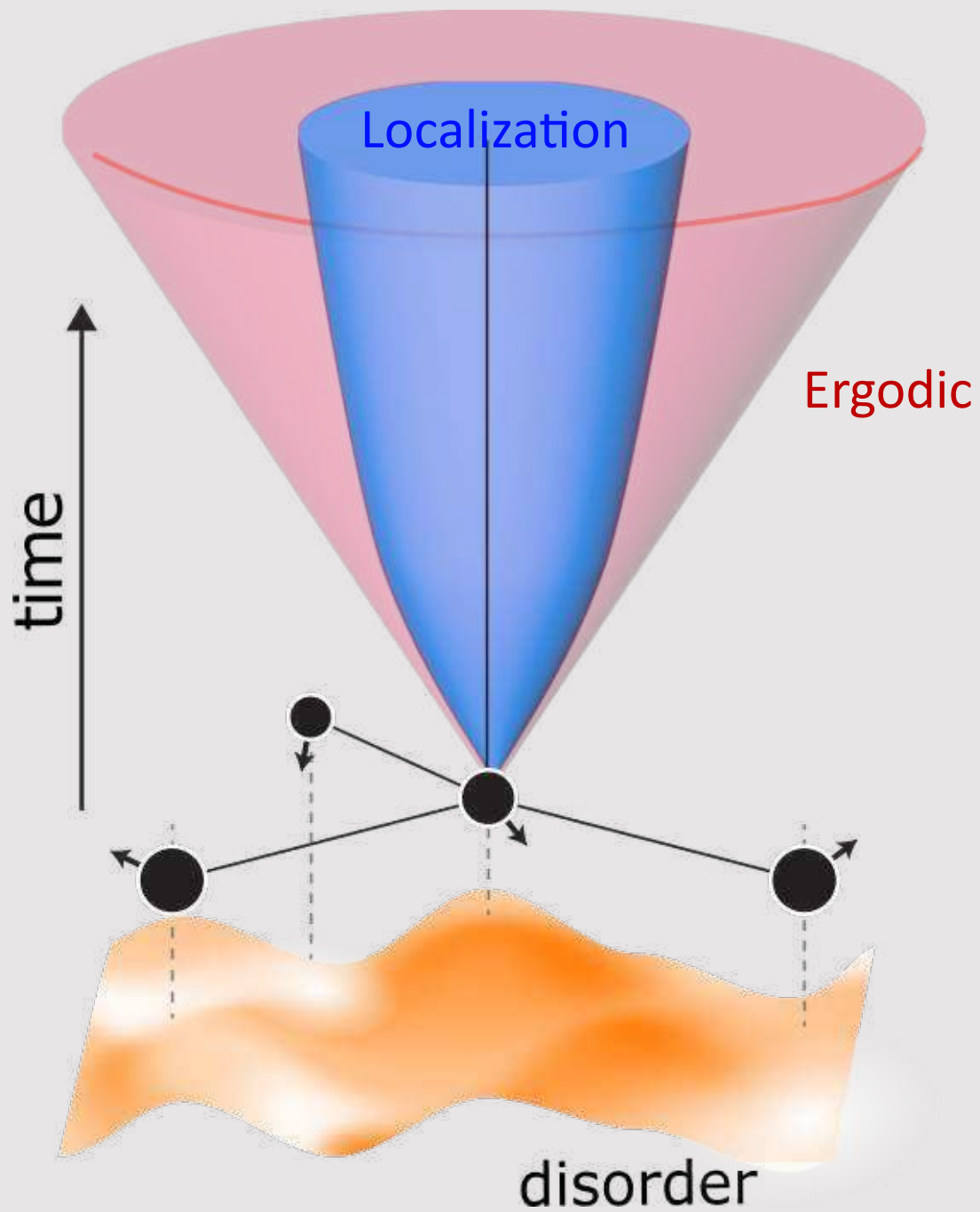


$$P(\phi_k) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi_k} \end{pmatrix}$$

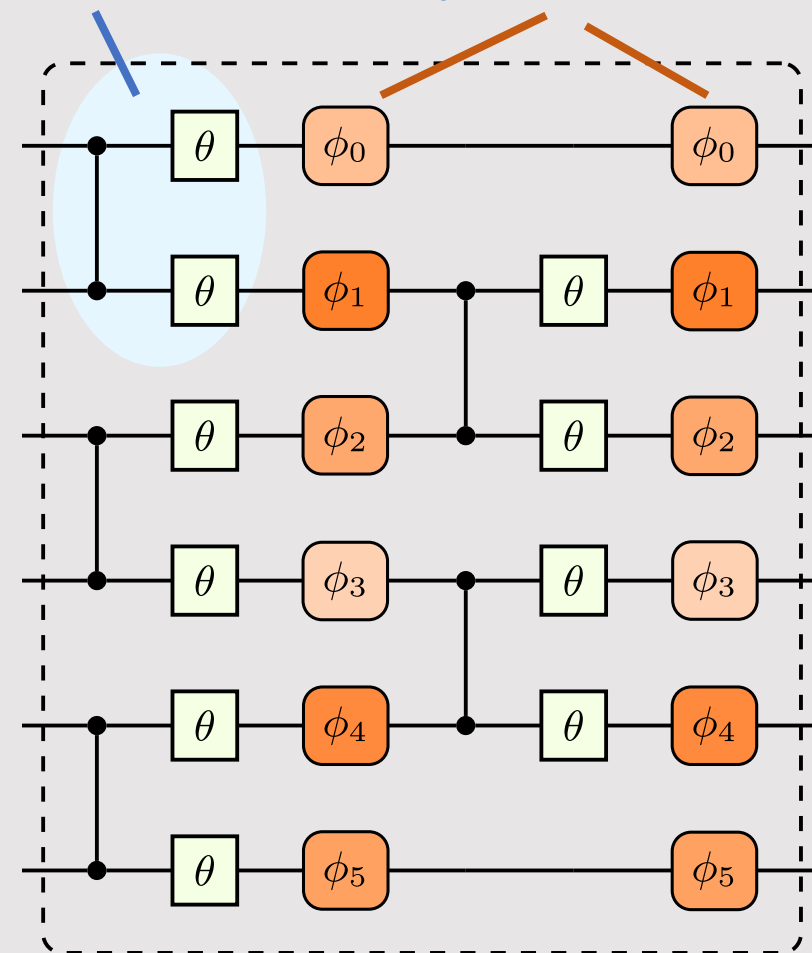
$$\phi_k \in [-\pi, \pi]$$

Uniformly sample
disorder

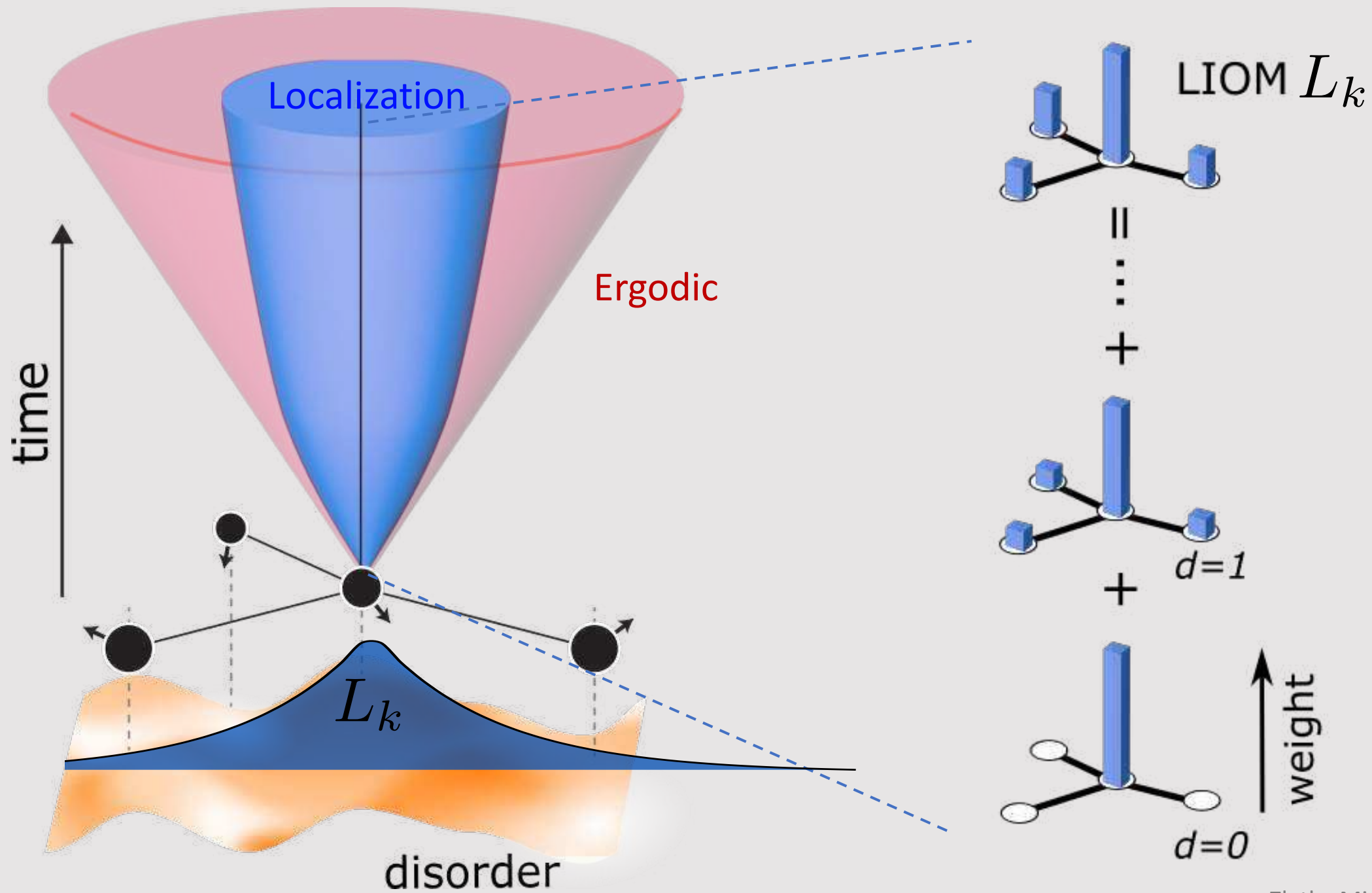
$$U(\theta) = \begin{pmatrix} \cos \theta/2 & \sin \theta/2 \\ \sin \theta/2 & -\cos \theta/2 \end{pmatrix}$$



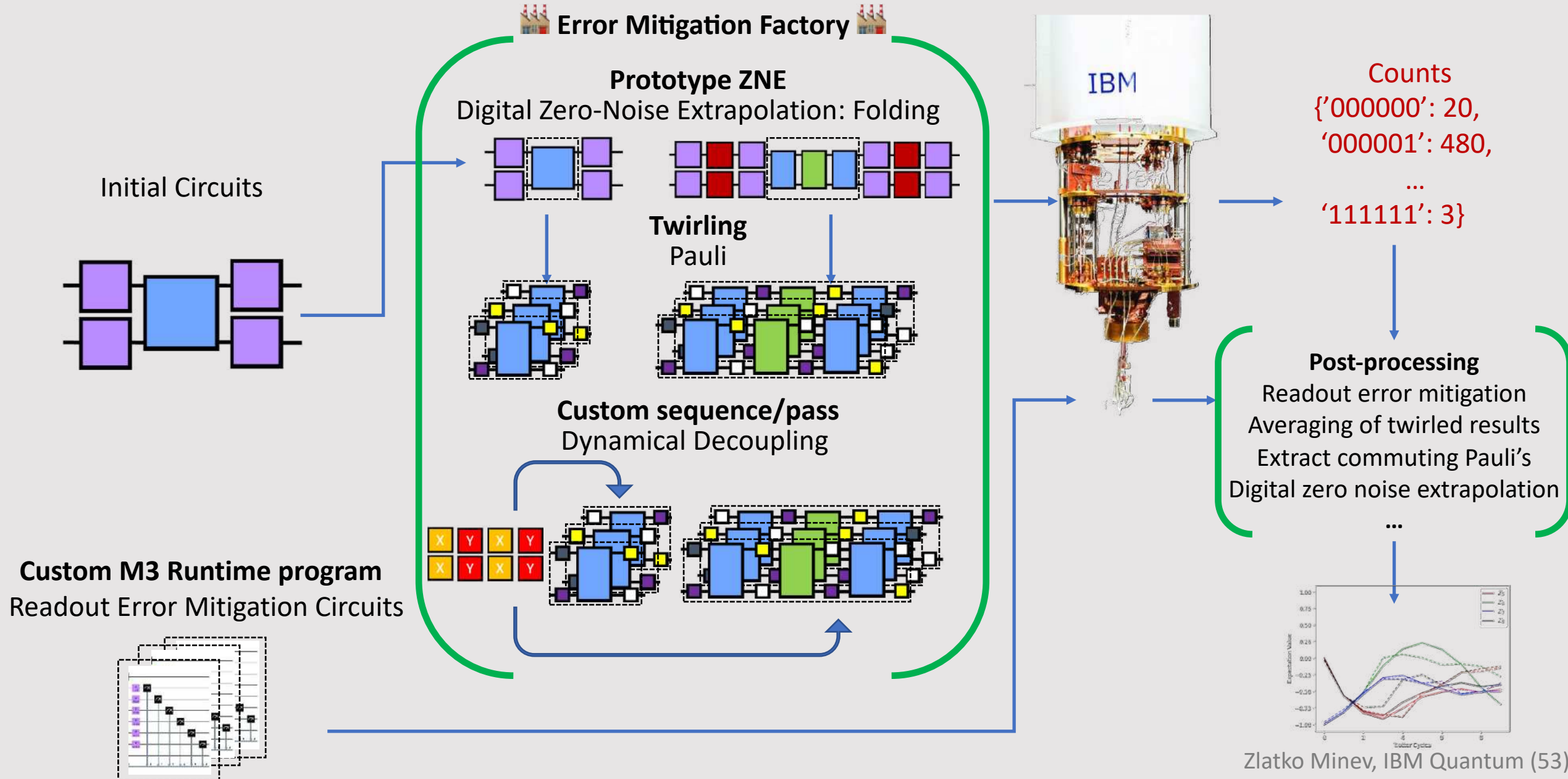
Kinetic term
+ Interactions Spatial disorder



Depends on parameters



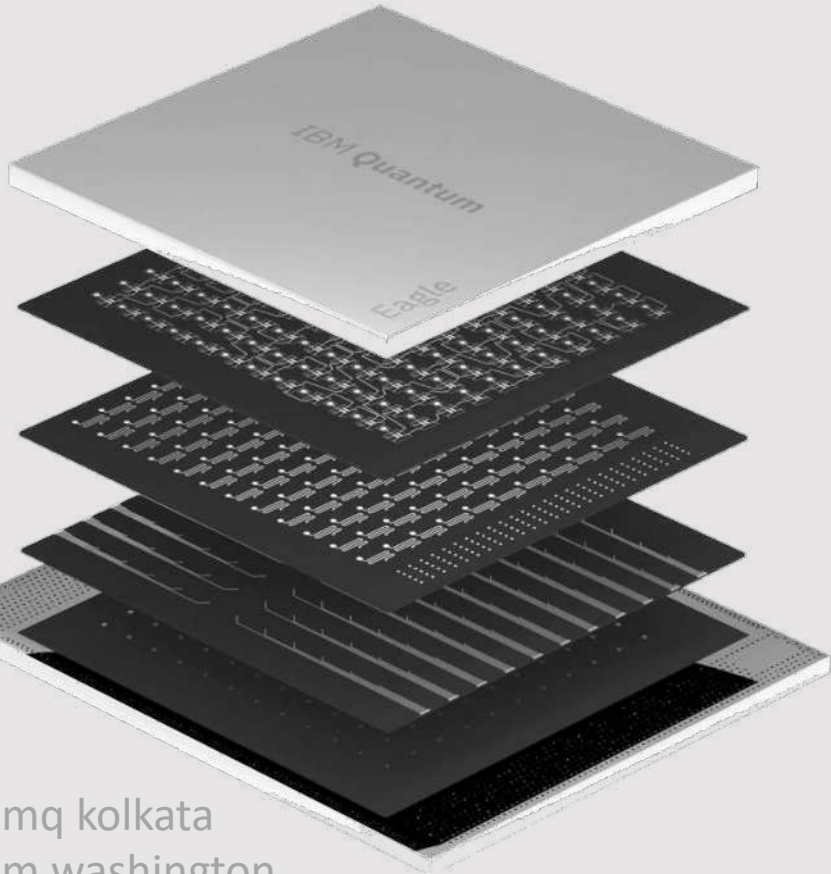
A composite error mitigation strategy



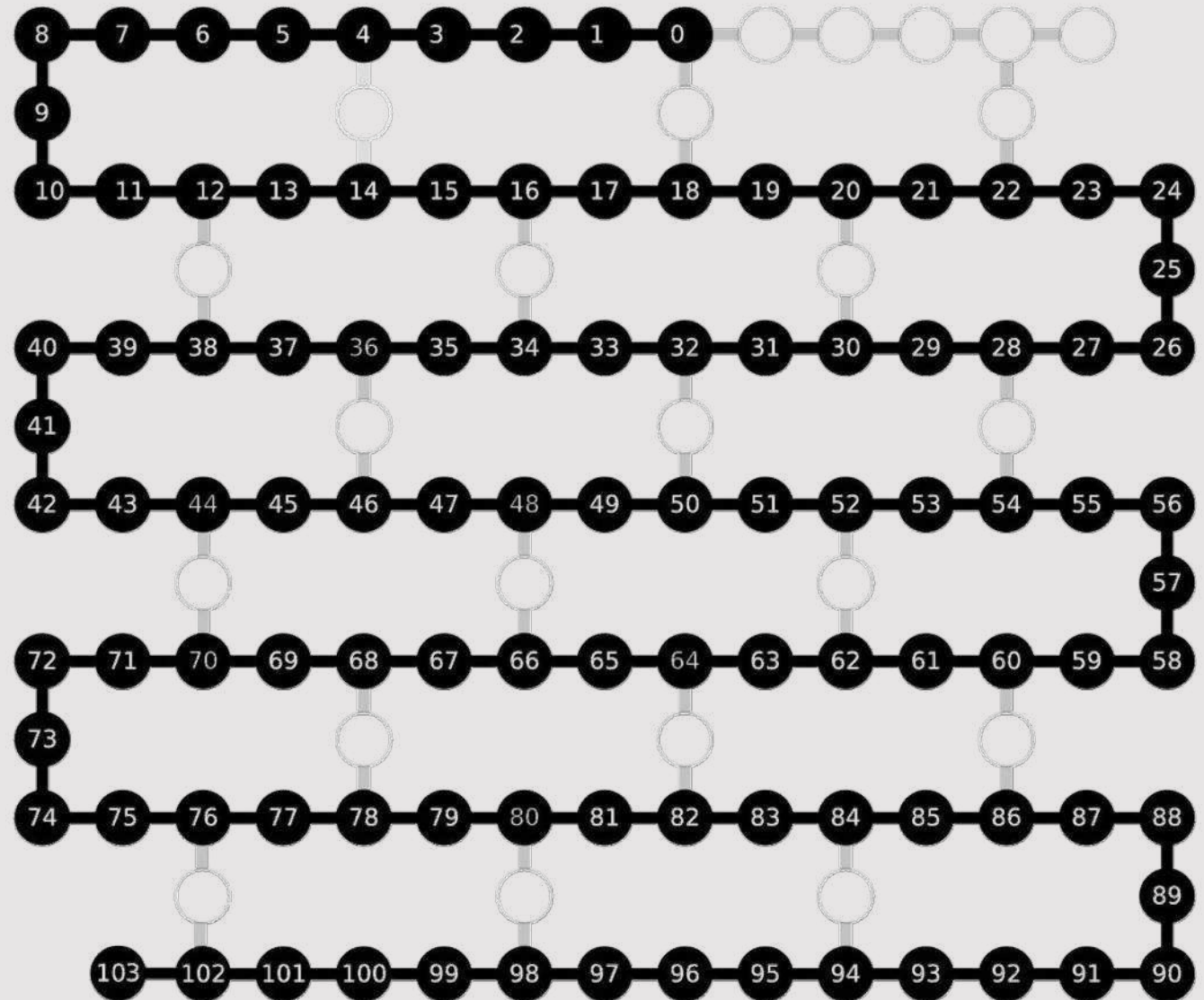
Benchmark quantum hardware and model

Start with 1D

1D spin chain



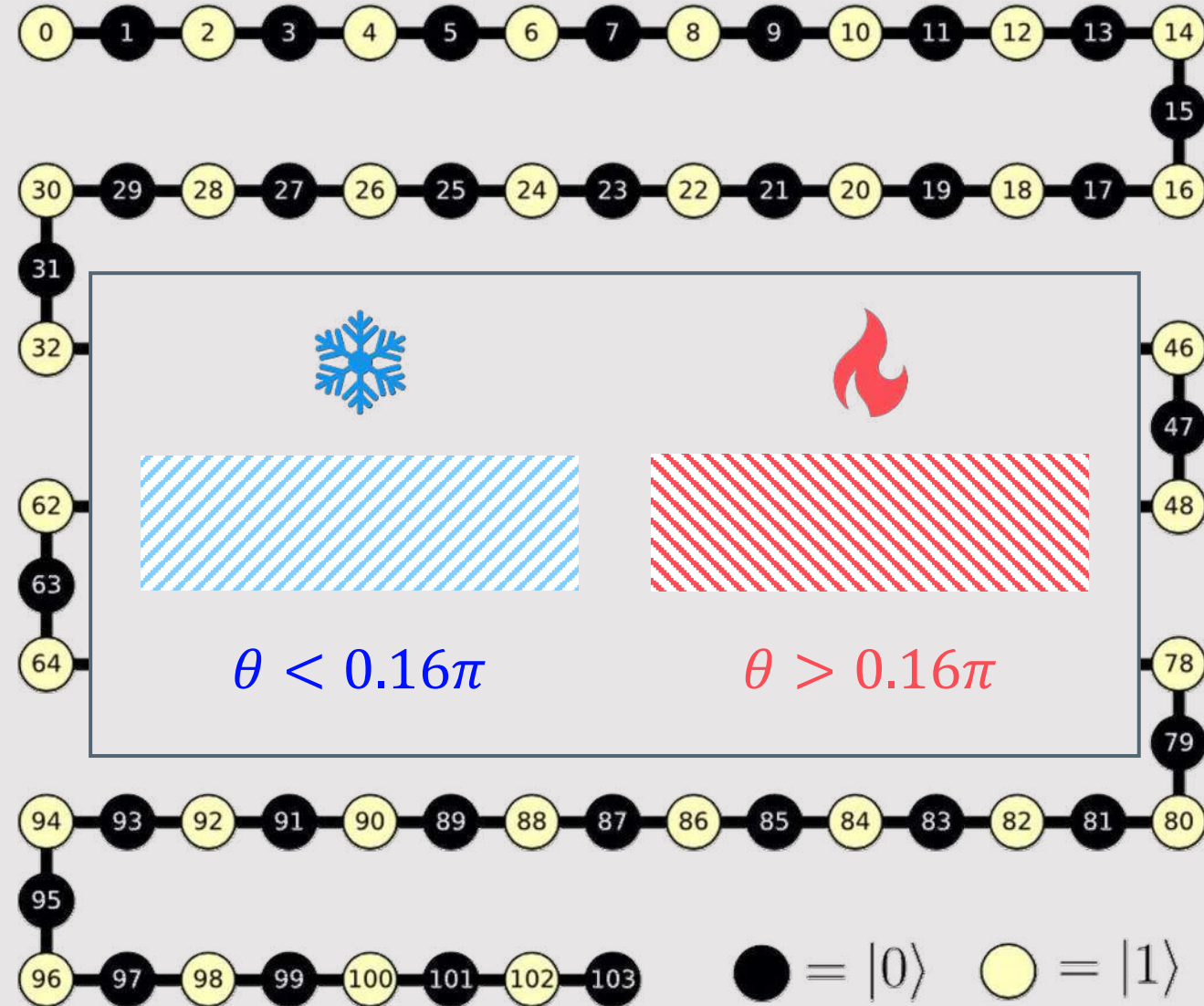
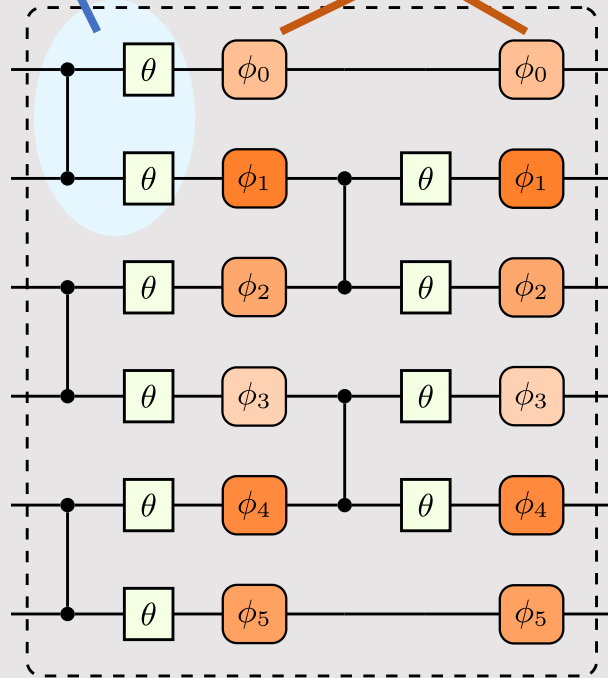
ibmq kolkata
ibm washington



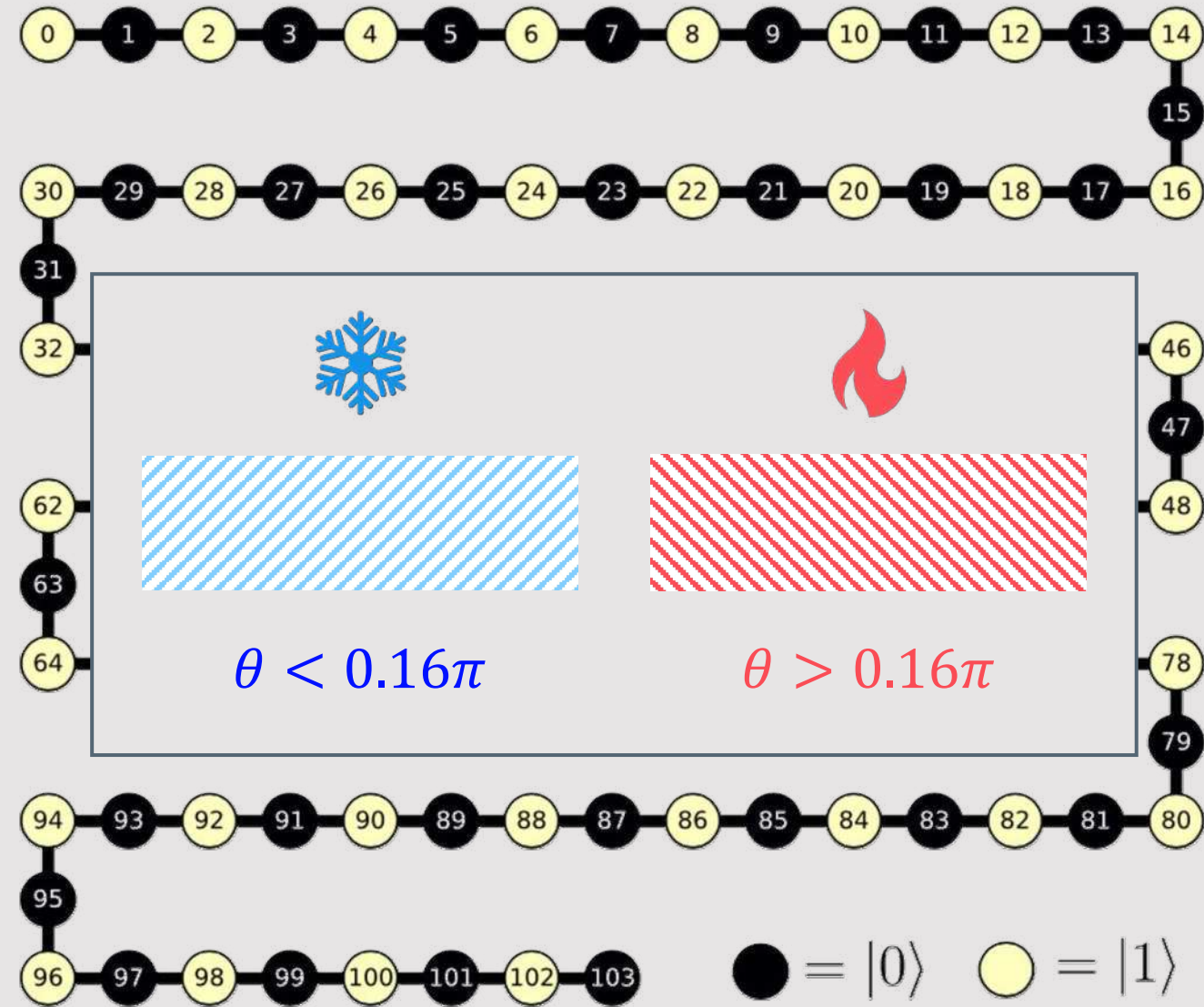
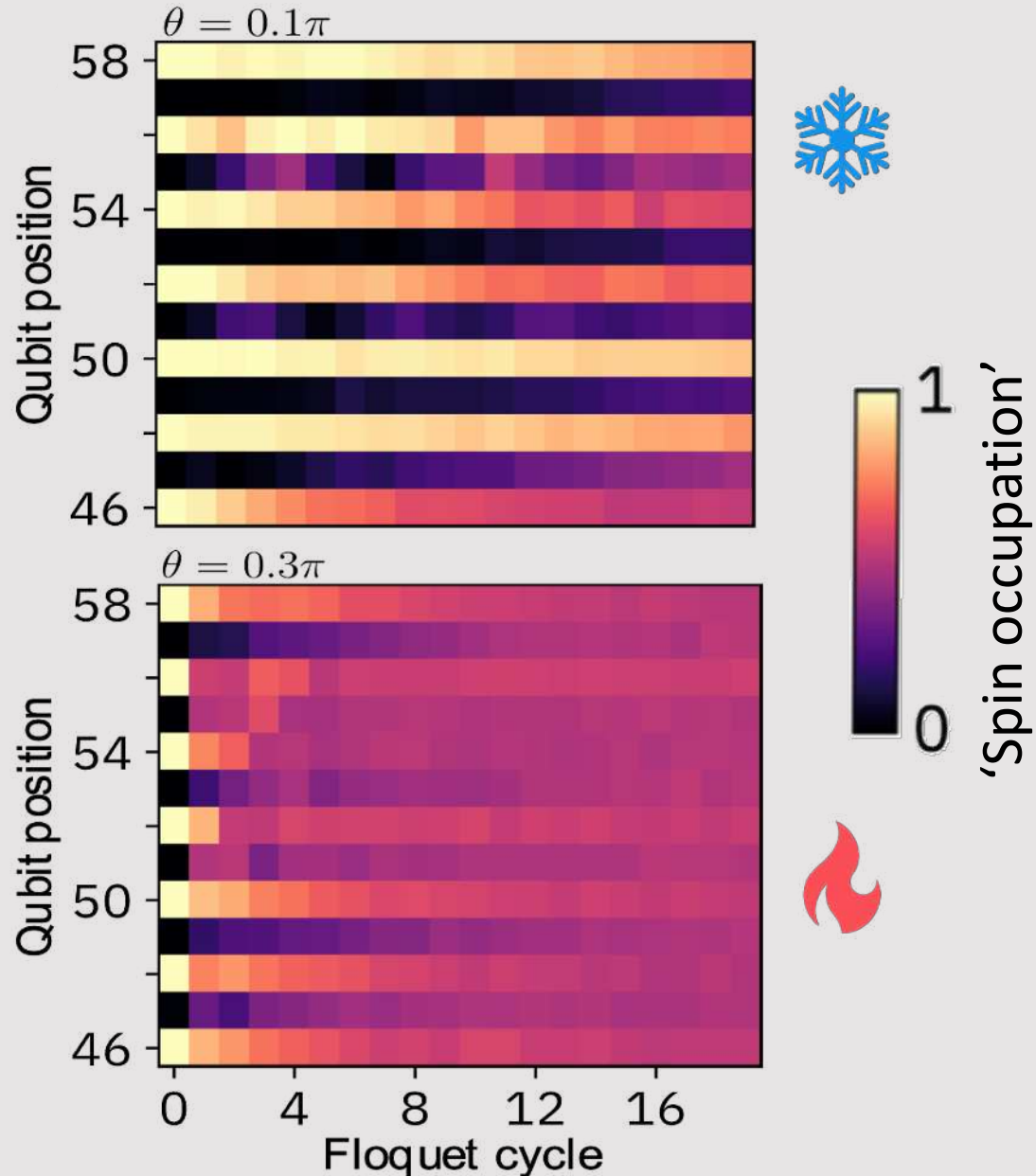
Initial antiferromagnetic state with spin imbalance

$$|\psi_0\rangle = |1\rangle|0\rangle|1\rangle|0\rangle|1\rangle|0\rangle \dots |1\rangle|0\rangle|1\rangle|0\rangle|1\rangle$$

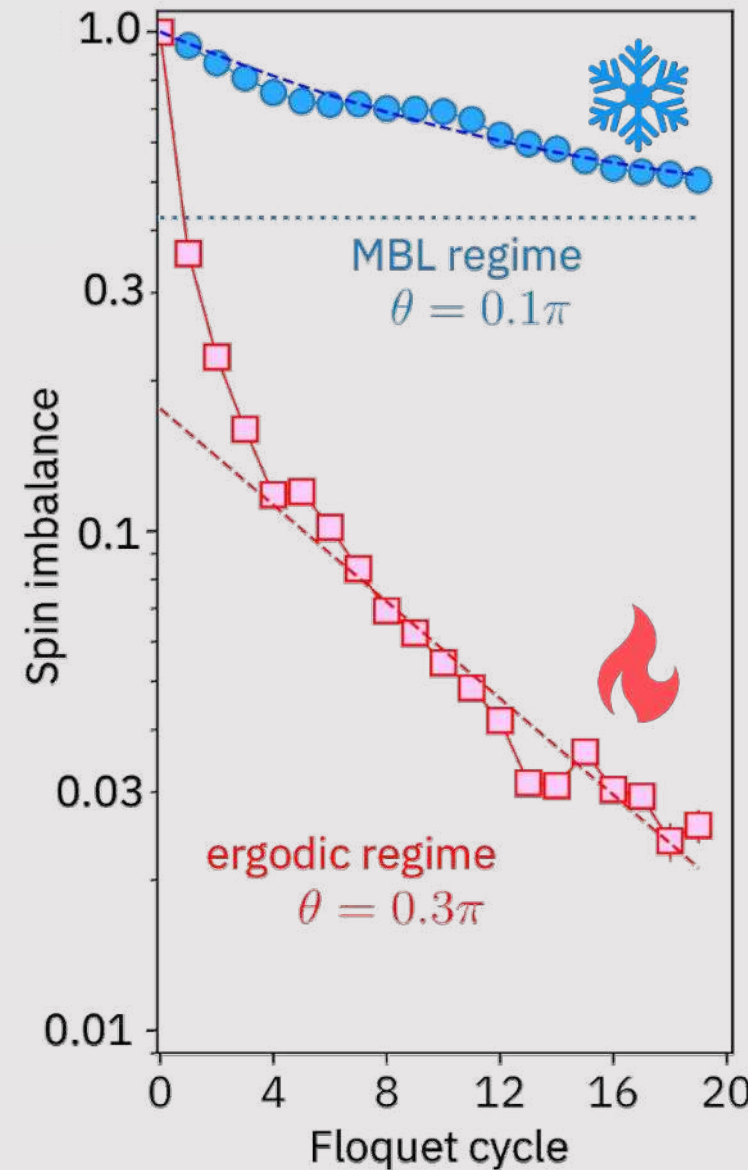
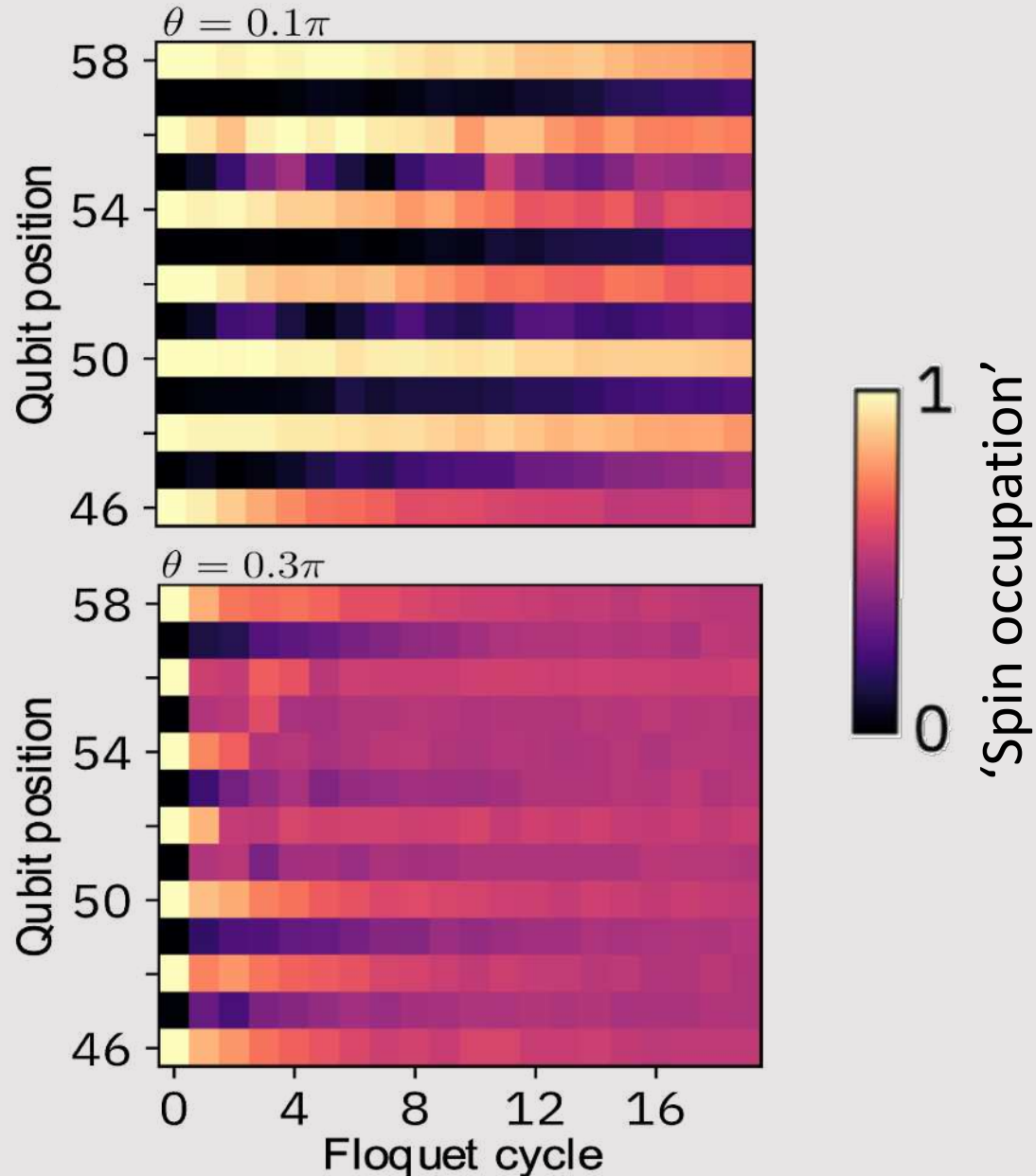
Kinetic term Spatial disorder



Time evolution of the antiferromagnetic state



Memory of the initial state



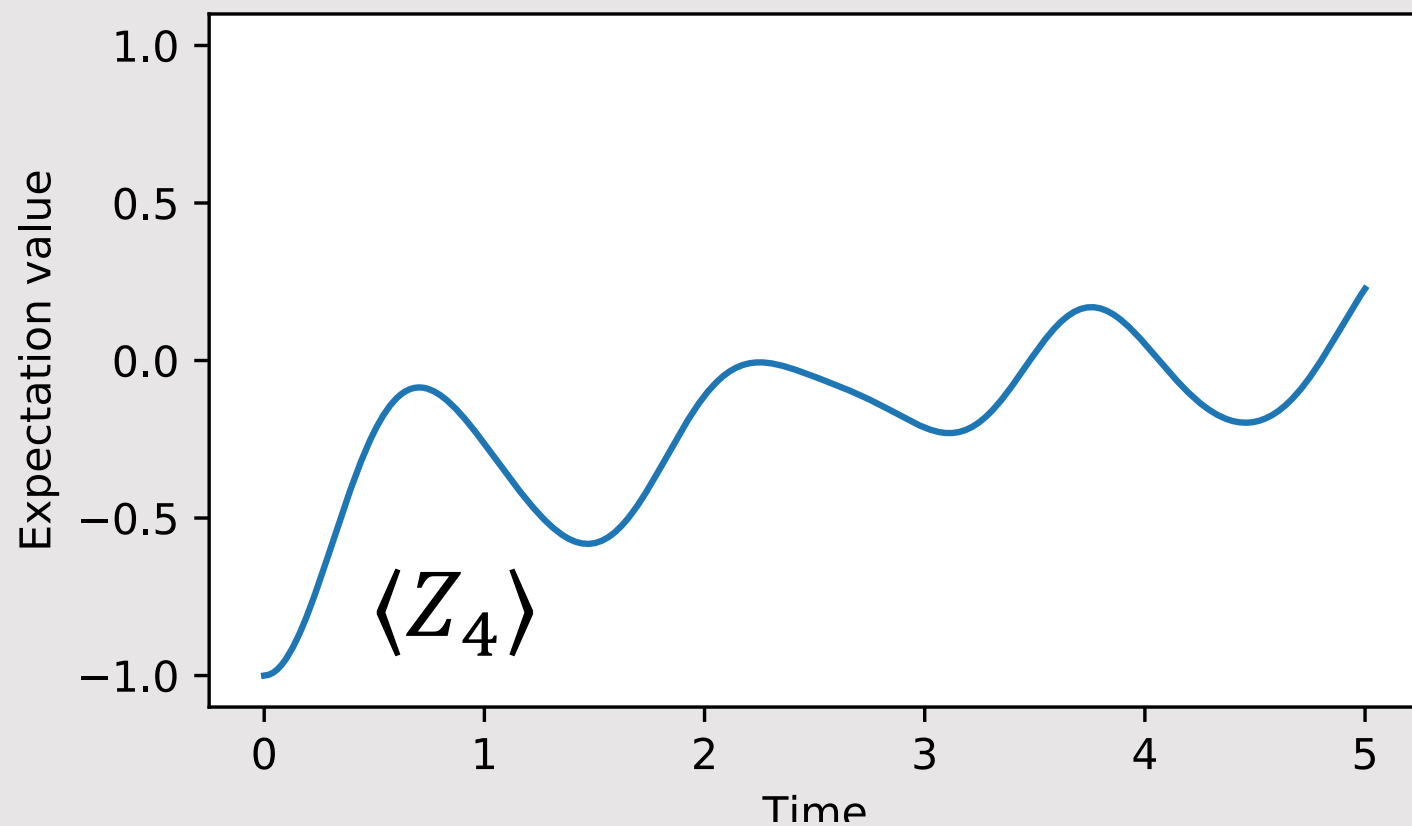
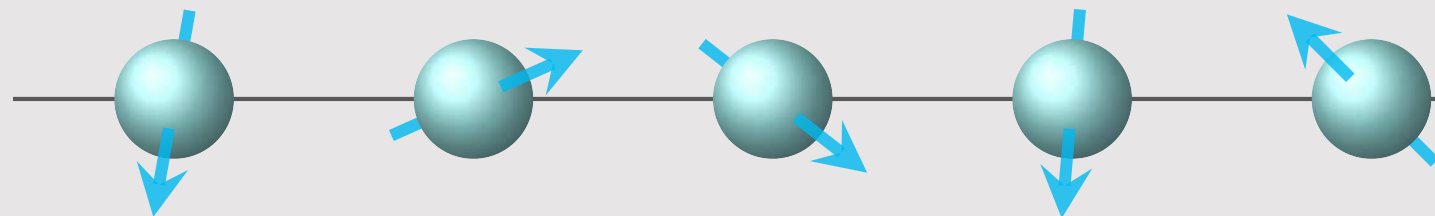
$$\mathcal{I} = \frac{n_1 - n_0}{n_1 + n_0}$$

$$n_k = \frac{1}{2} \mathbb{E}(1 - \langle Z_i \rangle)$$

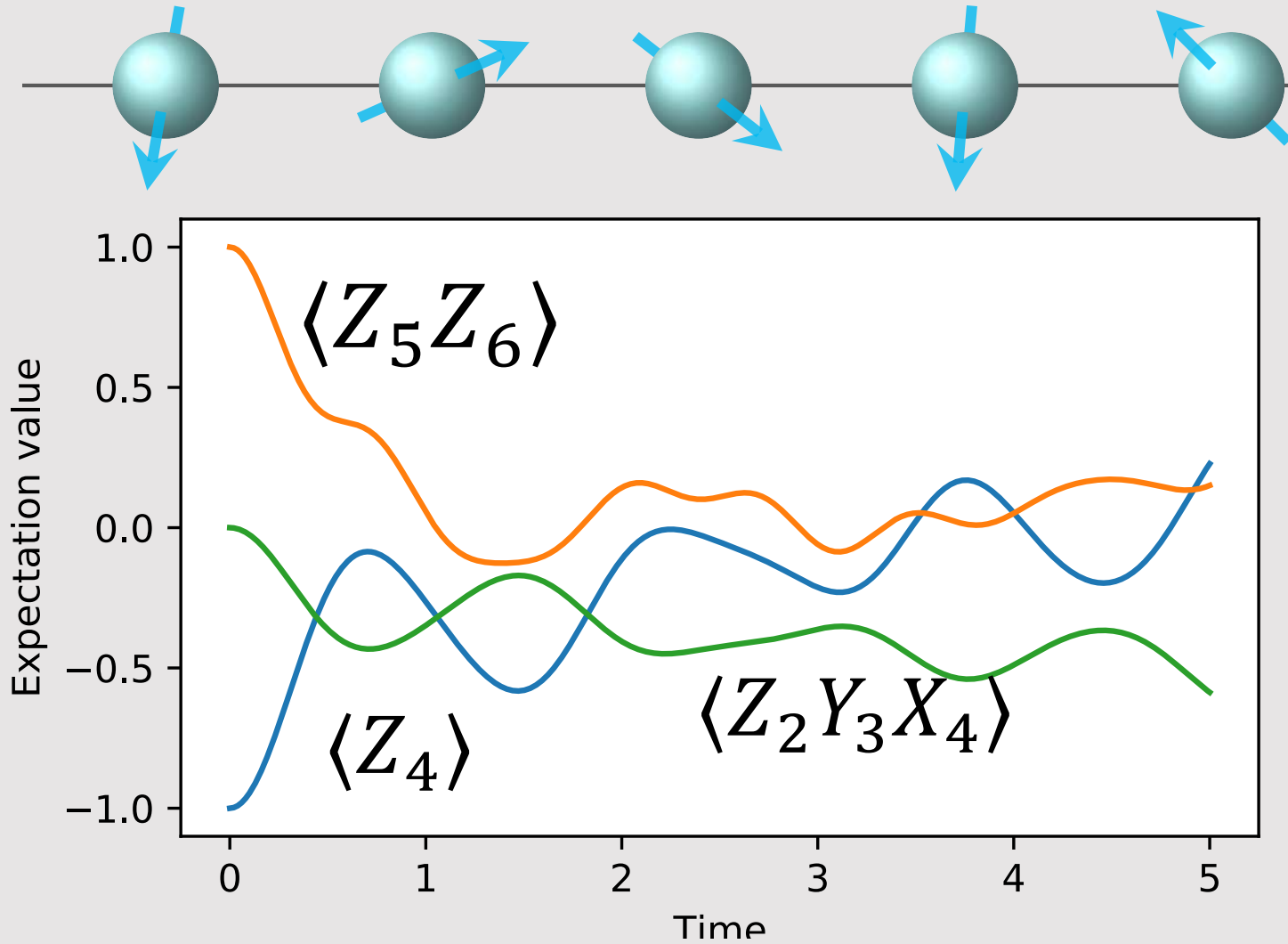
With error mitigation

Local integrals of motion (LIOMs)

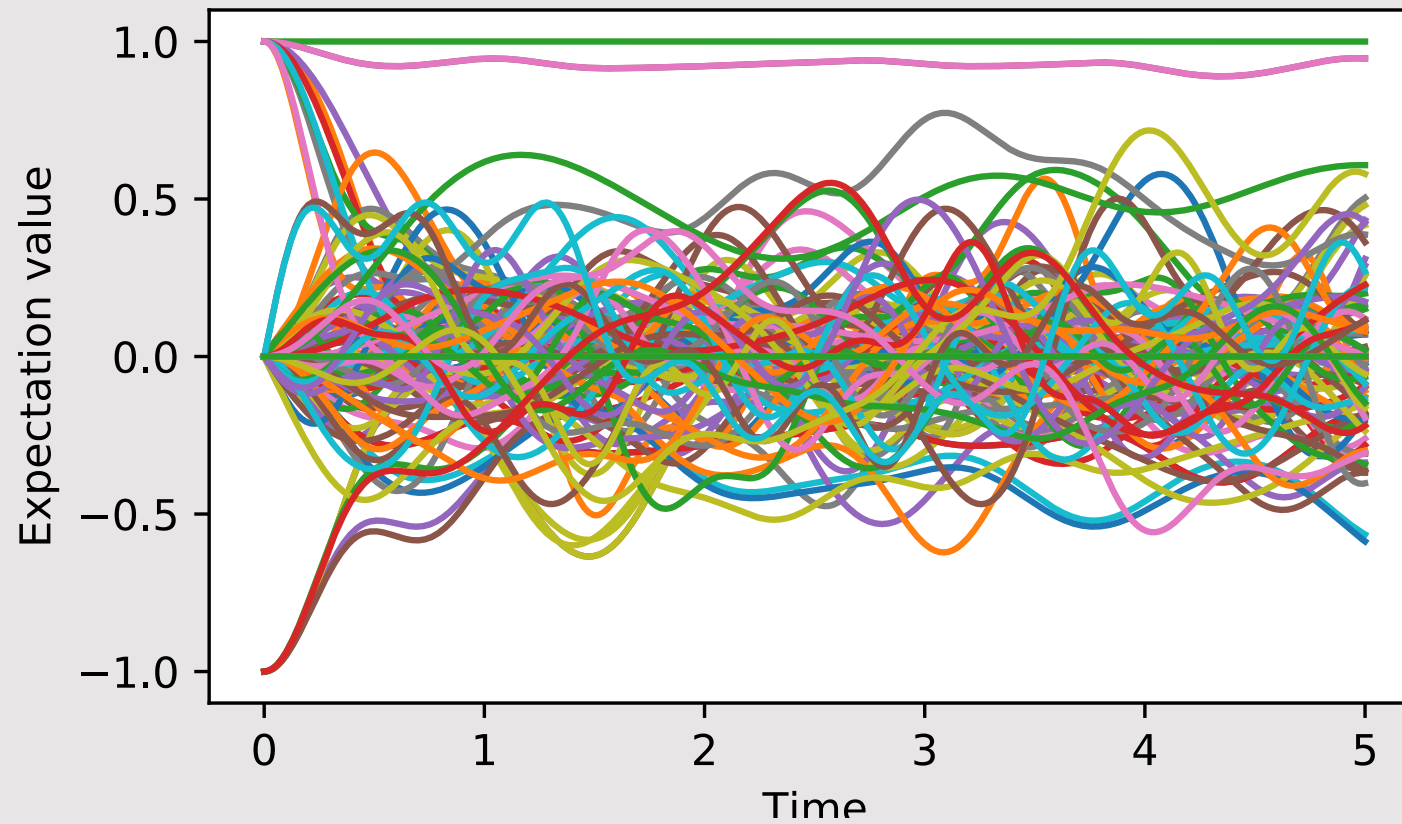
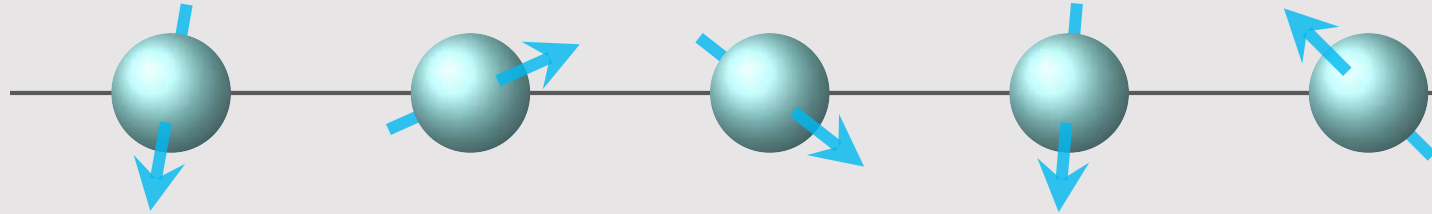
Consider some spin chain



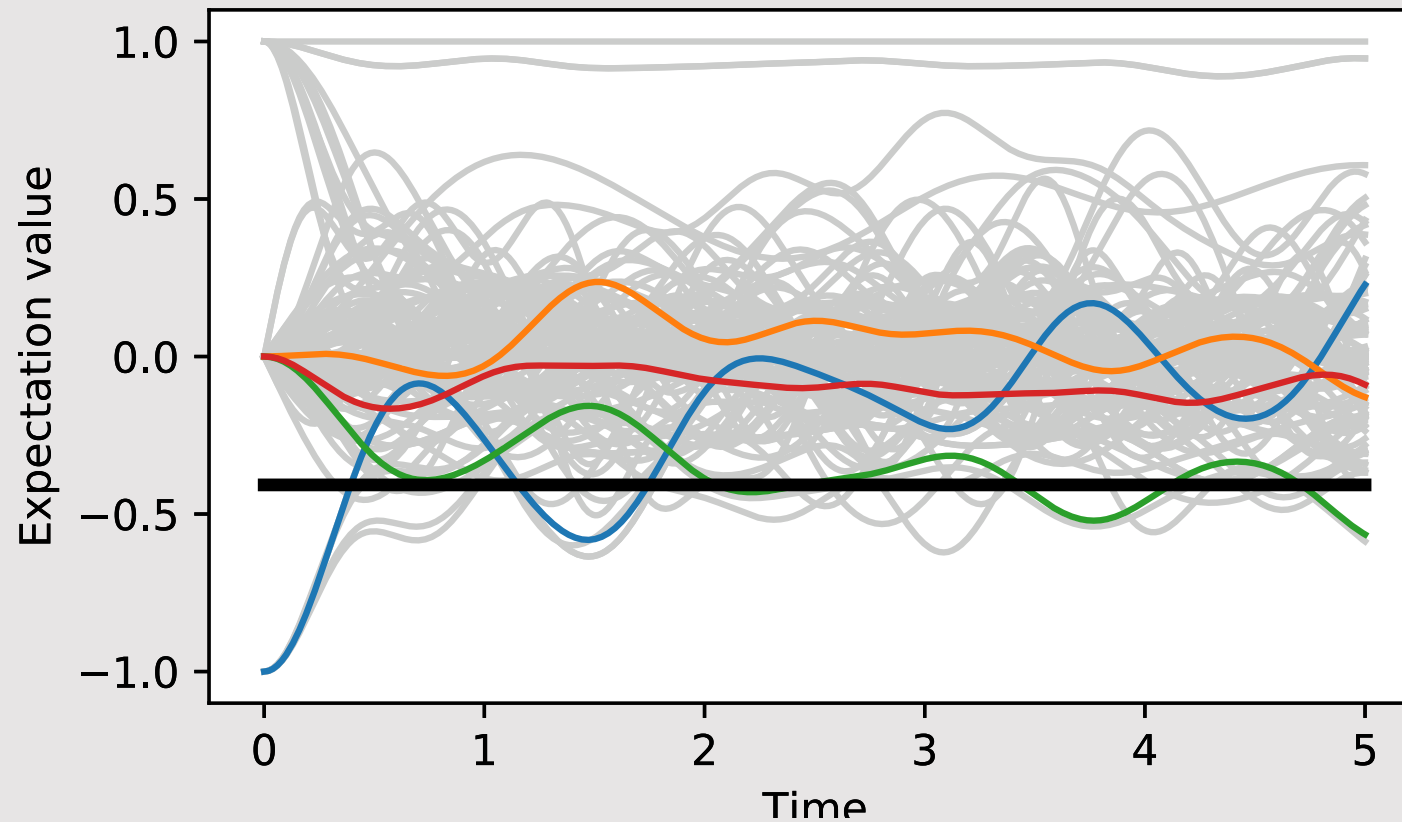
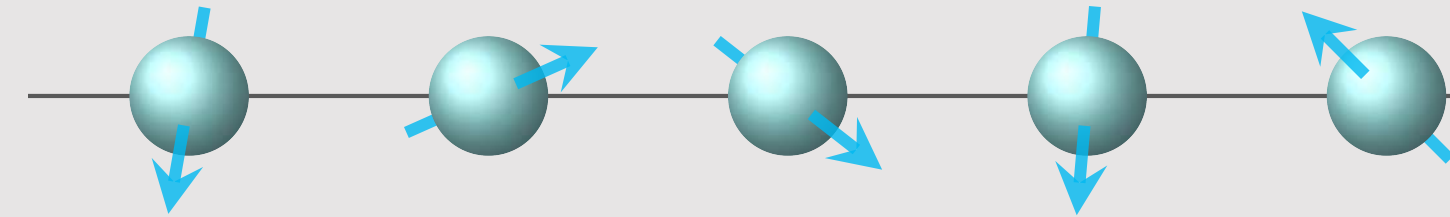
Measure time observables over time



Measure many of them



Find a constant of motion over the observation timescale



Repeat for
different initial
states and find
constant of
motion over
entire data set

$$a\langle Z_2 \rangle + b\langle X_1 Y_2 \rangle + c\langle Y_2 X_3 \rangle + d\langle X_1 Z_2 X_3 \rangle = \text{const}$$

Basis of protocol for measuring LIOMs

Given: Unitary operator U_F

Goal: Find an operator that commutes with the unitary $[U_F, L] = 0$

Solution. Follow the steps:

1. Start with a suitable local operator L_0
2. Evaluate the operator

$$L \propto L_0 + U_F L_0 U_F^\dagger + U_F^2 L_0 U_F^{2\dagger} + \dots + U_F^D L_0 U_F^{D\dagger}$$

In the limit $D \gg 1$,

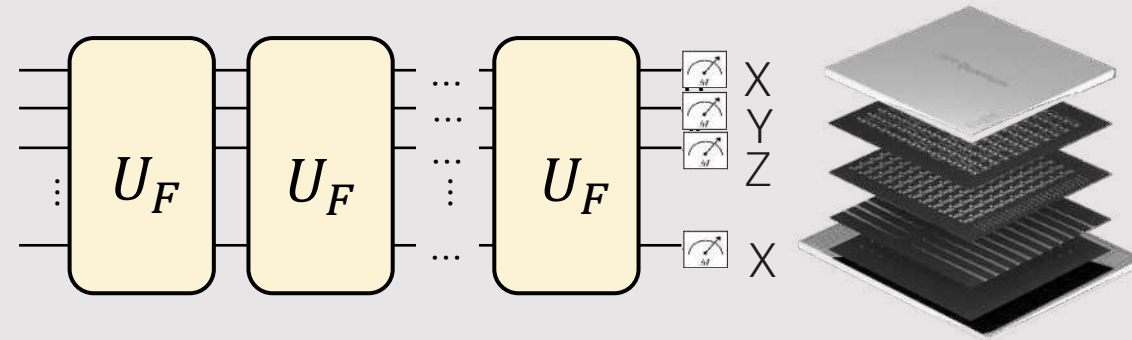
$$[U_F, L] \sim O(D^{-1})$$

Pauli decomposition of LIOM:

$$L = \sum_{\mu=1}^{4^n-1} a_\mu P_\mu$$

Coefficients are

$$a_\mu = \frac{1}{2^n} \frac{1}{D+1} \sum_{d=0}^D \text{Tr}(L_0 U^{d\dagger} P_\mu U^d)$$



Conserved quantity: LIOM

Integral of motion L

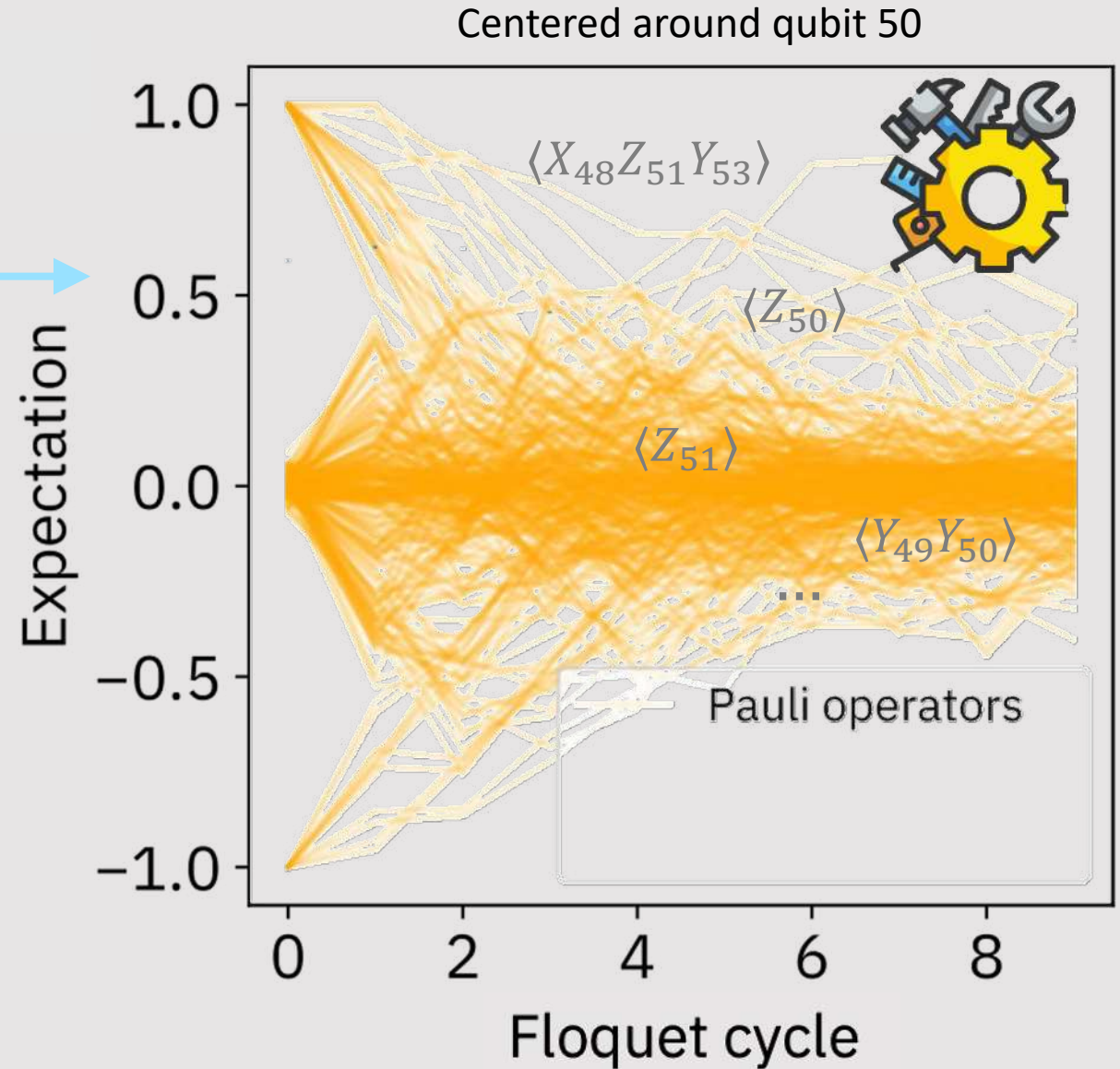
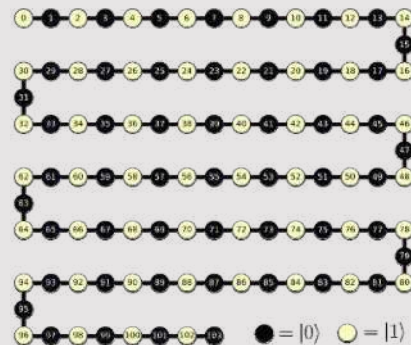
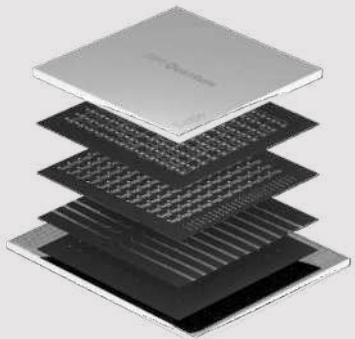
$$[U, L] = 0 \quad [H, L] = 0$$

$$\langle L \rangle = \text{const}$$

Local integral of motion (LIOM) L

$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_\mu P_\mu$$

local neighborhood



Conserved quantity: LIOM

Integral of motion L

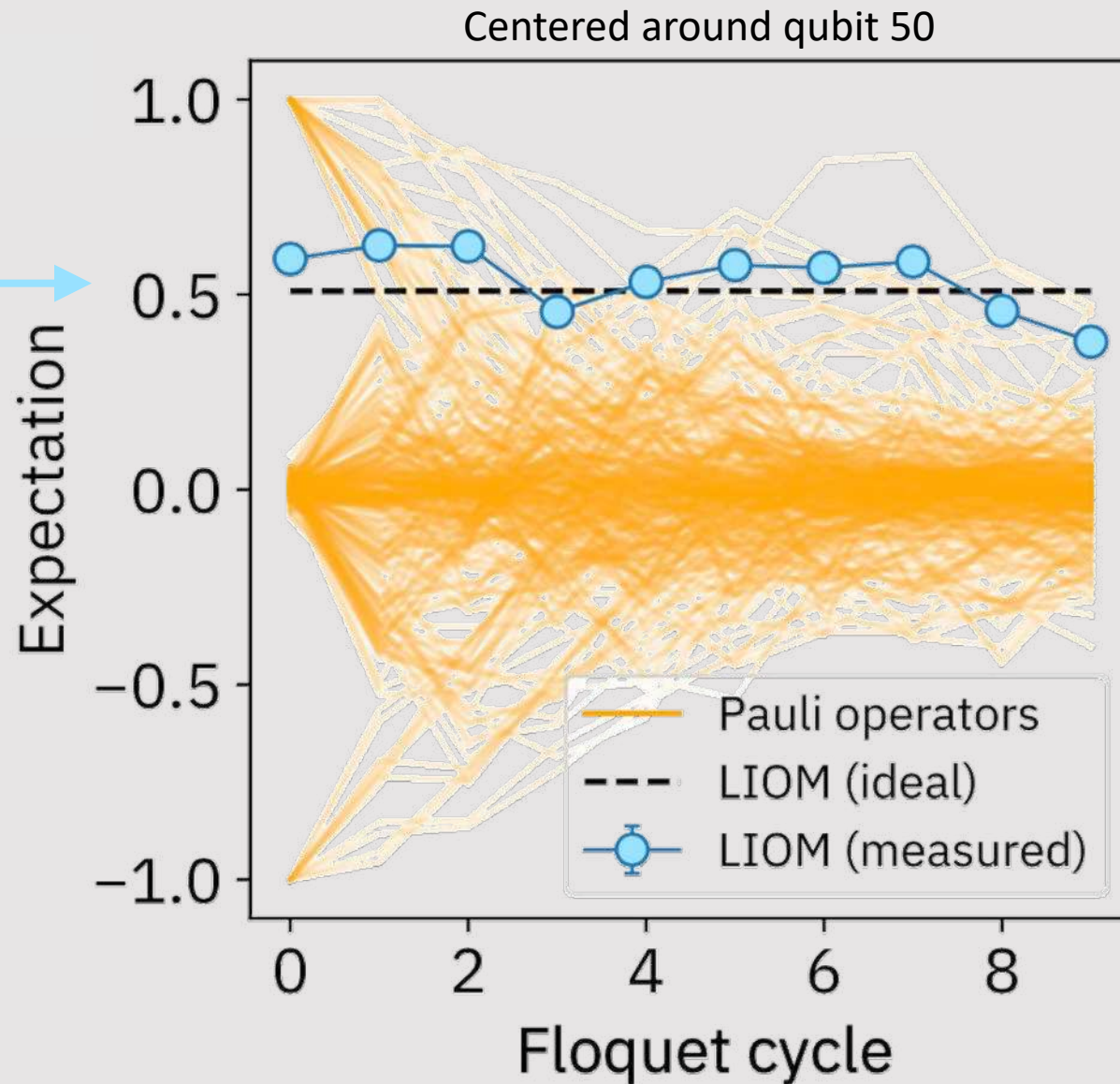
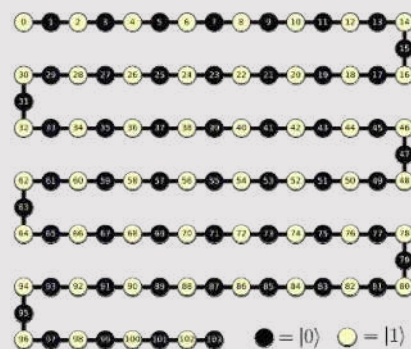
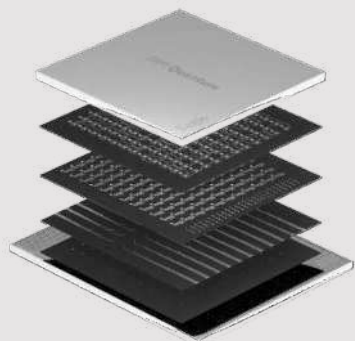
$$[U, L] = 0 \quad [H, L] = 0$$

$$\langle L \rangle = \text{const}$$

Local integral of motion (LIOM) L

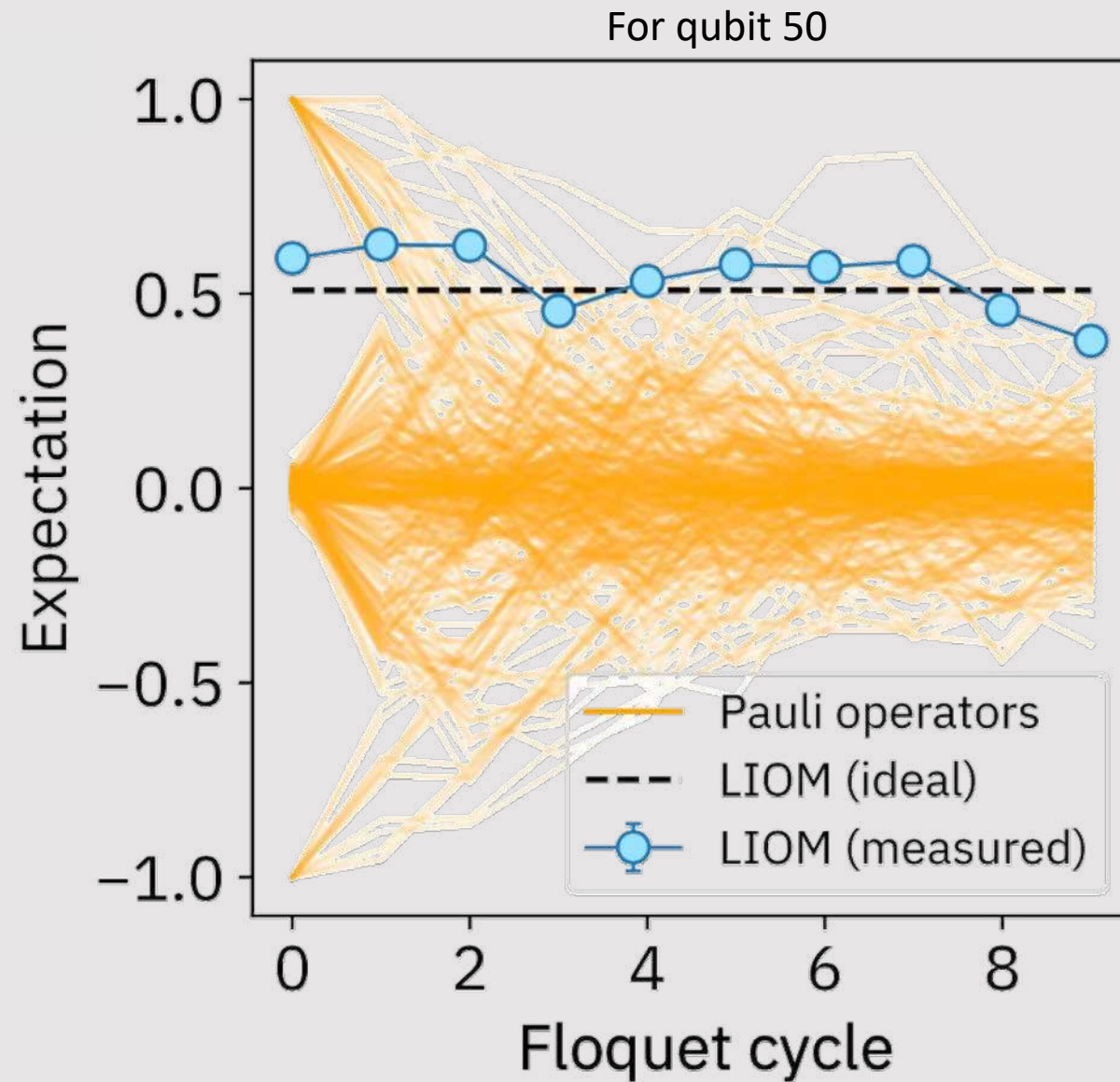
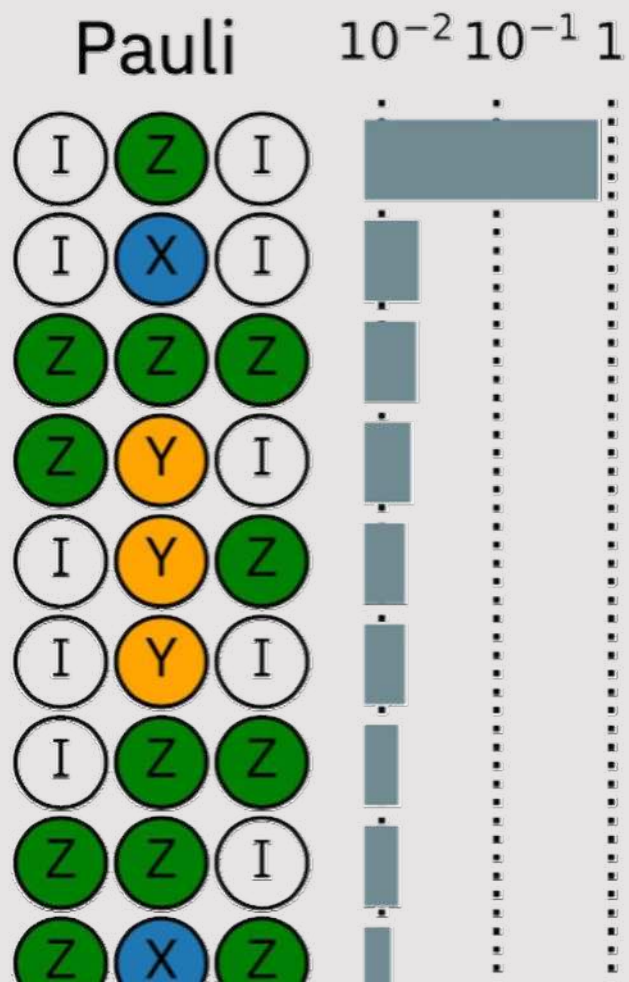
$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_{\mu} P_{\mu}$$

local neighborhood

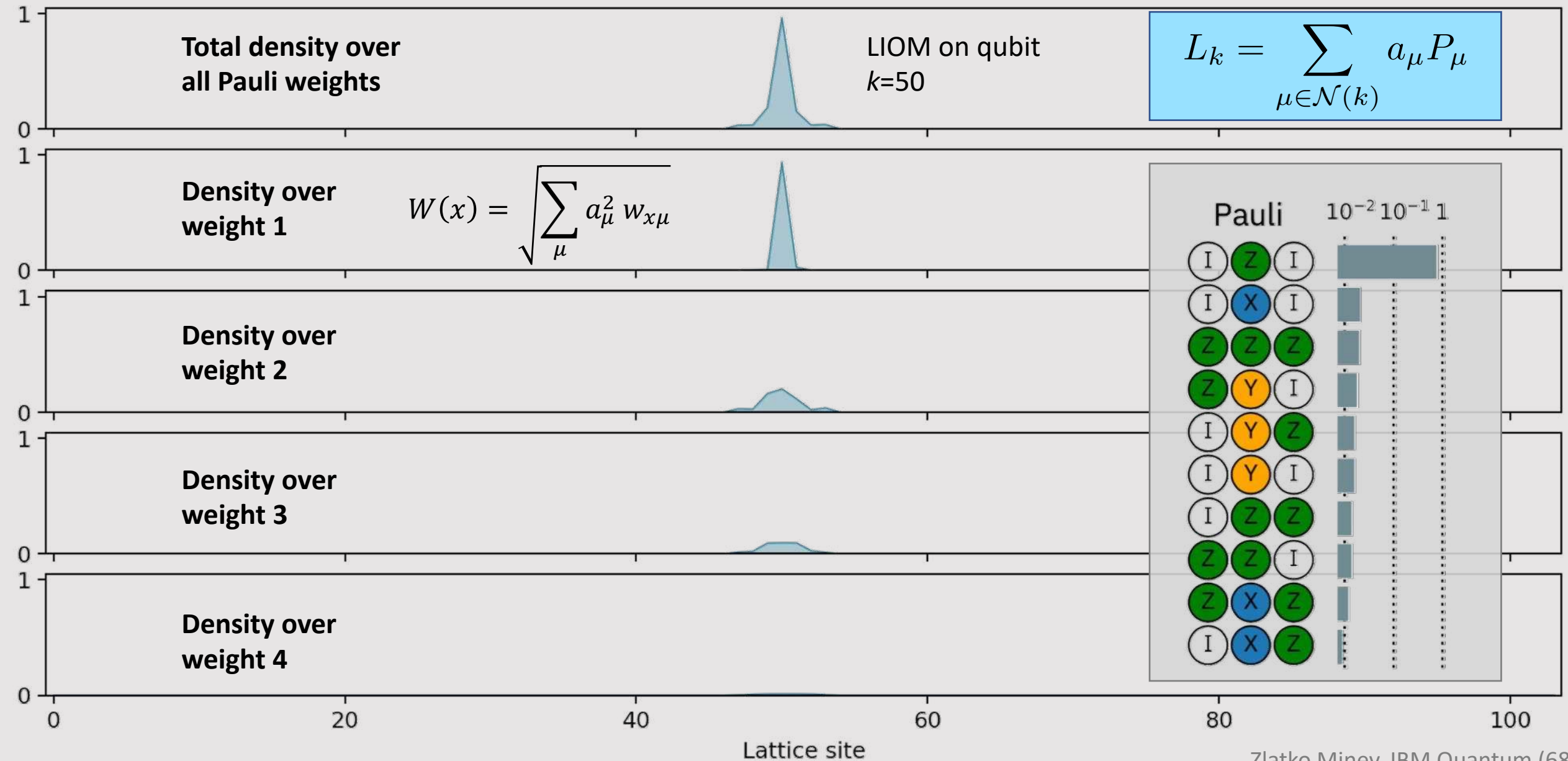


Conserved quantity: LIOM

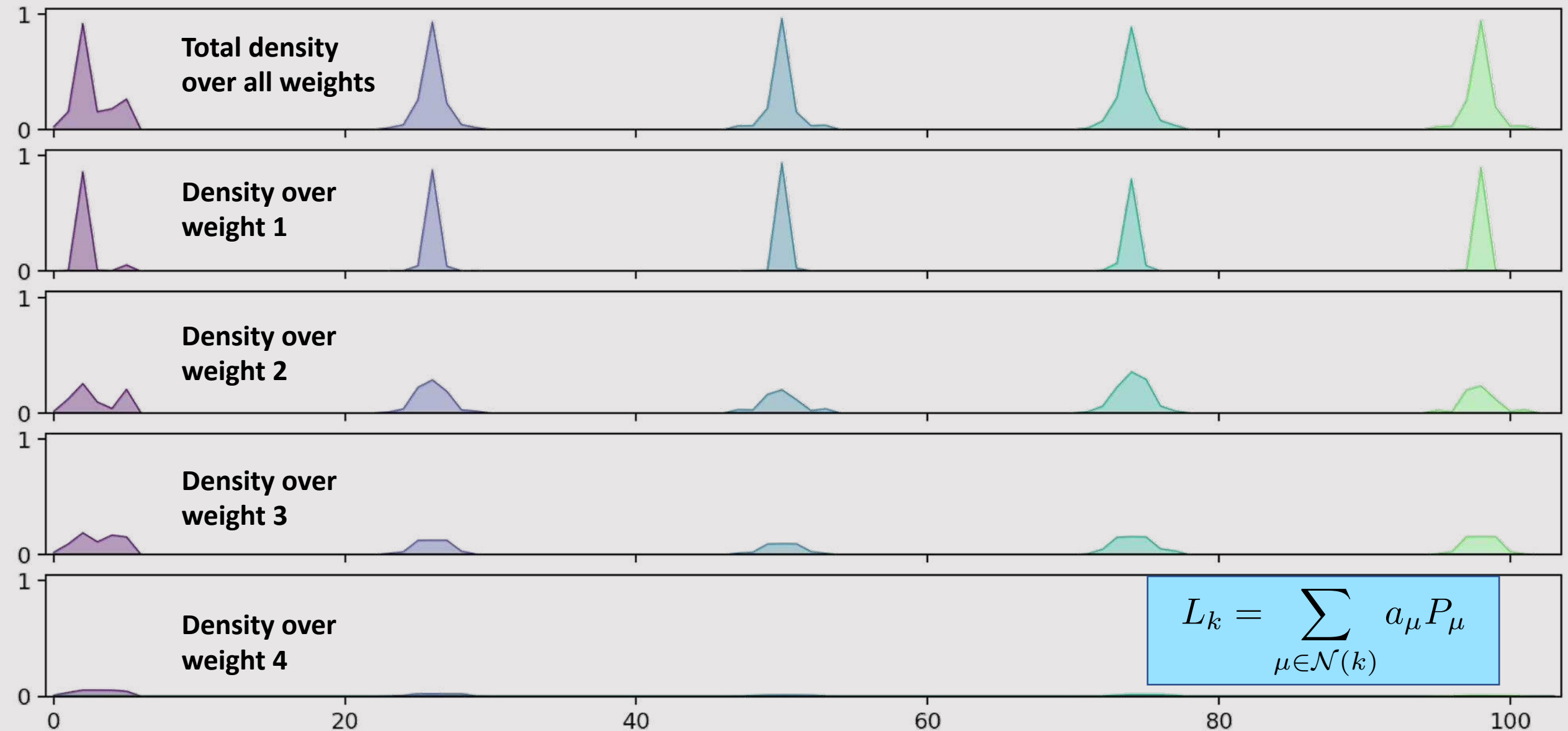
$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_{\mu} P_{\mu}$$



Operator density for LIOMs $\hat{L}_k$ in the 1D lattice

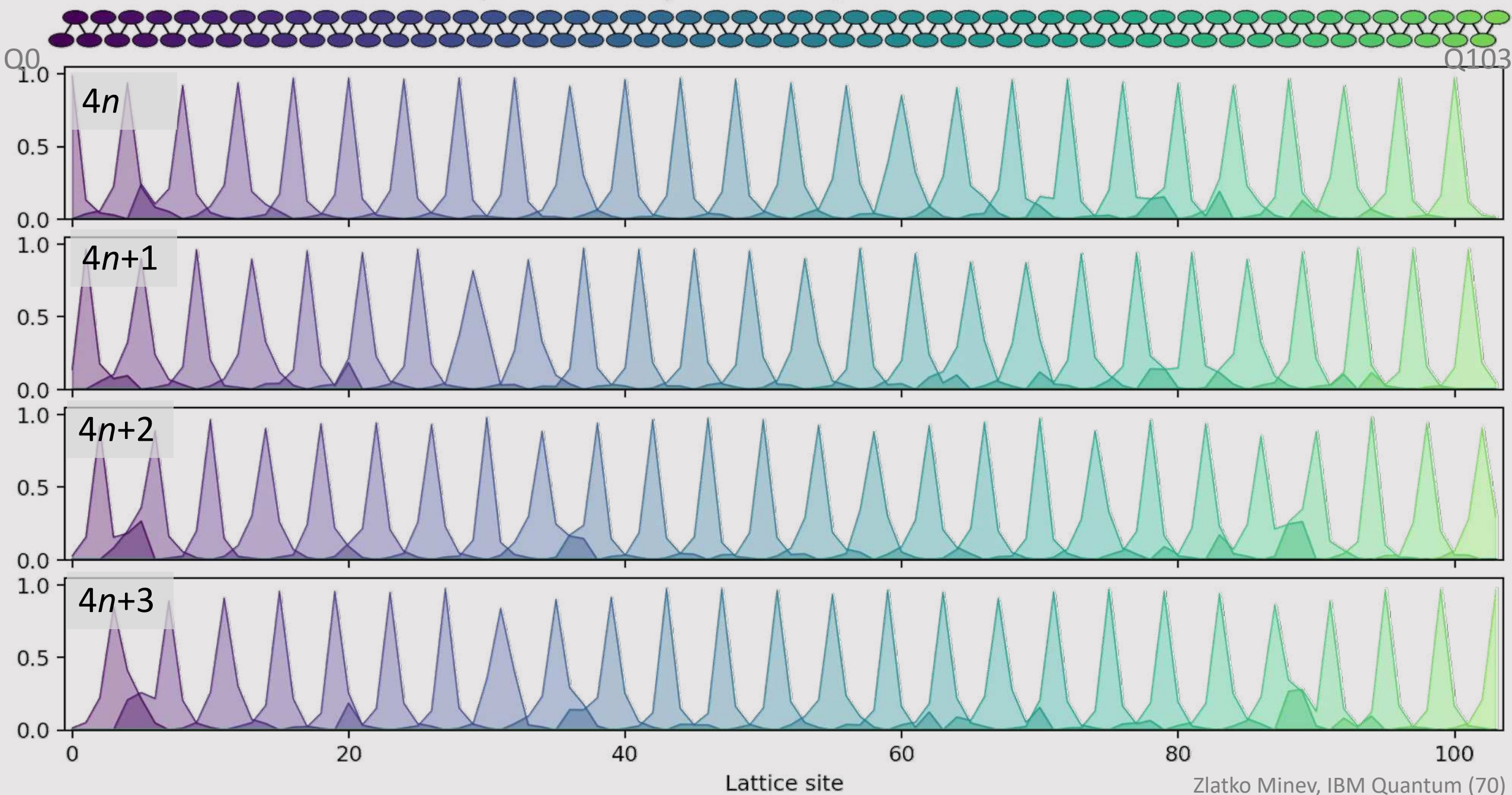


Operator density for LIOMs $\hat{L}_k$ in the 1D lattice



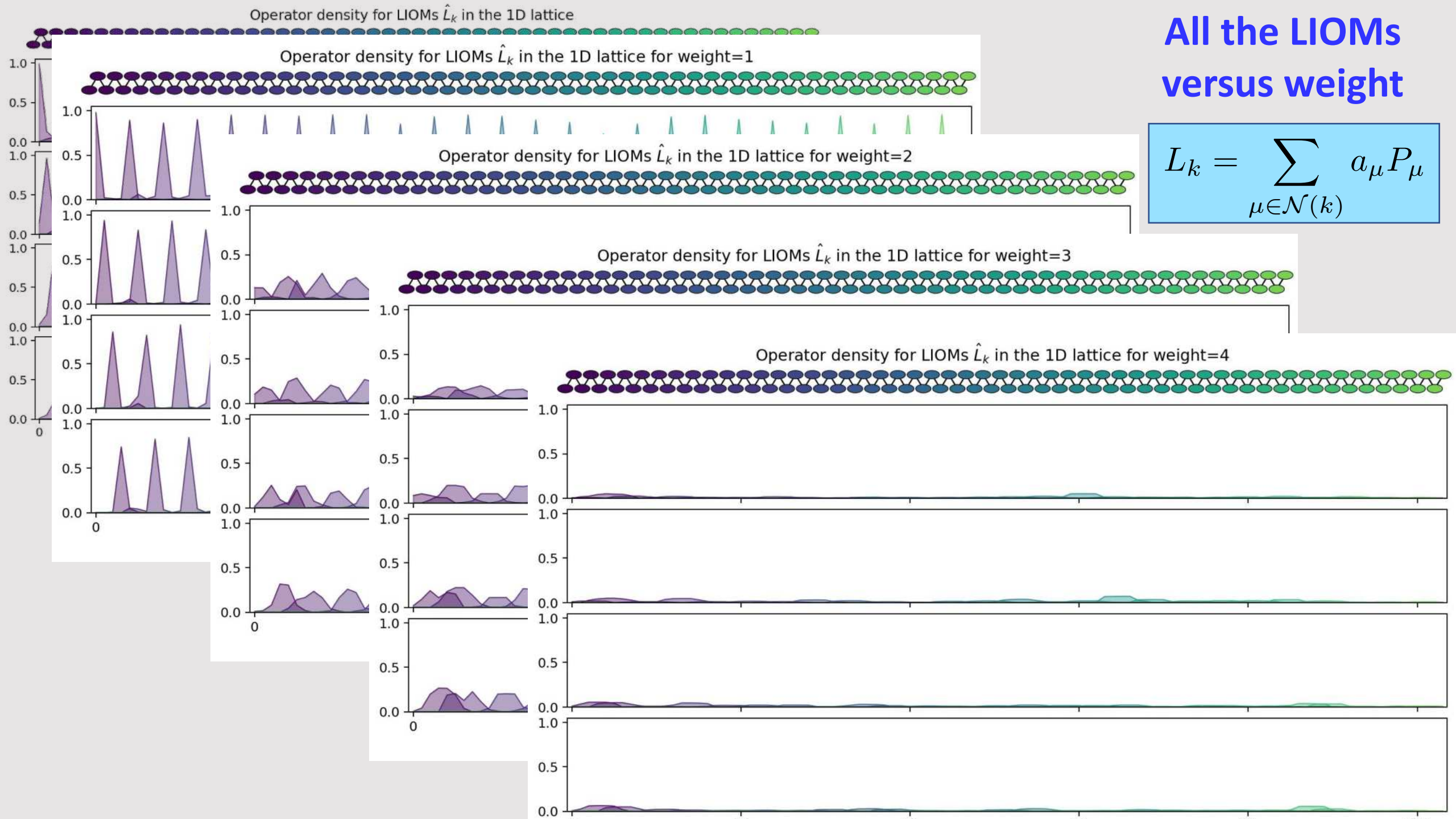
Operator density for LIOMs $\hat{L}_k$ in the 1D lattice

All the LIOMs

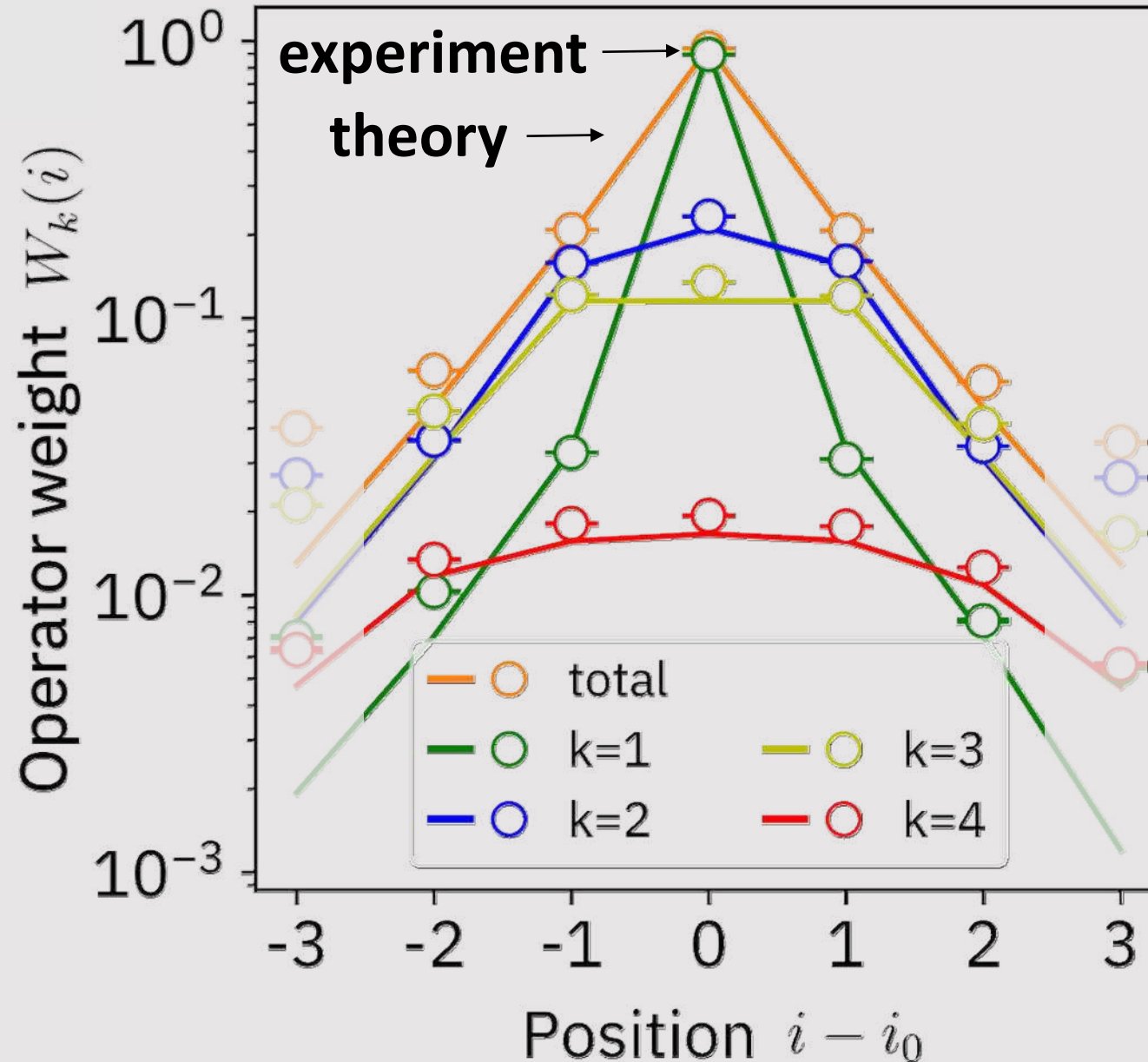
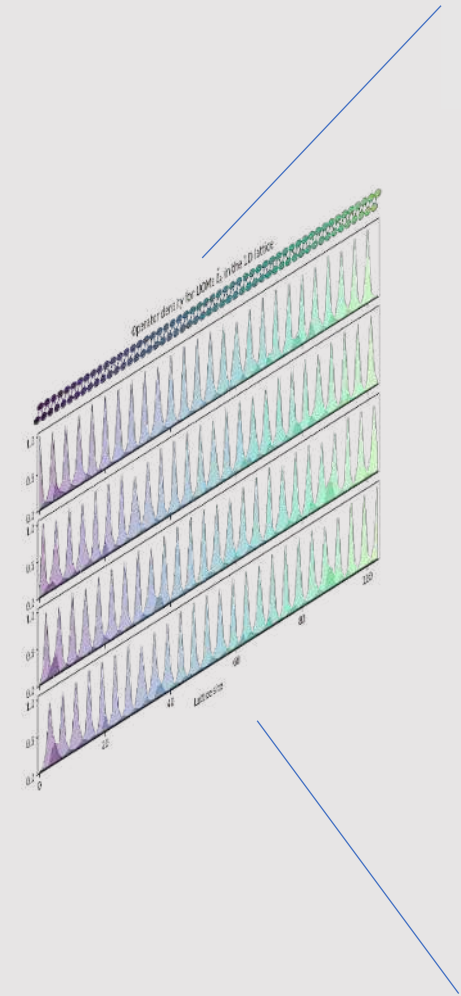


All the LIOMs versus weight

$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_\mu P_\mu$$



Average LIOM decomposed by weight: experiment vs. theory



$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_\mu P_\mu$$

total

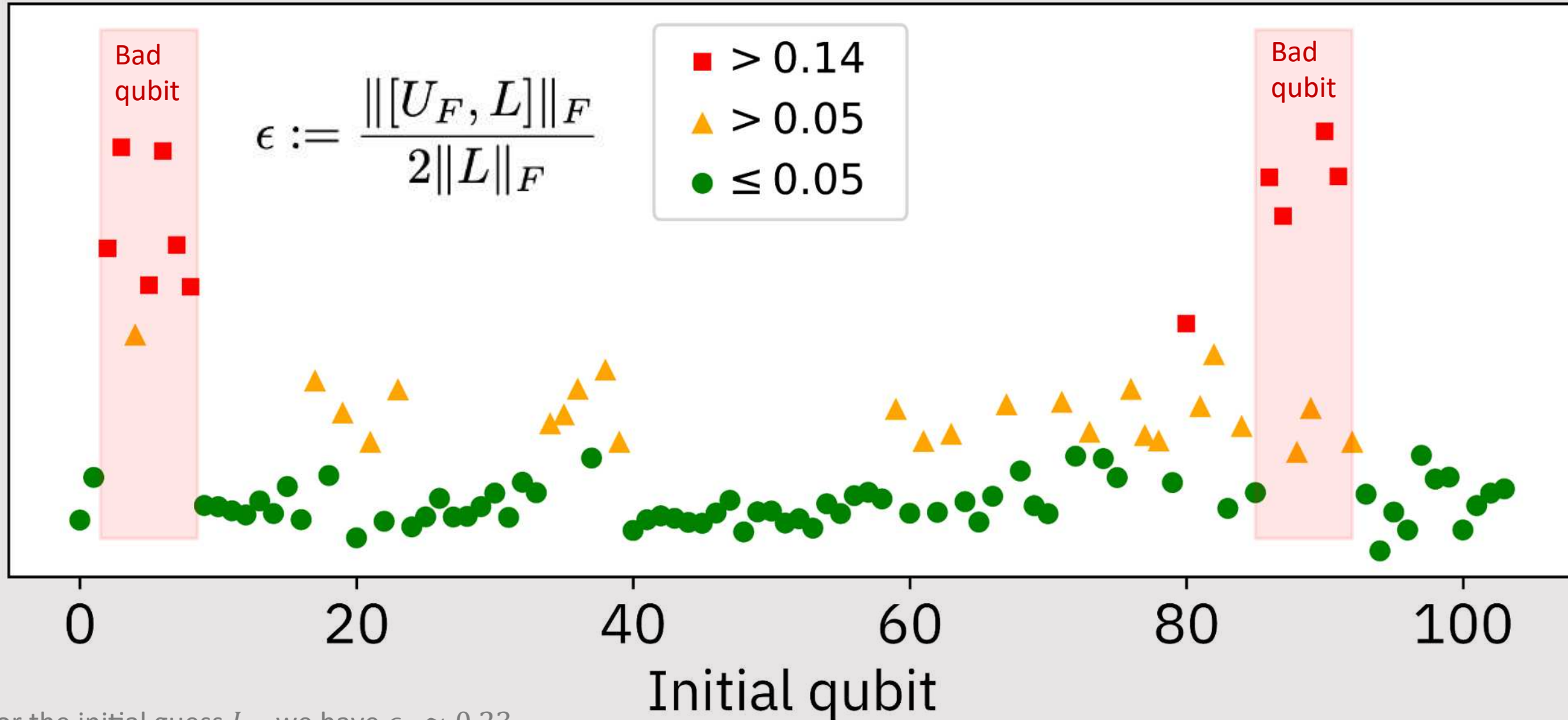
weight 1

weight 2

weight 3

weight 4

LIOMs error due to noise in device



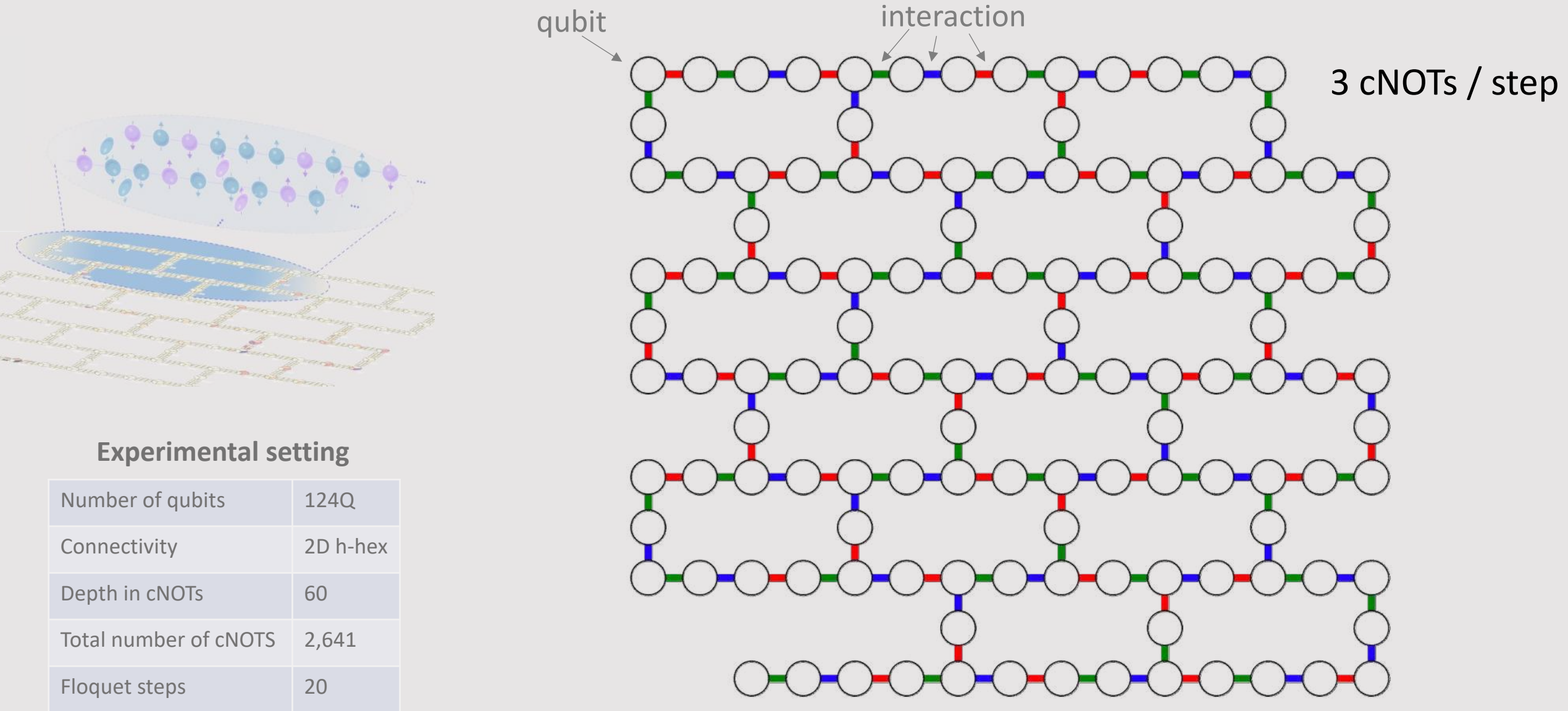
For the initial guess L_0 , we have $\epsilon_0 \approx 0.23$

2D

We study

- Spin imbalance
- OPDMs
- Scaling with system size
- LIOMs
- ...

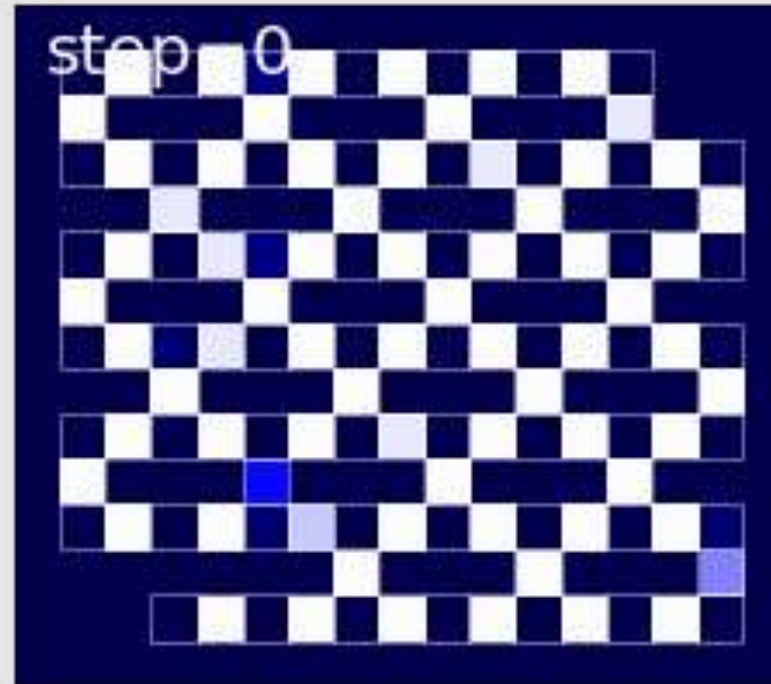
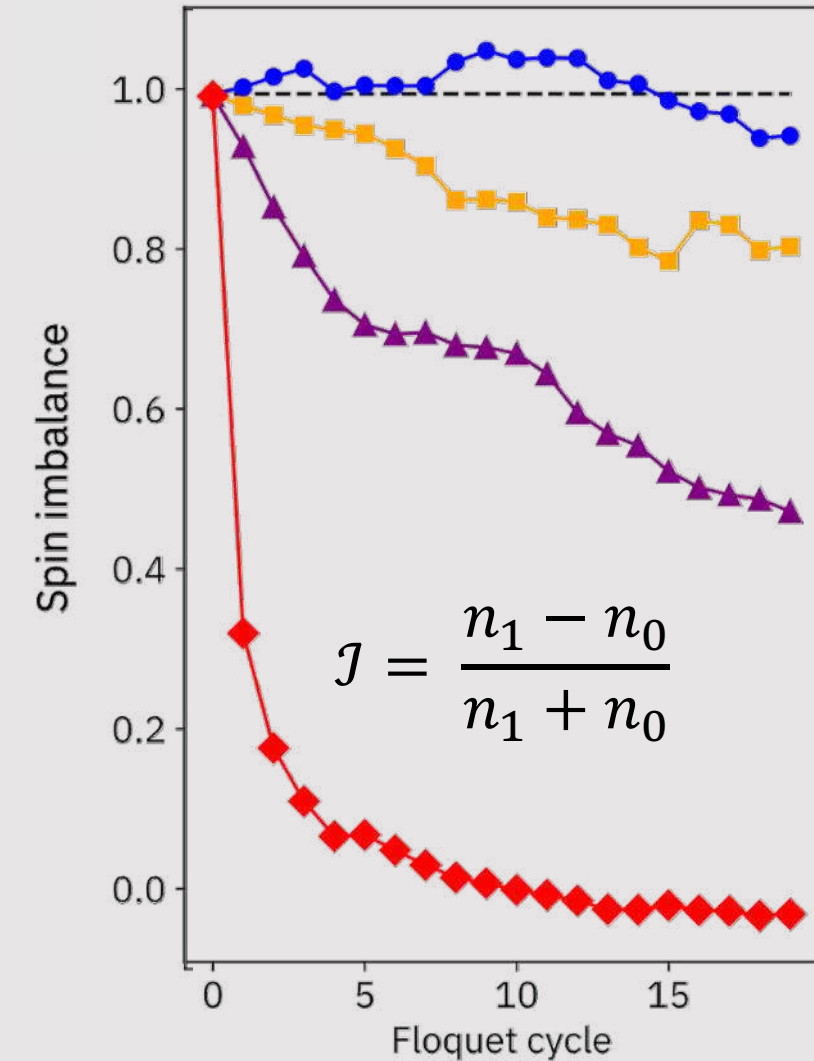
Interaction map and device layers



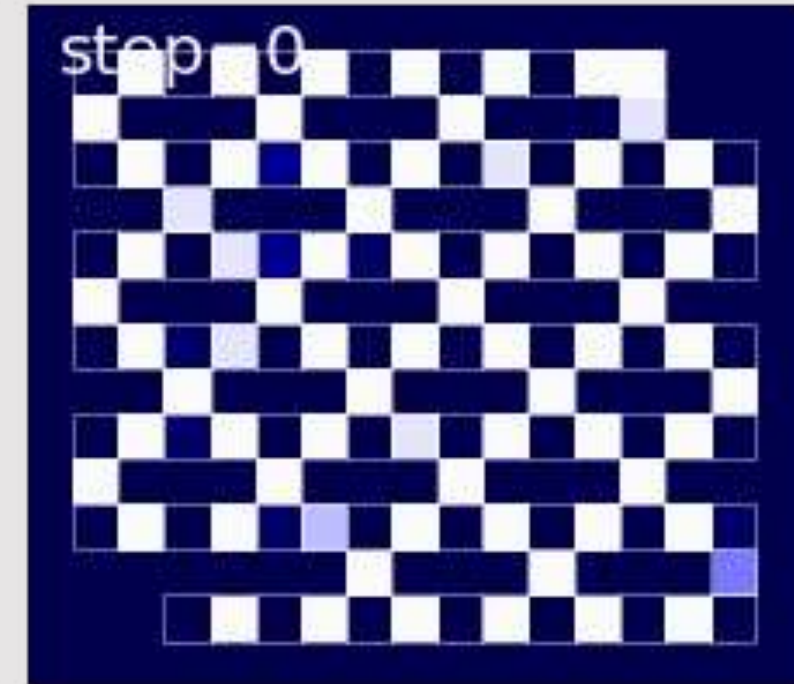
Color represent charge/spin polarization

IBM Quantum (76)

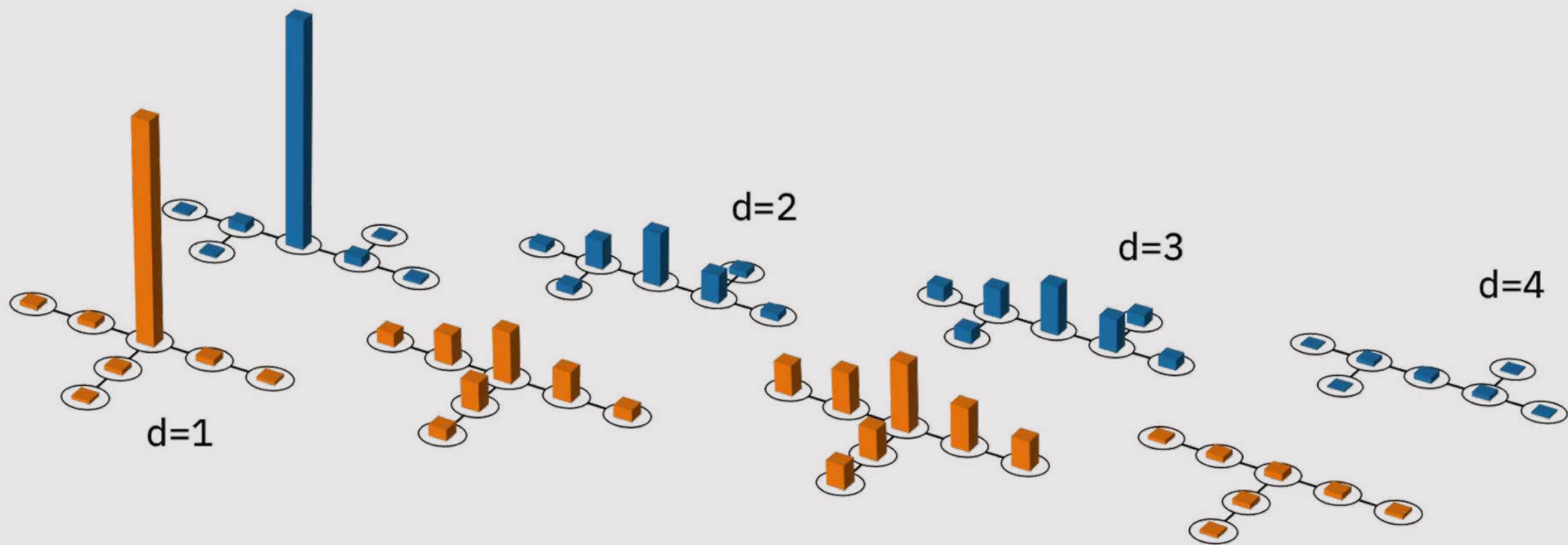
Spin imbalance for antiferromagnetic ordering



Thermalizing regime
 $\theta = 0.3\pi$



Prethermal regime
 $\theta = 0.1\pi$



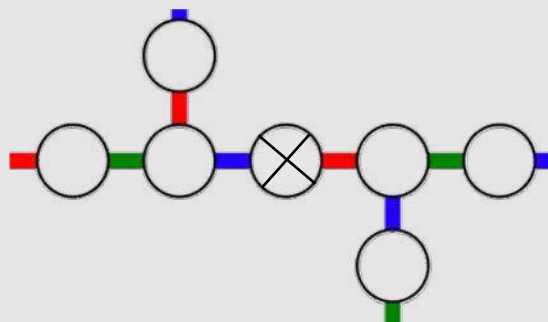
LIOM reconstruction in 2D

Prethermal LIOMS

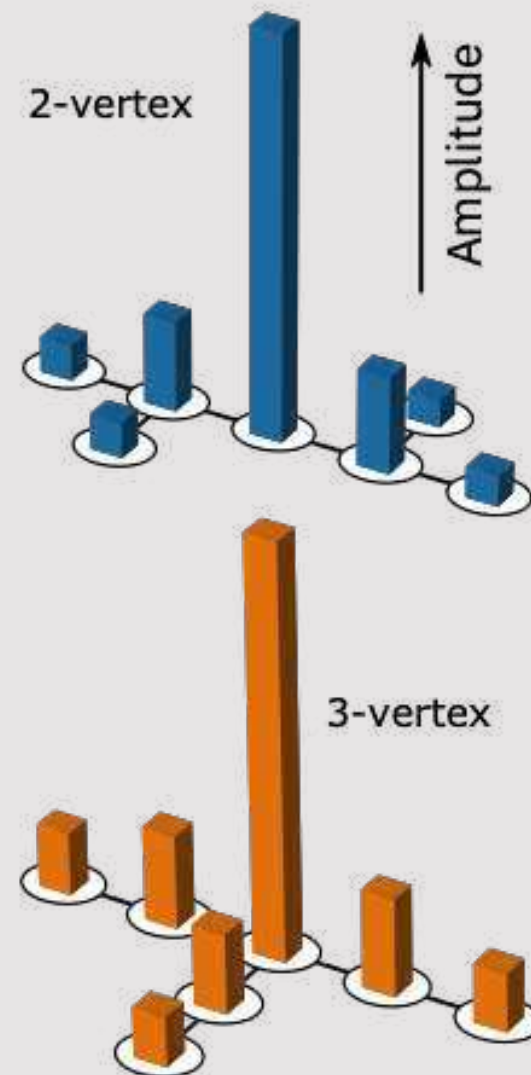
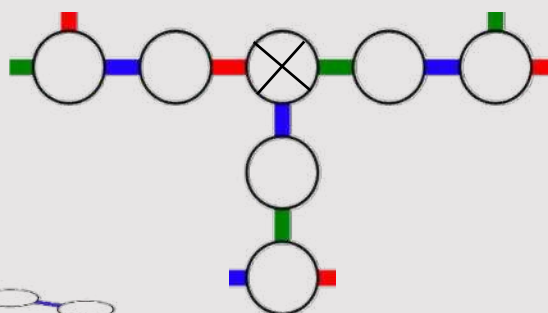
$$\left[e^{-iHt}, L_k \right] \approx 0$$

$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_\mu P_\mu$$

local neighborhood



⊗ LIOM center (position of the initial guess)



Discussion and future directions

Used 124Q, depth 60 circuits with error mitigation for many-body dynamics

Operationally restored a detailed portrait of a system's localized/prethermal dynamics in a new experimental regime

Explored a new model for MBL/ergodic phase transition

Use LIOMs for proper **calibration** of quantum hardware

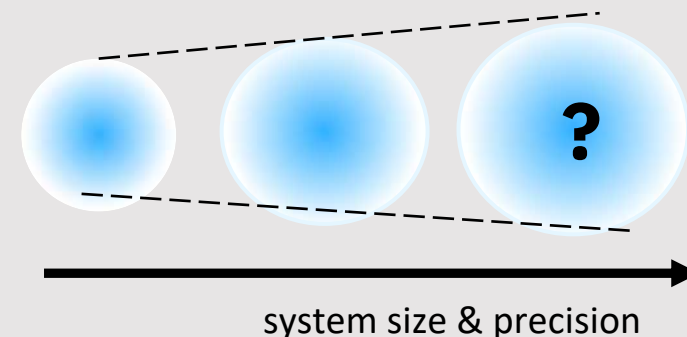
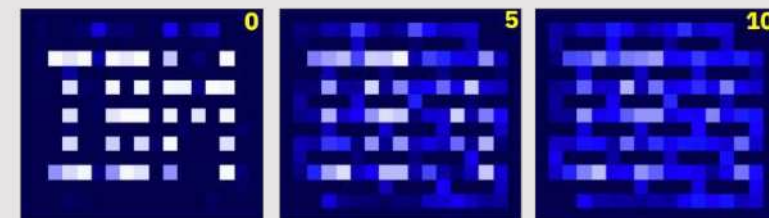
Use these systems as playground for **competition** quantum vs. classical

Deciding the **fate of MBL** in two dimensions?

Study other initial states, such as thermal

Improved error mitigation, further explore existing data, ...

...



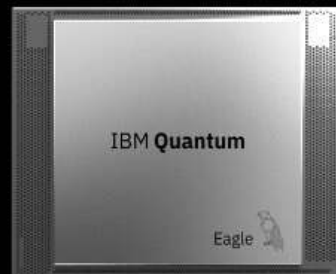
Scale and quality



2019
Falcon
27 Qubits



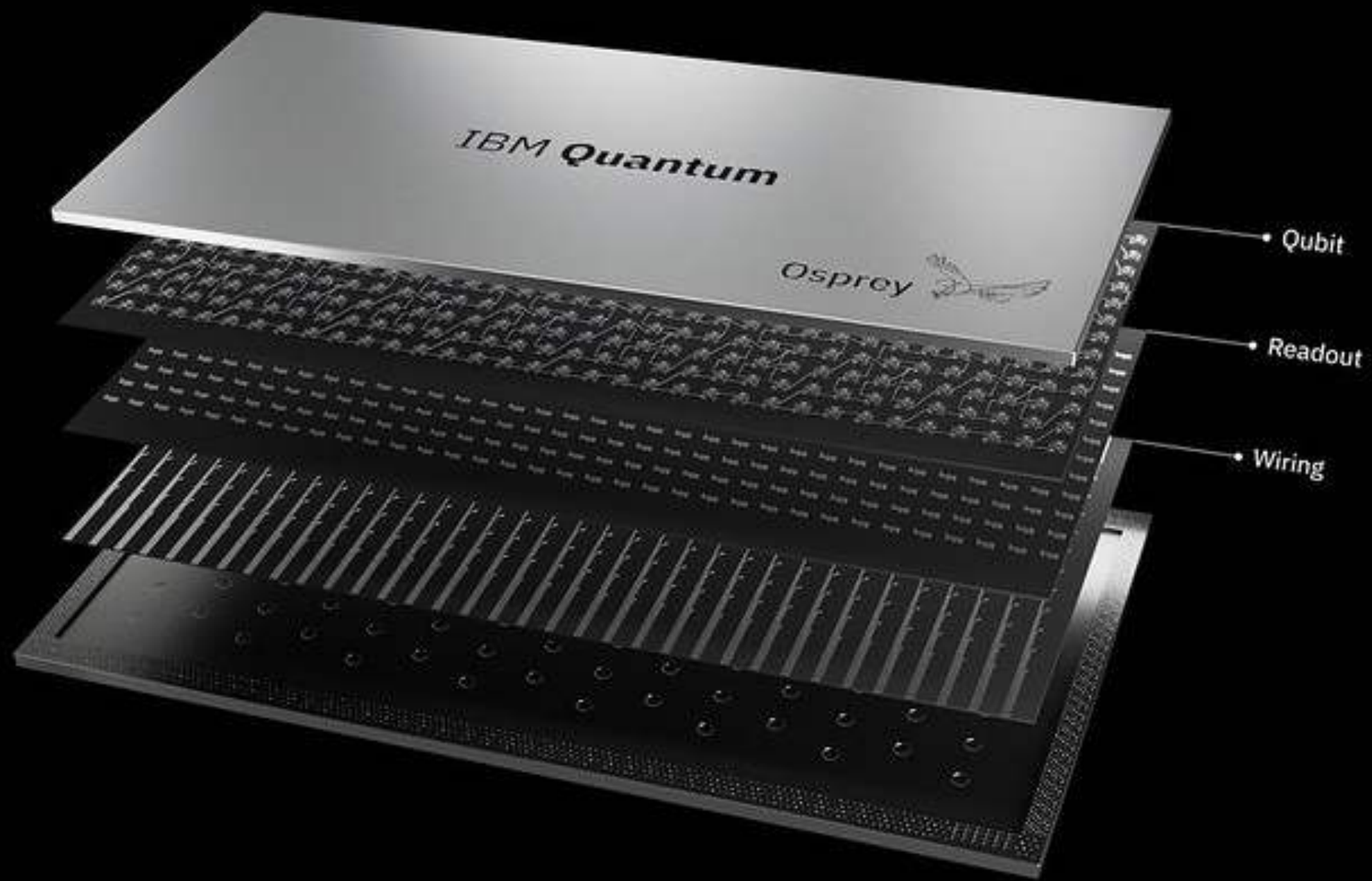
2020
Hummingbird
65 Qubits



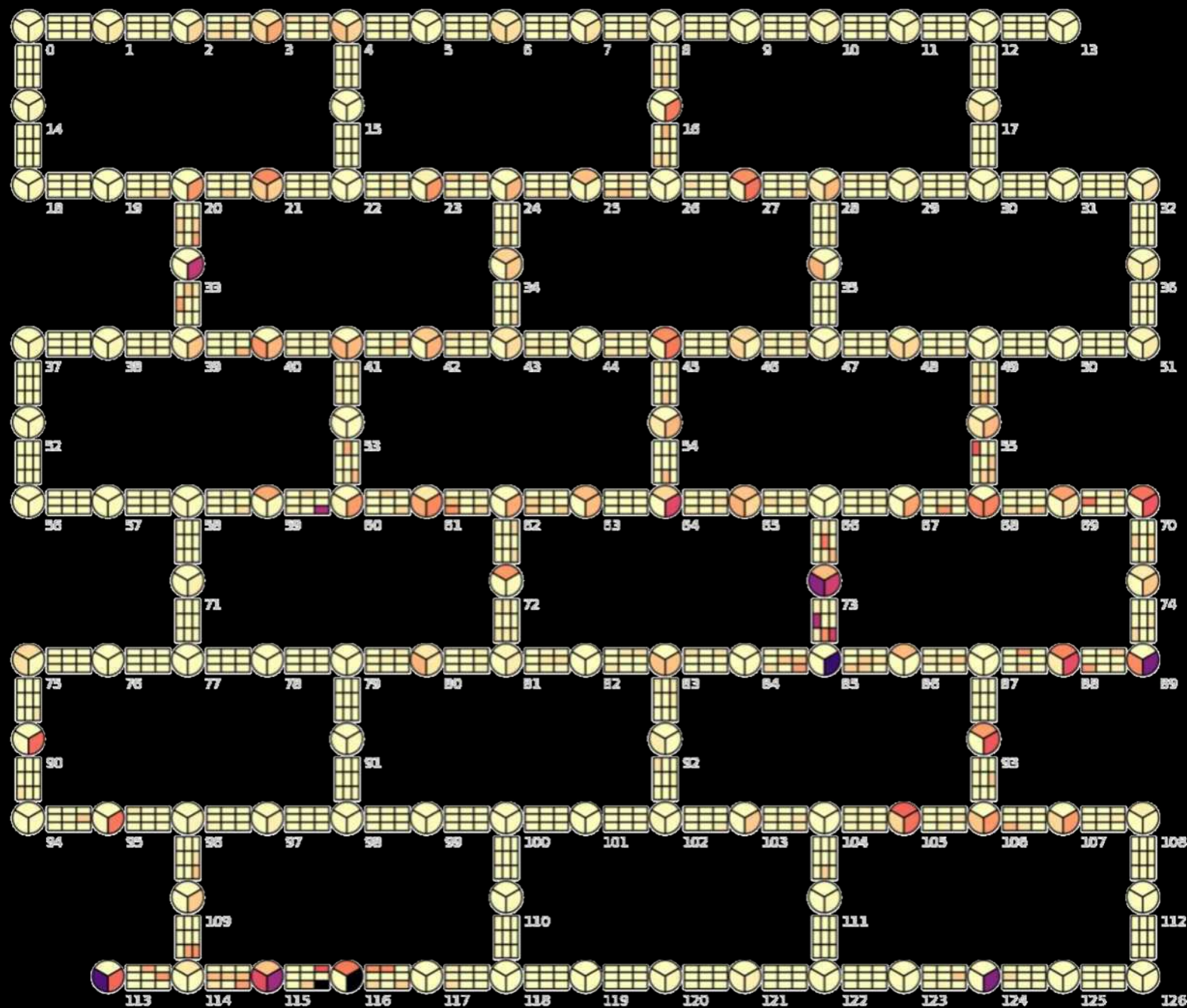
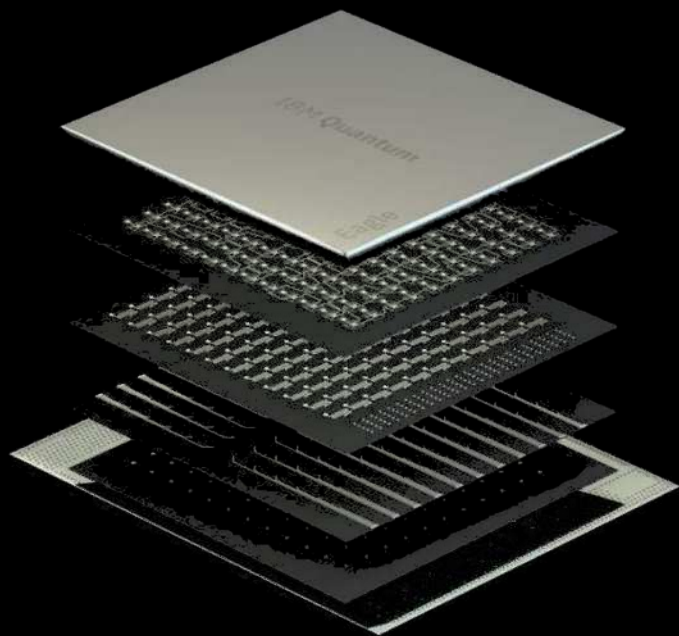
2021
Eagle
127 Qubits



2022
Osprey
433 Qubits



Quantum simulation



The important thing is not to stop questioning.
Curiosity has its own reason for existence.

One cannot help but be in awe when they
contemplate the mysteries of eternity, of life, of the
marvelous structure of reality.

It is enough if one tries merely to comprehend a
little of this mystery each day.

Albert Einstein

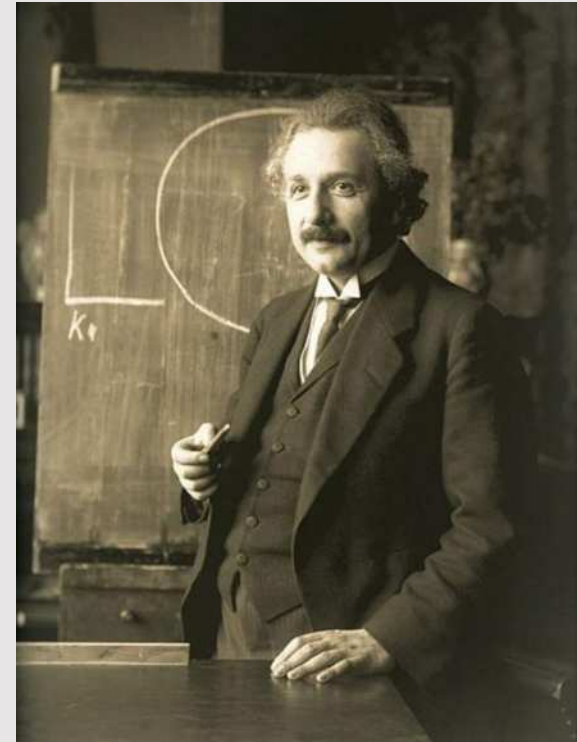


Photo: F. Schmutzer



@zlatko_minev



zlatko-minev.com

IBM Quantum